Antitrust, Market Power, and U.S. Economic Outcomes*

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November 7, 2026

Abstract

We assemble a narrative record of all federal antitrust case filings by the DoJ and FTC since the mid-1950s. We document a shift from civil non-merger and criminal enforcement toward civil merger cases in the early 1980s. Non-merger and criminal filings reduce defendants' markups while increasing sales and input expenditure. We construct sector-level enforcement proxies from publicly listed defendants' stock-market reactions to filings. Increased enforcement strengthens competition, raises output, R&D capital and patenting, and improves allocative efficiency. For civil non-merger and criminal cases, output gains are accompanied by greater input use. For civil merger cases, output rises without sustained increases in inputs, consistent with a role for productivity and resource allocation.

JEL Classification: C32, E30, L16, L40

Keywords: Antitrust, Market Power, Firm and Sector Outcomes.

*We are grateful for comments received from seminar audiences at the Hydra 2023 Conference, the 2024 Annual Meeting of the Society for Economic Dynamics, the 2023 Spanish Macroeconomics Network Meeting, IIES (Stockholm), LUISS, Paris School of Economics, University of Bristol, University of Manchester, University of Mannheim, the University of New South Wales, the University of Texas at Austin, and Sciences Po. We thank Alice Avenale, Marina Kotsa Or Koca and Diego Almonacid Lovera for excellent research assistance. Ravn acknowledges financial support from ERC Project BUCCAC - DLV 8845598.

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1 Introduction

Recent empirical research documents rising market power in the United States over recent decades (de Loecker, Eeckhout and Unger, 2020; Döpper et al., 2025). This trend raises concerns about higher prices, reduced consumer welfare, and distorted resource allocation. Antitrust enforcement seeks to address such concerns by protecting competition. However, the extent to which containing market power with antitrust interventions raises output, improves productivity, and stimulates innovation remains an empirical question.

In this paper, we show that curbing market power through antitrust enforcement stimulates economic activity. We establish these findings using a new dataset of federal antitrust case filings since the mid-1950s. First, firm-level estimates obtained using synthetic difference-in-differences show that defendants facing civil non-merger and criminal proceedings reduce their markups while expanding sales and input expenditure. Second, event studies reveal persistent declines in defendants' equity valuations following filings, with smaller losses for cases filed from 1980 onward. We use the abnormal returns estimated in narrow windows around these events to construct sector-level proxies of antitrust activity. Third, panel local projections show that civil non-merger and criminal enforcement reduces markups while raising output and input use. Merger enforcement also lowers markups and raises output, alongside improvements in productivity and resource allocation but without sustained input expansion. Both enforcement categories also stimulate innovation, as measured by increases in R&D capital and patent grants. Fourth, markup reductions are concentrated in the upper tail of the within-sector distribution, contributing to declines in both average markups and dispersion. Finally, historical counterfactual simulations suggest that enforcement has helped to restrain the rise in aggregate markups.

The empirical analysis draws on a new dataset of federal antitrust cases filed by the Federal Trade Commission (FTC) from 1954 and by the Antitrust Division (ATR) of the U.S. Department of Justice (DoJ) from 1957 through early 2026, constructed from the CCH Bluebook and information published on the antitrust authorities' websites. The dataset contains more than 6,600 cases, approximately two thirds initiated by the DoJ, involving close to 19,000 defen-

dant observations, including more than 13,300 involving firms. Despite the absence of a clear long-run trend in annual filings, the number of defendants involved each year has declined.

We show that the composition of federal antitrust cases brought against corporate defendants changed markedly around 1980. Until the late 1970s, antitrust authorities mostly pursued cases involving collusion, monopolization, and price discrimination. Since the early 1980s, enforcement has focused predominantly on anticompetitive mergers and acquisitions. This shift is particularly pronounced among publicly listed firms: merger-related filings account for 21.9 percent of listed corporate defendant observations before 1980 and 75.6 percent afterward. Publicly listed defendants also tend to be substantially larger than other listed firms in the same industry.

We leverage the antitrust data to study the impact of antitrust filings on firm-level outcomes. We use a synthetic difference-in-differences approach and focus on civil non-merger and criminal cases involving publicly listed companies with available accounting data. Averaged over the filing year and the following ten years, defendants' markups[1] decline significantly by 2.8 percent, while sales and spending on inputs rise by around 10 and 13 percent, respectively.

We extend the analysis to industry outcomes and merger enforcement using defendants' stock-market responses to case filings. We estimate daily abnormal equity returns in narrow windows around filings and interpret them as capturing the market assessment of profit changes associated with reduced market power. Combining these responses with defendants' pre-filing market capitalizations yields enforcement proxies that vary across sectors and over time and reflect differences in the economic importance of individual cases. We find that defendants' equity valuations fall significantly on the filing day and decline further on the following trading day. These losses persist: one trading month after filing, cumulative abnormal returns reach -1.9 percent for civil merger cases and approximately -1 percent for civil non-merger and criminal cases. Valuation losses are smaller for filings from 1980 onward, and responses to civil non-merger and criminal cases are no longer statistically distinguishable from zero.

We use the proxies to estimate the dynamic causal impact of curbing market power through antitrust enforcement on sectoral outcomes. We rely on a panel local projection estimator with both sector and time fixed effects and a large set of controls, and the results are robust to a

[1] We recover markups with a production function approach.

wide range of specification changes. We find that an increase in civil non-merger and criminal antitrust enforcement is associated with a decline in markups, firm concentration, and profit margins, consistent with antitrust improving competition. In parallel, real value added and employment rise very persistently while prices fall. There is no trade-off between competition and innovation as both R&D capital and patent grants rise. Finally, allocative efficiency, as measured by markup dispersion, improves.

Increased civil merger enforcement is also associated with stronger competition and higher sectoral output. Markups and profit margins fall, while real value added rises. In contrast to civil non-merger and criminal enforcement, however, we find less evidence of sustained increases in employment or input use. These responses are consistent with a role for higher productivity and improved resource allocation. R&D capital and patent grants also rise, indicating that procompetitive merger enforcement can be accompanied by greater innovation.

Antitrust enforcement reduces markups most strongly in the upper tail of the within-sector distribution. Nonparametric panel local projections indicate that declines in this part of the distribution account for reductions in both average markups and markup dispersion. To facilitate interpretation in terms of changes in market power, we use the antitrust proxies as instruments for average markups, finding that a one percent increase in markups lowers real value added by approximately 1.4–1.9 percent after five years, regardless of the proxy used. Finally, separate panel VAR counterfactual simulations indicate that enforcement has restrained the rise in market power: by 2024, aggregate markups would have been approximately 2.2 percentage points higher without civil non-merger and criminal enforcement and 4.1 percentage points higher without merger enforcement.

The remainder of the paper is structured as follows. In the next section, we describe the institutional background, present the dataset, and document patterns in antitrust enforcement. Section 3 estimates firm-level responses to filings. Section 4 examines equity-market reactions and constructs the enforcement proxies. Section 5 estimates sector-level responses to enforcement, including changes in the markup distribution. Section 6 uses the proxies as instruments for average markups and examines the extent to which enforcement has restrained the historical rise in market power. Section 7 concludes.

1.1 Related Literature

Our analysis significantly extends existing accounts of modern federal antitrust enforcement, combining DoJ filings from 1957 and FTC filings from 1954 through early 2026. Earlier contributions include Posner (1970), covering DoJ (1890–1969), FTC (1915–1954), and private (1890–1969) cases; Gallo et al. (2000), covering DoJ cases over 1955–1997; and Ghosal (2011), studying DoJ enforcement over 1958–2002. Our analysis thus adds two decades of data for the DoJ and seven decades of data for the FTC.

Our event-study analysis extends earlier work by providing a systematic account of the valuation effects of U.S. federal antitrust enforcement across case types and over time. Bosch and Eckard (1991) estimate abnormal stock returns for 127 firms charged with price fixing under Section 1 of the Sherman Act during 1962–1980.[2] Aguzzoni, Langus and Motta (2013) and Günster and van Dijk (2016) study the impact of unannounced inspections and infringement decisions of the European Commission on firm valuations. Our analysis covers a substantially larger sample, comprising 2,446 firm-filing events.

A recent literature argues that competition has declined over recent decades. A number of papers have estimated markups using the "production function approach," see, e.g., de Loecker, Eeckhout and Unger (2020), Traina (2018), and Hasenzagl and Pérez (2023). Other studies have applied a "demand-system" approach to estimate markups for a large number of products, e.g., Döpper et al. (2025). These studies find that markups have risen over the last few decades and that this rise has been accompanied by increased markup dispersion. Barkai (2020) and Karabarbounis and Neiman (2019) focus instead on the profit share and document that it has risen since the 1980s. Autor et al. (2020) exploit U.S. Census data for the 1982–2012 period and document rising firm concentration within 4-digit industries. They argue that this trend mainly reflects reallocation towards large and productive firms. Gutiérrez and Philippon (2017) document rising firm concentration at the economy-wide level.[3] Much of this literature

[2] Studying the 1904–1945 period, Bittlingmayer (1992) finds that the Dow Jones Industrial Average declines when the number of antitrust cases filed by the DoJ increases.

[3] Rossi-Hansberg, Sarte and Trachter (2021) find that "local market concentration" has declined. Berry, Gaynor and Scott Morton (2019) warn against interpreting correlations between output, or other outcomes, and market concentration measures as causal evidence on the impact of market power.

has focused on the measurement of market power, while we study the impact of antitrust on these indicators, and identify the causal effects of changes in measures of market power.

Most closely related to our study are Babina et al. (2023) and Besley et al. (2021). Using data on DoJ antitrust cases filed during 1971–2018, Babina et al. (2023) focus on non-traded industries, using variation in enforcement across states. They find that enforcement raises the labor share and business formation. Our study extends theirs in several dimensions. Their identification strategy excludes nationwide cases, restricting their ability to study enforcement against large firms operating across states. First, we examine all cases against publicly listed firms, and thus include firms whose scale and influence on competition give these cases particular economic importance. Second, our data cover a longer period and include both FTC and DoJ filings. Third, we examine merger challenges, which are excluded from their analysis of economic outcomes. Fourth, we examine a broader range of firm- and sector-level outcomes, including prices and indicators of productivity and innovation, and relate these responses to changes in the competitive environment.

Besley et al. (2021) combine firm data from 94 countries with an index of the scope of antitrust laws to examine the relationship between antitrust policy and profitability. They find lower profit margins in non-traded sectors in countries with stronger antitrust policies.[4] Our analysis uses observed enforcement actions to trace the dynamic responses of indicators of competition and economic activity, extending the evidence based on differences in legal frameworks across countries.

2 U.S. Antitrust

2.1 Legislative Background

Formal U.S. antitrust legislation dates back to the Interstate Commerce Act of 1887 introduced to regulate the U.S. railway industry, which was perceived as charging high prices because

[4]Similarly, using a panel of 22 countries, Buccirossi et al. (2013) find that stronger competition policy, measured by a competition policy index, is associated with higher productivity growth. Vaziri (2021) relates firm- and sector-level outcomes to agency budgets in the U.S. and an index of competition law in Europe, finding some evidence that larger budgets and stricter laws stimulate entry and productivity growth.

of limited competition and collusion.[5] Broader antitrust legislation followed three years later when Congress passed the Sherman Antitrust Act of 1890 to combat "business trusts." The next major development in U.S. antitrust legislation came in 1914, when the Clayton Antitrust Act and the Federal Trade Commission Act were enacted under President Woodrow Wilson. These acts still form the backbone of federal U.S. antitrust legislation, but their interpretation and implementation have evolved over time through the U.S. judicial system.[6]

Section 1 of the Sherman Act is concerned with collusion and prohibits "*every contract, combination, in the form of trust or otherwise, or conspiracy, in restraint of trade or commerce among the several States, or with foreign nations.*" Certain agreements among competitors are considered anticompetitive *per se* and are prohibited outright. These include agreements involving:

- *Price fixing*: Agreements between competitors regarding the prices of products or conditions offered;
- *Bid rigging*: Coordination of bidding behavior among competitors that undermines the bidding process;
- *Market or customer allocation*: Agreements among competitors not to compete for customers.

Certain concerted refusals to deal may also be unlawful per se, but not every boycott falls into that category. Tying arrangements, whereby the purchase of one product is conditional on the purchase of another, may also violate antitrust law, depending on the seller's market power and the extent to which the arrangement restricts competition. Vertical agreements and exclusive dealing are generally evaluated under the rule of reason, which weighs their potential restrictions on competition against any resulting efficiency gains.

Section 2 of the Sherman Act prohibits monopolization, attempted monopolization, and conspiracies to monopolize. Such conduct may be undertaken unilaterally or jointly with other firms in a coordinated manner (as for Section 1 violations). Section 2 addresses the acquisition

[5] In practice, the legislation also affected non-railroad shipping, see e.g. Gilligan, Marshall and Weingast (1989).

[6] In addition, U.S. states typically have enacted their own antitrust laws targeted at anti-competitive behavior within their states.

or maintenance of monopoly power through exclusionary conduct; possessing monopoly power is not itself unlawful.[7]

The Sherman Act was deliberately left vague in many of its details such that U.S. courts had considerable discretion regarding its interpretation and implementation. This lack of specificity was eventually viewed as a weakness and antitrust reform became an issue during the 1912 U.S. Presidential election. Following Woodrow Wilson's election, Congress passed the Federal Trade Commission Act and the Clayton Antitrust Act in 1914. These acts were aimed at making antitrust legislation more specific and forward-looking.

The Clayton Antitrust Act prohibits specific actions when they *may* lessen competition:

- *Price discrimination* between different buyers when it substantially lessens competition or tends to create a monopoly (Section 2);
- *Tying and exclusivity* of sales when these substantially lessen competition (Section 3);
- *Mergers and acquisitions* that may substantially lessen competition or tend to create a monopoly (Section 7);
- *Forming interlocking boards* on competing companies when a merger of those would violate antitrust (Section 8).

In an amendment to the Clayton Act that extends Section 2, the Robinson-Patman Act of 1936 further bans a set of price discriminatory actions. Moreover, while Section 7 of the Clayton Act specifically prohibits stock acquisitions whose likely effect is to lessen competition, it did not apply to asset acquisitions nor, somewhat surprisingly, to actual mergers. These issues were addressed with the 1950 Celler-Kefauver amendment which extended the legislation to asset purchases and mergers.

The Hart-Scott-Rodino Antitrust Improvements Act of 1976 introduced a number of amendments to U.S. antitrust legislation. Principally, the act requires firms to pay a filing fee and to pre-notify and register mergers and acquisitions that affect U.S. commerce and exceed certain limits on the size of the transactions, assets, or net sales of the parties involved. Before completing transactions above these limits, the parties must notify the antitrust authorities and

[7]Section 3 of the Sherman Act extends Sections 1 and 2 to the District of Columbia and to U.S. Territories.

wait until the applicable statutory waiting period expires or is terminated. Expiration of the waiting period does not constitute approval of the transaction. Failure to notify or premature completion of a transaction can trigger a civil penalty. The amendment came into force in September 1978.

We also include antitrust cases filed under Section 5 of the FTC Act which refers to "unfair methods of competition in or affecting commerce, and unfair or deceptive acts or practices in or affecting commerce." Section 5 is deliberately open-ended in its coverage, encompassing all violations of Sections 1 and 2 of the Sherman Act and of the Clayton Act. Section 5 also permits the FTC to challenge certain conduct that falls outside the prohibitions of the Sherman and Clayton Acts. For instance, such "standalone" Section 5 cases include invitations to collude that do not result in an agreement and therefore do not violate Section 1 of the Sherman Act.

2.2 Enforcement

Federal antitrust enforcement initially rested with the Attorney General and U.S. Attorneys. Responsibility is now shared by the Antitrust Division of the U.S. Department of Justice (ATR) and the Federal Trade Commission (FTC). State attorneys general also enforce federal and state antitrust laws. Private parties, including consumers and businesses, may bring civil actions seeking damages for injuries caused by antitrust violations or injunctions to prevent anticompetitive conduct. Our analysis focuses on the enforcement of federal antitrust laws by the FTC and the DoJ.

The DoJ has been involved with antitrust enforcement since the passage of the Sherman Act, but its role was limited in the early years due to scarce resources. A dedicated task force was formed in 1919 when the Attorney General introduced the ATR, see Werden (2018). The ATR has since been dedicated to antitrust enforcement, and it is now authorized to fill more than a thousand positions, which currently include more than 50 PhD economists and around 400 attorneys. The mission of the ATR is to "*promote economic competition through enforcing and providing guidance on antitrust laws and principles.*"

The DoJ enforces federal antitrust laws through both criminal prosecutions and civil actions against individuals and firms. It can ask courts to prohibit future violations and order remedies for past violations. Courts may impose fines and, in the case of individuals, impose a maximum

prison sentence of 10 years. The choice to pursue a case criminally or civilly depends on the applicable statute, the nature of the alleged offenses, and the available evidence. Criminal enforcement has primarily targeted cases involving price fixing, bid rigging, and market allocation violations. Sherman Act offenses, previously classified as misdemeanors, became felonies under the Antitrust Procedures and Penalties Act of 1974.

The FTC was created in 1914 upon enactment of the Federal Trade Commission Act. It is an independent agency of the U.S. government and enforces civil antitrust and consumer protection laws. Its mission is to "*prevent business practices that are anticompetitive or deceptive or unfair to consumers.*" The FTC investigates competition issues brought to its attention by reports from consumers or firms, from pre-merger notification filings, from congressional inquiries, or from the media. It can seek voluntary compliance, file administrative complaints, and initiate federal litigation. The FTC employs more than a thousand staff including around 300 lawyers and 75 PhD economists.[8]

The DoJ brings criminal prosecutions under Sections 1 and 2 of the Sherman Act and can initiate civil actions requesting injunctive relief and, where the United States itself has suffered antitrust injury, damages. The DoJ also enforces Sections 2 and 3 of the Clayton Act in federal court, while the FTC conducts administrative proceedings and may also seek injunctive relief in federal court. Sections 7 and 8 of the Clayton Act are enforced by both the FTC and the DoJ. The FTC may also challenge conduct that violates Sections 1 and 2 of the Sherman Act through Section 5 of the FTC Act, but cannot bring criminal prosecutions. The HSR Act's premerger notification provisions are enforced by both agencies. For these pre-consummation reviews, the agencies coordinate to determine which agency reviews the transaction and seeks any necessary remedies to address competition concerns.

2.3 Sources and Methodology

We collect data on the universe of federal antitrust cases filed by the FTC and the DoJ over 1954–2026 and 1957–2026, respectively. Unless otherwise indicated, descriptive statistics exclude filings from the incomplete 2026 calendar year. The primary source of information is the

[8] Jointly, the ATR and the FTC therefore possess significant analytical and technical expertise in the area of antitrust. However, their joint analytical capacity and budgets are not large when compared to, e.g., the Federal Reserve system. For instance, the Federal Reserve Board employs more than 400 PhD economists.

Commerce Clearing House Trade Regulation Reporter (the CCH Trade Reporter), commonly referred to as the CCH Bluebook. As noted by Posner (1970) and Gallo et al. (2000), the CCH Trade Regulation Reporter frequently contains several updates on the same case as proceedings unfold. Counting such updates as separate cases would lead to substantial double counting. In some instances, though, further defendants are added as a case progresses. We consolidate reports referring to the same case and date each case by its initial filing; defendants added later are assigned the date they entered the proceedings.[9]

We supplement this source with case records and legal documents collected from the websites of the DoJ and the FTC. This allows us to identify additional cases, obtain further details on existing cases, and cross-check the data contained in the CCH Trade Reporter. We reconcile these sources and conduct targeted internet searches using defendant names, docket numbers, and case dates to recover missing information and resolve discrepancies between sources.

For each antitrust case filed by the DoJ or the FTC, we record the following information, where available:

- the docket number and court or administrative forum;
- the case type: civil non-merger, criminal, or civil merger;
- the names of defendants and whether they are firms or individuals;
- the alleged anticompetitive conduct and the statutory provisions invoked;
- the industry involved and the volume of commerce affected;
- the start and end dates of the alleged conduct;
- the filing date, dates of subsequent charges and significant case developments, and resolution dates;
- for merger cases, the transaction type, merger classification, identity of the acquirer, and transaction value;
- case outcomes and the nature of any structural or behavioral remedies imposed;

[9]We retain related proceedings as separate cases, including criminal proceedings against employees arising from investigations of corporations. Our quantitative analysis focuses on corporate defendants.

- criminal fines, civil monetary payments, prison sentences, probation terms, and community service requirements;
- whether defendants received leniency and references to related cases.

Outcomes, sanctions, and resolution dates are recorded separately for individual defendants where applicable.

For corporate defendants, we establish whether they were publicly traded at the filing date and link listed firms to their CRSP identifiers and, where available, Compustat accounting data. We retain only defendants matched to ordinary common shares, excluding American depositary receipts (ADRs), preferred shares, and other security types.[10]

2.4 Broad Trends

Our sample contains 6,674 federal antitrust cases through 2025, with the DoJ accounting for approximately two thirds of the total. Table 1 reports the breakdown of cases and defendants by agency and case type.

Our dataset covers major cases across all three categories of antitrust enforcement. Criminal cases include the 1960 electrical-equipment cartel prosecutions against General Electric, Westinghouse, and other manufacturers, which led to prison sentences for several executives; and the 2015 foreign-exchange cartel prosecutions against Citicorp, JPMorgan Chase, Barclays, and Royal Bank of Scotland, which resulted in guilty pleas and combined fines of $2.52 billion. Among civil non-merger cases, examples include the 1969 monopolization case against IBM in the computer industry, which was withdrawn in 1982, and the 2020 case against Google for monopolization of general search and search text advertising, which resulted in a finding of liability in 2024. Merger cases include the FTC's 1999 challenge to the Exxon–Mobil merger, which resulted in extensive divestitures of petroleum refining and gasoline marketing assets; and the DoJ's 2011 challenge to AT&T's proposed acquisition of T-Mobile, after which the parties abandoned the transaction. Table A.1 in Appendix A.1.1 reports a broader selection of major cases against publicly listed firms.

[10]Matching firms named in antitrust cases to publicly traded companies requires substantial manual verification. Case documents may use abbreviated or historical company names, contain typographical errors, or name subsidiaries whose ownership links to listed parent companies must be established at the filing date. To minimize mistakes, each identified case has been checked at least three times.

The annual number of federal antitrust cases is relatively stable over time, with some notable exceptions (Figure 1, left panel). The FTC and DoJ samples cover 1954–2025 and 1957–2025, respectively. First, the peaks in the late 1950s and early 1960s reflect the large number of price discrimination cases initiated by the FTC.[11] Second, DoJ filings increased sharply in the early 1980s following a rise in criminal prosecutions for bid rigging in highway and other construction industries. Third, the decline in cases in the last part of the sample is driven by a fall in criminal filings. Finally, there are no clear systematic differences in antitrust activity between Democrat and Republican administrations (see Appendix Figure A.1).

		Defendants				
				Firms		
	Cases	**Total**	**Individuals**	Total	Non-Listed	Listed
Panel A: Cases by Agency						
FTC	2,440	6,563	1,685	4,878	3,570	1,308
DoJ	4,240	12,251	3,791	8,460	6,324	2,136
Panel B: Cases by Type						
Civil Non-Merger	2,459	8,097	2,043	6,054	4,784	1,270
Criminal	2,619	7,299	3,334	3,965	3,375	590
Civil Merger	1,562	3,271	75	3,196	1,660	1,536
Mixed	34	138	23	115	70	45
Total	6,674	18,805	5,475	13,330	9,889	3,441

Table 1: Antitrust Case Filings: Summary Statistics

Notes: The FTC sample covers the period 1954–2025; the DoJ sample covers 1957–2025. The first column reports the number of antitrust cases initiated by antitrust authorities. The other columns report the number of defendants charged in antitrust cases. "Listed" identifies corporate defendants linked to eligible CRSP ordinary common shares at filing. The remaining firms include unlisted companies and firms without such a match. Agency rows overlap for proceedings recorded under both agencies and therefore need not sum to the total. Source: Own calculations, see text.

Corporations represent over 70 percent of the 18,805 defendant observations recorded in our database through 2025 (13,330 in total). Publicly listed firms account for approximately 26 percent of corporate defendant observations, despite representing a very small share of U.S. firms.[12] The number of corporate defendants has declined markedly over time, while case filings

[11] Between 1957 and 1963, over 85% of the 683 cases brought by the FTC involved price discrimination offenses.

[12] For comparison, World Bank data record 4,642 listed domestic companies in 2022, while the U.S. Census Bureau reports approximately 5.9 million employer firms in that year, implying a share of roughly 0.08 percent.

have remained relatively stable (Figure 1, right panel). An average of 455 firms were involved in antitrust cases each year during 1957–1964, declining to 250.3 during 1965–1980 and 117.5 during 1981–2025. This mostly reflects a decline in corporate defendants per case, from 2.84 in 1957–1964 to 1.39 in 1981–2025, particularly within criminal cases. This pattern is corroborated by Gallo et al. (2000), who document a substantial decline in firms per DoJ case after 1980.

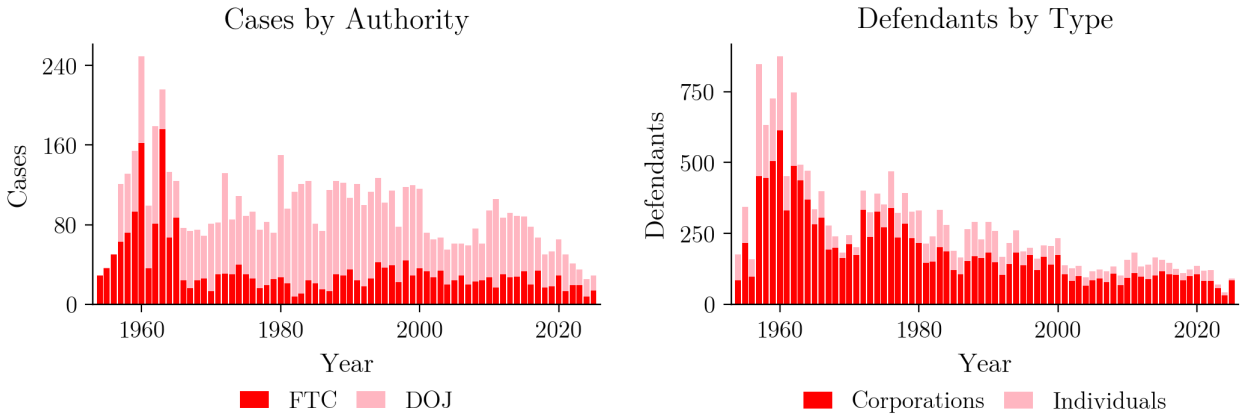


Figure 1: Antitrust Trends

Notes: The left panel shows the number of federal antitrust cases filed by the FTC and the DoJ each year since 1954. The sample for the DoJ starts in 1957. The right panel shows the number of defendants in federal antitrust cases filed in each year. Source: Own calculations, see text.

Over the entire sample, civil non-merger cases account for 36.8 percent of cases, criminal cases for 39.2 percent, civil merger cases for 23.4 percent, and mixed cases for 0.5 percent.[13] Among corporate defendant observations, civil non-merger cases account for 45.4 percent, criminal cases for 29.7 percent, and merger cases for 24.0 percent. Among listed corporate defendants, merger cases are more prevalent and account for 44.6 percent of observations.

From the early 1980s, civil non-merger cases become less frequent, while the number of criminal cases rises sharply (Figure 2, left panel).[14] For corporate defendants, shown in the right panel, the structural break around 1980 remains, with a clear shift towards merger-related antitrust violations, while both civil non-merger and criminal cases decline in relative importance. The shift towards merger-related enforcement around 1980 is particularly pronounced

[13]Mixed cases involve both non-merger and merger violations.

[14]Similarly, Ghosal (2011) finds a structural break in antitrust enforcement in 1979 with a shift towards criminal cases.

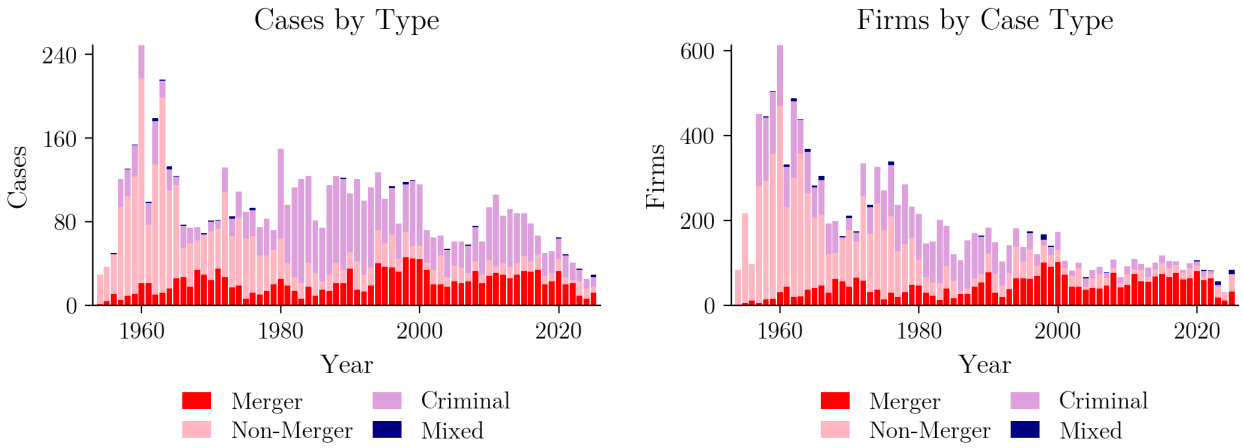


Figure 2: Antitrust Trends by Case Type

Notes: The left panel shows the number of federal cases filed by the FTC and the DoJ each year since 1954 when decomposed into civil non-merger violations, criminal violations, civil merger violations, or mixed violations. The right panel shows the same decomposition for the number of corporate defendants. Source: Own calculations, see text.

among *publicly listed companies* (Figure 3). Merger-related violations account for 21.9 percent of listed corporate defendant observations in 1957–1980, rising to 75.6 percent in 1981–2025. Over the same periods, the share involving horizontal violations falls from 48.2 to 9.1 percent, while that involving monopolization and exclusion-related cases declines from 14.2 to 5.1 percent. The number of listed defendants in DoJ cases has also declined over recent decades, while the corresponding number for FTC cases has remained relatively stable.

The vast majority of merger cases concern horizontal transactions (Figure 4). The same figure also shows a shift toward challenges before transactions are completed. Among reviewed cases with verified status at filing, proposed transactions account for 27.5 percent of cases through 1977, compared with 88.4 percent from 1979 onward. This pattern is consistent with the introduction of the HSR Act, implemented in 1978, which facilitated intervention before the completion of merger transactions.

Antitrust authorities obtain documented relief in more than 90 percent of cases with known outcomes (Table 2). The shares are 92.7 percent for civil merger cases, 91.8 percent for civil non-merger cases, and 94.5 percent for criminal cases. Outcomes are known for 90.8, 78.7, and 86.7 percent of filings in these categories, respectively. These high success rates suggest that

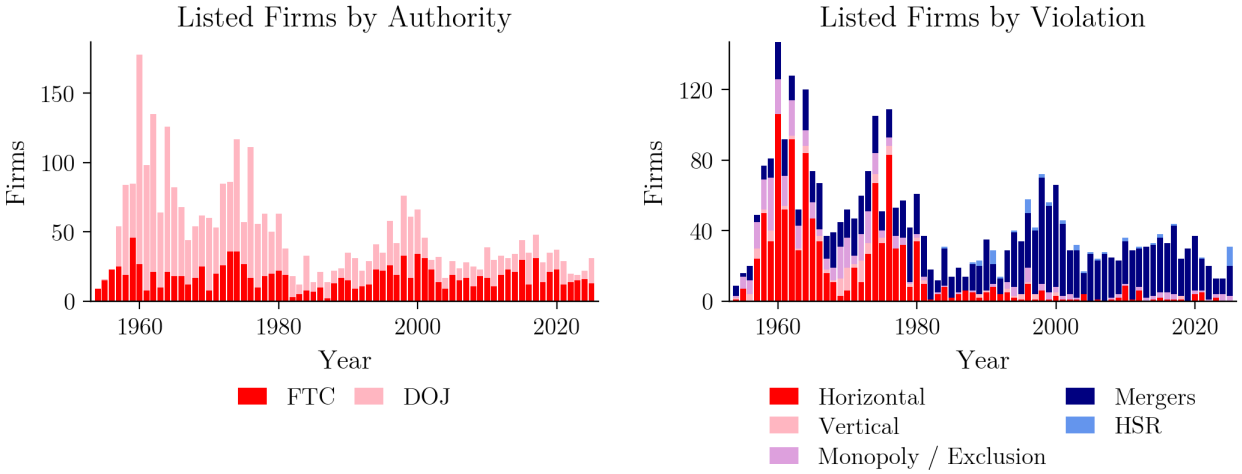


Figure 3: Publicly Listed Firms

Notes: The left panel shows the number of publicly listed firms involved in federal antitrust cases since 1954 for the FTC, and since 1957 for the DoJ. The right panel shows the number of publicly listed defendants across antitrust violation types. Source: Own calculations, see text.

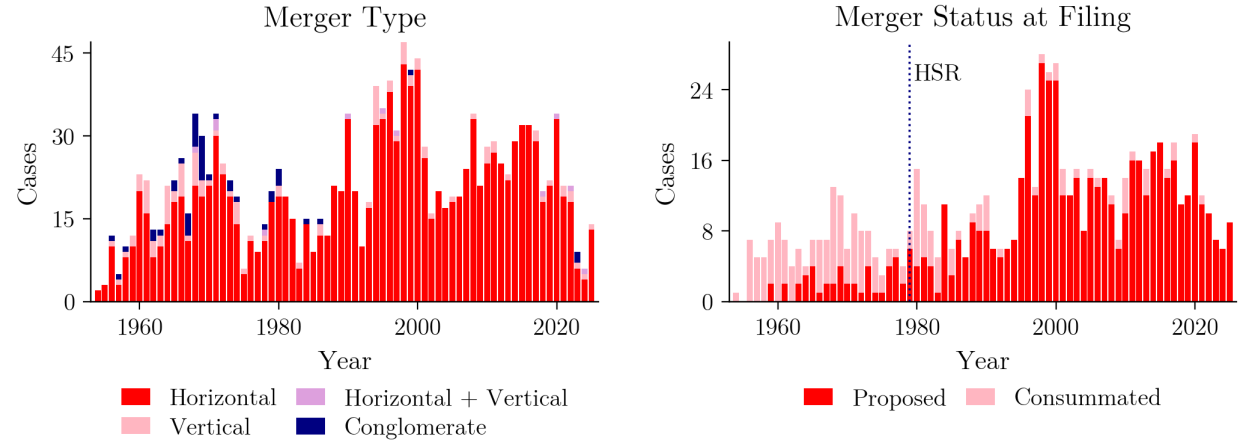


Figure 4: Merger Cases by Type and Transaction Status

Notes: Left: merger cases with a recorded transaction type. Right: transaction status at the initial public filing for 752 of 1,034 reviewed merger cases linked to the event-study sample (570 proposed and 182 consummated). Cases with unknown or mixed status or without verified initial filing dates are excluded from the right panel. The vertical line marks the implementation of the HSR notification regime in 1978. Sources: CCH Bluebook, FTC and DoJ, and own calculations.

filings signal a credible prospect of intervention, whose anticipated consequences can affect firm behavior and valuations before cases are resolved.

	Civil Merger			Civil Non-Merger			Criminal		
	Success (%)	Known Cases	Coverage (%)	Success (%)	Known Cases	Coverage (%)	Success (%)	Known Cases	Coverage (%)
1954–1959	91.9	37	90.2	86.3	357	89.5	85.5	69	83.1
1960–1969	86.7	196	91.6	91.9	568	64.4	87.5	152	80.9
1970–1979	77.8	126	68.5	92.0	398	80.1	93.5	139	65.0
1980–1989	90.9	143	88.3	88.6	166	88.3	94.8	650	84.6
1990–1999	95.9	296	94.0	97.9	188	89.5	95.9	533	94.0
2000–2009	96.6	234	96.7	98.4	122	96.8	95.3	274	89.8
2010–2019	98.1	269	98.2	96.2	104	98.1	98.7	371	91.6
2020–2026	92.6	94	90.4	91.2	34	64.2	83.8	80	92.0
All Years	92.7	1,395	90.8	91.8	1,937	78.7	94.5	2,268	86.7

Table 2: Antitrust Success Rates and Outcome Coverage by Filing Decade

Notes: "Success" is the percentage of classified cases in which the authority obtained documented relief against at least one defendant. Relief includes consent and cease-and-desist orders, settlements, guilty or no-contest pleas, blocked or abandoned transactions, and favorable judgments; partial relief and mixed outcomes with surviving relief also qualify. This is not the percentage of contested trials won. A case without documented relief is classified as unsuccessful only when all defendant records show proceedings concluded without relief, including nonmerits closures. Otherwise it remains unclassified. "Known Cases" counts classified cases; some defendants in a case with relief may remain unresolved. "Coverage" is the classified share of all eligible filings. Cases use persistent identifiers and their earliest filing date. The sample covers 1954–2026, with partial coverage of 2026; collateral proceedings without an antitrust disposition and 34 cases spanning multiple case types are excluded. Source: Own calculations.

Larger firms face higher average annual filing rates and account for a disproportionate share of defendants (Figure 5). Civil non-merger and criminal defendants, merger acquirers, and merger targets are concentrated in the upper deciles of the sales distribution of Compustat firms in industry-years with antitrust filings, and merger targets are generally smaller than other defendants. The relationship is similar when size is measured by employment or Compustat market share. Table A.2 in Appendix A.1.2 provides additional comparisons.

3 Firm-Level Outcomes: Civil Non-Merger and Criminal Cases

We start by examining the impact of antitrust filings on defendants' markups, sales, spending on inputs, and employment using synthetic difference-in-differences. We focus on civil non-merger and criminal cases, comparing defendants with firms whose pre-filing outcomes indicate

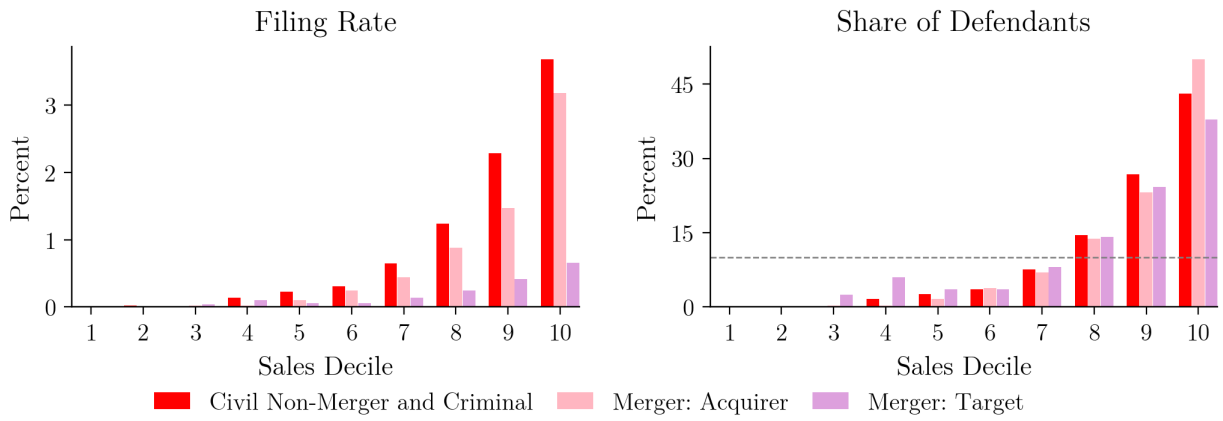


Figure 5: Firm Size and Antitrust Filings

Notes: Both panels use the same pre-filing real-sales deciles. The left panel divides each group's defendant firm-year count in a decile by all eligible firm-year observations in that decile. The right divides that count by the group's total defendant firm-years, so shares sum to 100 percent within each group. Its dashed line at 10 percent represents an even distribution across deciles. Deciles pool positive, finite sales observations from three-digit NAICS industry-years with at least one linked defendant. Sales and industry refer to the preceding accounting year; sales are deflated using the GDP deflator. Merger roles use reviewed issuer links; unknown roles do not enter either role's defendant counts. A firm is counted at most once per group and year. No post-filing accounts are required.

similar underlying trajectories. For merger cases, completing the proposed transaction could alter defendants' subsequent outcomes, even when their pre-filing histories resemble those of the comparison firms. Constructing a credible comparison group is therefore difficult, since it would require firms proposing similar transactions that were not challenged.

3.1 Outcomes and Measurement

We use Compustat accounts covering 1962–2025 to measure defendants' markups, sales, expenditure on variable inputs, and employment. Sales and costs of goods sold (COGS) are deflated using the GDP deflator. Employment is the reported number of employees, while COGS measures expenditure on variable inputs. All four outcomes enter in logarithms, and estimated responses are multiplied by 100 and interpreted as approximate percentage changes. Appendix A.2.2 describes the accounting variables and data construction.

Following Hall (1988) and de Loecker and Warzynski (2012), we recover markups from firms' cost-minimization conditions. Let the production function be given by

$$q_{j,t} = f_l(x^d_{j,t}, x^v_{j,t}) + z^p_{j,t} + z^{tr}_{j,t}, \tag{1}$$

where $q_{j,t}$, $x^d_{j,t}$, and $x^v_{j,t}$ denote the logarithms of output, capital, and variable input use, respectively. For estimation, we proxy these quantities with deflated sales, deflated net property, plant and equipment, and deflated COGS. $z^p_{j,t}$ and $z^{tr}_{j,t}$ capture persistent productivity and a transitory error, respectively. We specify f_l as translog, with common parameters within two-digit NAICS industry l, and estimate it using the two-stage version of the Olley and Pakes (1996) control-function approach suggested by Ackerberg, Caves and Frazer (2015). The firm's markup is then given by

$$\mu_{j,t} = \frac{\theta^v_{j,t}}{\alpha^v_{j,t}}, \qquad \theta^v_{j,t} = \frac{\partial f_l(x^d_{j,t}, x^v_{j,t})}{\partial x^v_{j,t}}, \qquad \alpha^v_{j,t} = \frac{\text{COGS}_{j,t}}{\text{Sales}_{j,t}}, \tag{2}$$

where $\theta^v_{j,t}$ is the output elasticity with respect to variable input use and $\alpha^v_{j,t}$ is the share of revenue spent on that input. The translog specification allows the elasticity to vary across firms and over time, and we use Cobb–Douglas markups as a robustness check. Appendix A.3.1 provides the estimation details, and Appendix Figure A.2 compares the resulting aggregate markup measures.

3.2 Estimation Approach

We use each firm's first observed antitrust filing year to define treatment, retaining firms facing civil non-merger or criminal proceedings in that year. Each treated firm contributes a single treatment event, irrespective of subsequent filings. The comparison group consists of never-treated Compustat firms with no recorded antitrust filing of any category over the entire sample.[15]

[15] Multiple filings against a firm in its first filing year constitute one treatment event. We also retain firms facing additional cases during the period following the first filing. Appendix Figure A.4 shows that the results remain similar when firms with additional filings through each reported horizon are excluded.

For each filing cohort g, we compare defendants with never-treated controls in the same BEA sector and sales quintile, measured in year $g-1$. Larger firms are more likely to face antitrust filings, as shown in Figure 5, and may follow different growth trajectories from smaller firms. To accommodate staggered filing dates, we estimate the following weighted regression separately within each cohort–sector–size cell c and at each horizon $h = 0, \ldots, 10$, following Arkhangelsky et al. (2021):

$$y_{j,t} = \alpha_{j,c,h} + \delta_{t,c,h} + \tau_{c,h} D_j \mathbf{1}\{t = g + h\} + \varepsilon_{j,t,c,h}, \tag{3}$$

where g denotes the filing year and D_j identifies defendants. Each regression uses the five pre-filing years and year $g+h$, and includes firm and calendar-year fixed effects. The coefficient $\tau_{c,h}$ measures the effect of antitrust filings at horizon h within cell c. We estimate the regression by weighted least squares, assigning observation (j,t) the product $\hat{\omega}_{j,c,h}\hat{\lambda}_{t,c,h}$ of firm and time weights.

Firm weights construct a comparison group with a pre-filing outcome path similar to that of defendants, allowing for differences in levels, while time weights construct the pre-filing reference period against which subsequent changes are measured. Using controls alone, time weights select the combination of pre-filing years that best approximates outcomes in year $g+h$, up to a common shift in levels. Both control-firm and pre-filing time weights are non-negative, with each set summing to one. Neither set is estimated using defendants' post-filing outcomes.

The estimated effect is the post-filing treated–control difference minus the time-weighted pre-filing difference. We aggregate cell estimates in proportion to their numbers of eligible defendants. Together with the sector and size restrictions, synthetic weights reduce bias from differences in firms' exposure to changing economic conditions that firm and year fixed effects alone may not absorb. Identification requires the weighted comparisons to recover defendants' changes in outcomes had the filing not occurred. We obtain pointwise 95 percent confidence intervals from 1,000 firm-pairs bootstrap draws, refitting the synthetic weights in every draw. Appendix A.3.2 provides the weighting and inference details.

The estimation sample varies across outcomes and horizons, since each estimate requires five years of pre-filing data and an observation in the relevant post-filing year. Observations

in the intervening post-filing years are not required. We retain comparison cells with at least five eligible controls. For all four outcomes, defendants and controls must have markups above one, positive sales, a positive outcome, and available sector information in every required year, including the post-filing endpoint. The first eligible filing cohort is therefore 1967. At year five, the markup, sales, and COGS estimates each use 134 defendants and 1,789 distinct eligible control firms across 40 filing cohorts. The employment estimate uses 120 defendants and 1,576 distinct controls across 40 cohorts.

3.3 Results

First, we assess the comparison group's ability to reproduce defendants' pre-treatment outcomes. The fitted pre-treatment differences in Figure 6 are close to zero and statistically insignificant. In a separate validation exercise, we fit the synthetic weights using years -5 through -3 and find that the held-out defendant prediction errors in years -2 and -1 are also statistically insignificant (Appendix Figure A.5). Local-projection difference-in-differences estimates following Dube et al. (2025) show similar patterns (Appendix Figure A.3).

The estimates indicate that antitrust enforcement has procompetitive effects (Figure 6). Defendants' markups fall by approximately 2.8 percent five years after filing, a decline significant at the five percent level. The estimates are statistically significant from the third through the ninth year. The markup responses are similar under the Cobb–Douglas specification (Appendix Figure A.4, left panel). Input expenditure rises faster than sales, increasing the share of revenue spent on variable inputs and reducing markups for a given output elasticity (equation (2)). After five years, real COGS rises by approximately 16.7 percent, exceeding the 12.9 percent increase in real sales. These responses are significant at the one and five percent levels, respectively. Employment also rises by an estimated 7.6 percent, but the response is not statistically significant.

Averaging the estimated responses over the filing year and the following ten years (Table 3), markups decline by 2.8 percent, while sales and input expenditure increase by 9.8 and 13.1 percent, respectively. The markup and input expenditure responses are significant at the one percent level, and the sales response at the five percent level. Employment rises by an estimated 6.9 percent, although this effect is not statistically significant. These findings are consistent

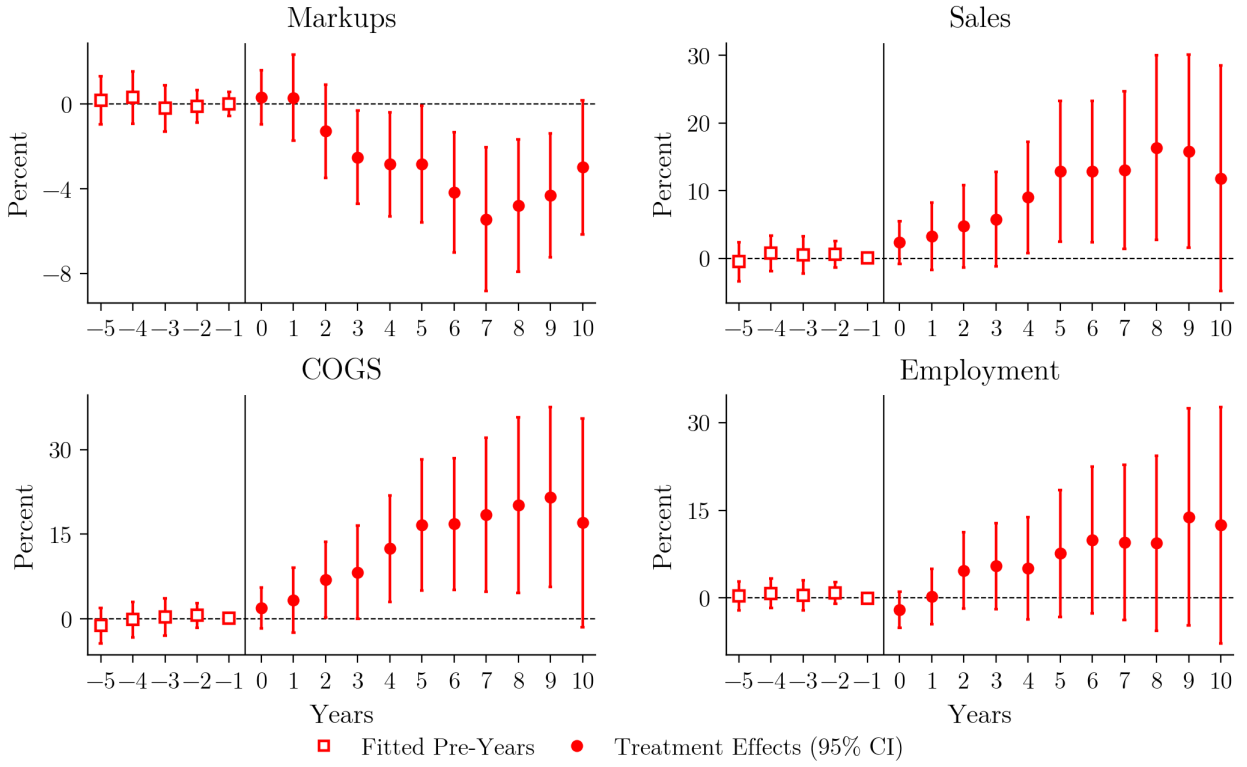


Figure 6: Firm-Level Impact of Civil Non-Merger and Criminal Case Filings

Notes: Firm-level responses to civil non-merger and criminal filings, estimated by synthetic difference-in-differences; see Appendix A.3.2. Year zero denotes the filing year. Pre-treatment points are fitted gaps. Whiskers are pointwise 95 percent normal confidence intervals based on 1,000 global firm-pairs bootstrap refits of the synthetic weights. Outcomes are measured as 100 times logarithms; reported effects use the percent approximation.

with the effects of curbing horizontal restraints in a standard collusion model (Motta, 2004, ch. 4). Such violations form the largest category of cases in the estimation sample. Disrupting coordination encourages firms to expand production and input use while lowering prices, with the latter dampening revenue growth.

The preceding analysis focuses only on the subset of civil non-merger and criminal cases for which we have the necessary accounting data. We now examine the response of daily stock returns to case filings, allowing us to extend the analysis to include merger cases and to derive enforcement proxies that account for the market's assessment of the effects of each case.

	Markups	Sales	COGS	Employment
Average Effect	-2.77^{***}	9.80^{**}	13.06^{***}	6.93
	(0.95)	(4.11)	(4.59)	(4.63)

Table 3: Average Firm-Level Effects, Years 0–10

Notes: Equally weighted averages of the annual synthetic difference-in-differences effects in Figure 6, from the filing year through year 10. Each horizon retains its eligible sample. Outcomes are measured as 100 times logarithms; effects use the percent approximation. Standard errors are in parentheses, computed from 1,000 global firm-pairs bootstrap refits across all horizons and accounting for dependence across annual estimates. Significance stars use the bootstrap p-values. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels, respectively. See Appendix A.3.2 for details.

4 Equity-Market Responses and Antitrust Enforcement Proxies

4.1 The Impact of Antitrust Case Filings on Firm Valuations: Methodology

We calculate abnormal returns around a case filing as the difference between realized returns and returns predicted by a factor model estimated over a pre-filing sample. The event-study sample comprises publicly listed defendants matched to ordinary common shares, as described in Section 2. We obtain daily equity returns from CRSP and use filing announcements as a consistently observed public signal, assigning filings on non-trading days to the next trading day. To limit contamination of the return-estimation period by earlier cases, we exclude filings occurring within 400 calendar days of a previous filing against the same firm. The resulting sample contains 2,446 firm-filing events.

For each firm-filing event, we use the three-factor Fama–French model to estimate counterfactual expected returns had the filing not occurred. Factor loadings are estimated over trading days -272 through -21 prior to the filing. Ending the estimation window one trading month before the filing reduces the risk that the counterfactual is contaminated by information about the forthcoming case. Specifically, let $t^a_{i,j}$ denote the announcement date of case i involving firm j, $r_{j,t}$ the firm's equity return, and r^f_t the risk-free rate. The coefficient vector $\alpha_{i,j}$ contains the intercept and factor loadings. We estimate the following equation using least squares:

$$r_{j,t} - r^f_t = \alpha'_{i,j} X_t + \varepsilon_{j,t}, \qquad t - t^a_{i,j} \in \{-272, \ldots, -21\}. \tag{4}$$

Here, $X_t = (1, r_t^{\text{mkt}}, smb_t, hml_t)'$, where r_t^{mkt} is the market return in excess of the risk-free rate. The size factor smb_t measures the return difference between portfolios of small and large firms, while the value factor hml_t measures the return difference between portfolios of firms with high and low book-to-market ratios.[16]

We study daily abnormal returns from five trading days before filing through one trading month afterward. We cumulate abnormal returns from two trading days before filing through trading day s, allowing for information leaks and differences between the timing of public disclosure and the recorded filing date:

$$AR_{i,j,s} = r_{j,t^a_{i,j}+s} - r^f_{t^a_{i,j}+s} - \widehat{\alpha}'_{i,j} X_{t^a_{i,j}+s}, \qquad s \in \{-5, \ldots, 21\}, \tag{5}$$

$$CAR_{i,j,s} = \sum_{h=-2}^{s} AR_{i,j,h}, \qquad s \in \{-2, \ldots, 21\}. \tag{6}$$

For each group of cases z (e.g., civil merger cases, civil non-merger and criminal cases) and horizon s, we take equally weighted averages across available firm-filing events:

$$\overline{AR}_{z,s} = \frac{1}{N^{AR}_{z,s}} \sum_{(i,j)\in\mathcal{E}^{AR}_{z,s}} AR_{i,j,s}, \tag{7}$$

$$\overline{CAR}_{z,s} = \frac{1}{N^{CAR}_{z,s}} \sum_{(i,j)\in\mathcal{E}^{CAR}_{z,s}} CAR_{i,j,s}. \tag{8}$$

The sets $\mathcal{E}^{AR}_{z,s}$ and $\mathcal{E}^{CAR}_{z,s}$ contain the available firm-filing events for the relevant return and horizon, with respective counts $N^{AR}_{z,s}$ and $N^{CAR}_{z,s}$.

Inference is based on the Kolari and Pynnönen (2010) adjustment to the Boehmer, Musumeci and Poulsen (1991) test, which accounts for event-induced volatility and cross-sectional de-

[16] We use the three-factor model because the five-factor specification is unavailable for the early sample. Results are very similar with a simpler market model and, over the period with five-factor coverage, with the five-factor specification. Appendix Figure A.6 reports the distribution and evolution of pre-filing factor-model R^2 statistics. The mean R^2 is 21.2% and relatively stable over the sample.

pendence.[17] We construct pointwise 95 percent confidence intervals using 5,000 wild-cluster-bootstrap draws, clustering events by their public filing date.[18]

4.2 Results

	All Cases			Civil Non-Merger and Criminal			Civil Merger		
Day/ Window	N	Return	p-value	N	Return	p-value	N	Return	p-value
Panel A: Average Abnormal Returns									
-2	2,445	-0.11	0.004	1,152	-0.08	0.039	1,261	-0.15	0.045
-1	2,445	-0.04	0.605	1,152	-0.07	0.374	1,261	-0.01	0.991
0	2,437	-0.46	<0.001	1,149	-0.47	<0.001	1,256	-0.40	<0.001
1	2,409	-0.38	<0.001	1,148	-0.31	<0.001	1,229	-0.43	<0.001
2	2,390	0.03	0.908	1,150	-0.04	0.525	1,208	0.09	0.577
Panel B: Average Cumulative Abnormal Returns									
$[-2,-1]$	2,445	-0.15	0.015	1,152	-0.16	0.031	1,261	-0.16	0.153
$[-2,0]$	2,445	-0.62	<0.001	1,152	-0.63	<0.001	1,261	-0.57	<0.001
$[-2,1]$	2,418	-1.04	<0.001	1,152	-0.94	<0.001	1,234	-1.08	<0.001
$[-2,5]$	2,350	-1.21	<0.001	1,152	-1.02	<0.001	1,166	-1.30	<0.001
$[-2,10]$	2,323	-1.28	<0.001	1,151	-0.92	<0.001	1,140	-1.51	<0.001
$[-2,21]$	2,290	-1.55	<0.001	1,150	-1.03	<0.001	1,108	-1.93	<0.001

Table 4: The Impact of Antitrust Case Filings on Equity Returns

Notes: Panel A reports average abnormal returns on the indicated trading days relative to filing (day 0). Panel B reports average cumulative abnormal returns over the indicated windows. Returns are equally weighted across firm-filing events and expressed in percent. N denotes the number of observations contributing to each estimate. Cases involving both merger and non-merger allegations are included only in the All Cases sample. Reported p-values are based on the Boehmer, Musumeci and Poulsen (1991) test with the Kolari and Pynnönen (2010) adjustment for cross-sectional dependence.

Case filings are followed by immediate equity-market losses (Table 4). On the filing day, average abnormal returns are -0.40 percent for civil merger cases and -0.47 percent for civil non-merger and criminal cases. The following trading day brings further abnormal losses of 0.43 and 0.31 percent, respectively. All four estimates are statistically significant at the one percent level. Thereafter, daily abnormal returns fluctuate around zero and are generally statistically

[17] In unreported exercises, the results remain robust to a wide range of alternative parametric and nonparametric tests.

[18] The Kolari–Pynnönen procedure provides an adjusted test statistic but does not directly yield confidence intervals for unstandardized average returns. We therefore use the bootstrap to construct intervals in return units.

insignificant. Smaller declines are also observed two trading days before filing, significant at the five percent level for both case types.

The initial valuation losses persist beyond a trading month following case filings. Cumulative responses are similar across case types at short horizons. Subsequently, the civil non-merger and criminal response stabilizes near -1 percent, while the merger response continues to drift downwards and settles at -2 percent. Both cumulative responses remain statistically significant at the 1 percent level at every horizon from the filing day through the end of the trading month. These cumulative responses are close to earlier estimates. Bosch and Eckard (1991) report a cumulative abnormal return of -0.78 percent over trading days $[-5, +5]$ for 127 firms charged with price fixing during 1962–1980. Over the same event window, our estimate for Sherman Act Section 1 cases is -0.93 percent.

Estimated abnormal returns differ markedly between cases filed prior to 1980 and those filed afterwards (Figure 7). Before 1980, cumulative abnormal returns are significantly negative for both case types. Merger responses remain negative in the later period but are smaller in magnitude. In contrast, the later-period response for civil non-merger and criminal cases is not statistically different from zero at any horizon considered.

Several explanations are consistent with the weakening of antitrust enforcement around 1980. Changes in antitrust thinking, notably those associated with Bork (1978), challenged the view that large firms and concentrated markets necessarily warranted intervention. Bork advocated consumer welfare as the sole objective of antitrust, emphasizing economic efficiency and criticizing policies that protected smaller competitors at the expense of more efficient firms (see Crane (2014)). Beginning in 1976, the Manne Economics Institute for Federal Judges provided intensive economic training, which Ash, Chen and Naidu (2026) argue affected judicial rulings.[19] Our analysis in Section 4.4 suggests that the 1976 HSR amendment may also have played a role. Because these three developments occurred around the same time, their separate contributions are difficult to establish.

Point estimates of valuation losses are larger for DoJ filings than for FTC filings. After twenty-one trading days, cumulative abnormal returns are approximately -1.8 and -1.3 per-

[19] Lancieri et al. (2022) argue that weaker enforcement is the result of influence by large and self-interested corporations.

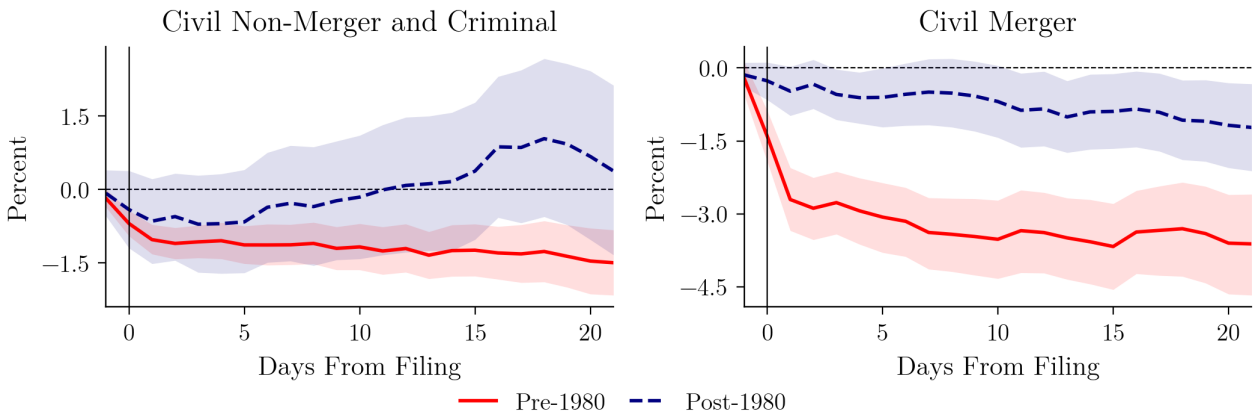


Figure 7: Cumulative Abnormal Equity Returns Before and After 1980

Notes: The left panel shows civil non-merger and criminal cases; the right panel shows civil merger cases. Each panel compares filings before 1980 (red solid line) with filings from 1980 onward (navy dashed line). Returns are cumulated from trading day −2 through the day on the horizontal axis and expressed in percent. Shaded areas are pointwise 95 percent wild-cluster-bootstrap confidence intervals.

cent, respectively (Appendix Table A.3). The difference is concentrated in merger cases, for which cumulative abnormal returns are approximately −3.1 percent following DoJ filings and −1.1 percent following FTC filings, and this difference is statistically significant at the 1 percent level (Appendix Table A.8).

We also compare valuation responses across presidential parties, violation categories, and merger parties (Appendix Tables A.4–A.6). At twenty-one trading days, valuation losses are larger for merger cases and for cases involving monopolization, exclusion, or price discrimination than for horizontal violations, with differences significant at the 1 and 5 percent levels, respectively (Appendix Table A.8). Merger targets experience larger losses than acquirers (−7.4 versus −1.4 percent, significant at the one percent level), consistent with a lower probability that target shareholders receive the takeover premium.

We also examine defendants' closest competitors using the smaller sample available for 1988–2023 (Appendix Table A.7). The estimates provide weak evidence of positive valuation spillovers following civil non-merger and criminal filings, but no clear response following merger filings.

4.3 The Antitrust Enforcement Proxies

We derive sector-level enforcement proxies by weighting defendants' cumulative abnormal returns by their market capitalizations. We retain defendants whose cumulative abnormal returns over trading days $[-2, +10]$ are non-positive. Since enforcement is expected to reduce defendants' profits, positive abnormal returns may reflect unrelated stock-return noise that obscures the underlying valuation loss. Cumulating returns through the tenth trading day also limits the influence of short-lived valuation reactions and daily stock return noise, helping the proxy capture changes more likely to matter for annual sector-level outcomes.

For each case, we construct a market-value-weighted portfolio for retained defendants and divide its cumulative abnormal return by its estimated standard deviation. This standard deviation is derived from the pre-filing Fama–French estimation window and accounts for cross-correlations among co-defendants' abnormal returns. The resulting measure gives greater weight to valuation losses estimated with greater precision:

$$T_c = \frac{\sum_{j \in S_c} w_{c,j} CAR_{c,j,10}}{\sqrt{w_c' \widehat{\Sigma}_c w_c}}. \tag{9}$$

Here, S_c denotes the retained defendants, $w_{c,j}$ their normalized pre-filing market-capitalization weights, and $\widehat{\Sigma}_c$ the estimated covariance matrix of their cumulative abnormal returns.[20] Market capitalization is measured five business days before filing, using the most recent earlier observation if data for that date are unavailable.

We multiply T_c by the retained defendants' total pre-filing market capitalization as a share of total U.S. market capitalization in the filing year. We redistribute this case contribution across sectors according to defendants' shares of that capitalization, and across the filing and following years according to the filing date.[21]

[20] For the proxy, we exclude an observation if the same issuer has a preceding filing within 60 calendar days in the sample with available returns and market capitalization. We construct $\widehat{\Sigma}_c$ from individual CAR standard deviations and correlations between co-defendants' pre-filing abnormal returns on common dates. Correlations are estimated before the CAR sign restriction and then restricted to retained defendants. If residual histories are missing or fewer than 50 common dates are available, cross-defendant correlations are set to zero.

[21] This allocation adapts the temporal-aggregation logic of Kilian (2024). A filing occurring a fraction u through the year receives weight $1 - u$ in the filing year and u in the following year.



Figure 8: Antitrust Activity Proxies

Notes: The upper-left (upper-right) panel shows the antitrust indicator for civil non-merger and criminal cases (civil merger cases). Each aggregate series is divided by its own standard deviation before applying a five-year moving average. Shaded areas denote NBER recessions. The lower panels show the corresponding sector-level proxies. Each heatmap displays the magnitude of the non-positive proxy divided by its pooled standard deviation. White cells indicate zero or unavailable observations. Both heatmaps use the common scale below them, with pale-to-dark red indicating increasing enforcement magnitude. The scale saturates at the 98th percentile of the pooled nonzero standardized magnitudes across both panels.

The proxies partially mirror the shift in the composition of cases described in Section 2. The civil non-merger and criminal proxy is more negative in the early decades and generally closer to zero after 1980. The merger proxy fluctuates without a comparable trend, with pronounced troughs in the late 1950s and 1960s, the mid-1980s, and the late 1990s (Figure 8).

The sectoral composition of exposure evolves with changes in enforcement priorities and the economic importance of different industries, as shown by the heatmaps in Figure 8. Petroleum cases in the late 1950s and the electrical-equipment prosecutions of 1960 contribute to the early concentration in manufacturing and mining. Later exposure extends to information and finance, including cases concerning digital platforms and payment networks, while major retail acquisitions generate distinct episodes of merger exposure. These patterns are consistent with historical accounts of industrial deconcentration campaigns in the 1970s and increasing attention to high-technology markets in the 1990s (see Scheffman, Coate and Silvia (2003) and Balto (1999)).

4.4 Predictability of Antitrust Enforcement Proxies

An important issue for our analysis is whether the proxies retain sufficient variation to identify the effects of antitrust enforcement after accounting for its predictable determinants. We consider four potential determinants of antitrust enforcement. First, the budgets of antitrust authorities may affect their ability to pursue violations. Figure A.8 in Appendix A.1.6 shows that real appropriations for the DoJ Antitrust Division and the FTC (deflated by the GDP deflator) have risen steadily over the sample. Second, higher industry concentration may attract regulatory scrutiny, given antitrust authorities' use of concentration measures when evaluating competition concerns, particularly for merger cases. We therefore consider the sector-level HHI measure described in Appendix A.2.6. Third, merger activity fluctuates substantially over time and may affect the number of transactions subject to antitrust review. Figure A.8 reports our indicator of aggregate U.S. merger activity, constructed from historical transaction-count series as described in Appendix A.2.5. Fourth, the HSR Act of 1976 creates a direct link between merger activity and agency workloads through its premerger notification requirements. To capture its implementation in late 1978, we include a binary indicator equal to one from 1979 onward and its interaction with lagged merger activity.

Table 5 reports pooled least squares regressions of the sector-level enforcement proxies on these potential determinants, without sector or year fixed effects. We also include a linear time trend and, in alternating columns, the lagged enforcement proxy. Concentration and lagged enforcement vary across sectors and years; the other predictors vary only over time and

	Civil Non-Merger and Criminal				Civil Merger			
	1957–2024		1962–2024		1957–2024		1962–2024	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	2.94*	2.09*	2.49*	1.91	−0.38	−0.42	−1.17	−1.27
	(1.50)	(1.06)	(1.48)	(1.18)	(1.03)	(0.82)	(1.24)	(0.89)
DoJ Budget$_{t-1}$	−0.21	−0.10	−0.25	−0.04	0.11	−0.04	0.08	−0.08
	(0.18)	(0.14)	(0.22)	(0.17)	(0.25)	(0.22)	(0.29)	(0.25)
FTC Budget$_{t-1}$	−0.12	−0.16	0.06	−0.15	0.14	0.22	0.29	0.38
	(0.21)	(0.16)	(0.25)	(0.23)	(0.25)	(0.22)	(0.29)	(0.25)
HHI$_{i,t-1}$	0.19**	0.10**	0.27***	0.14**	0.29***	0.19***	0.31***	0.19***
	(0.08)	(0.04)	(0.10)	(0.07)	(0.04)	(0.04)	(0.04)	(0.03)
Merger Activity$_{t-1}$	−0.10	−0.03	−0.09	−0.03	−0.13***	−0.09**	−0.10***	−0.08**
	(0.06)	(0.04)	(0.07)	(0.04)	(0.05)	(0.04)	(0.04)	(0.03)
HSR Regime	−0.18	−0.10	−0.35**	−0.24*	0.06	0.11	0.11	0.16
	(0.12)	(0.09)	(0.15)	(0.13)	(0.15)	(0.13)	(0.16)	(0.13)
HSR × Merger Activity$_{t-1}$	0.04	−0.03	0.04	−0.06	0.28***	0.27***	0.32***	0.32***
	(0.10)	(0.07)	(0.12)	(0.10)	(0.10)	(0.09)	(0.11)	(0.10)
Time Trend	0.09*	0.07*	0.06	0.07	−0.08**	−0.08**	−0.12***	−0.12***
	(0.05)	(0.04)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)	(0.04)
Enforcement$_{t-1}$	–	0.51***	–	0.46***	–	0.34***	–	0.38***
		(0.12)		(0.14)		(0.05)		(0.05)
R^2	0.05	0.30	0.07	0.27	0.04	0.16	0.05	0.18
Observations	1,281	1,281	1,194	1,194	1,281	1,281	1,194	1,194

Table 5: Predictability of the Enforcement Proxies at the Industry Level

Notes: Pooled OLS estimates for 19 two-digit BEA sectors. Enforcement equals the negative of the proxy divided by its pooled standard deviation over the indicated sample period, without demeaning. Each pair of columns uses the same sample of observations with all required variables available. Budgets and HHI enter in logarithms, and merger activity is a standardized filtered factor. The HSR Regime indicator equals one from 1979 onward and zero otherwise. Budget construction accounts for changes in the federal fiscal year; see Appendix A.2.5. Parentheses report Driscoll and Kraay (1998) standard errors with eight lags. *, **, and *** denote statistical significance at the 10, 5, and 1 percent levels, respectively.

are common to all sectors. We reverse the sign of the raw proxies and divide them by their pooled standard deviations, such that higher values indicate stronger enforcement. We examine samples beginning in 1957 and 1962, both ending in 2024. The latter corresponds to the start of our analysis for Compustat outcomes and excludes part of the early period of unusually high antitrust activity. For inference, we use Driscoll–Kraay standard errors with eight lags.

Higher industry concentration is positively correlated with both enforcement proxies, with coefficients significant at the five percent level in every specification. Under the HSR regime,

higher aggregate merger activity predicts a higher merger-enforcement proxy, with this relationship significant at the five percent level throughout. The relationship differs from that before 1979 at the one percent level in every specification. The HSR indicator is also negatively correlated with civil non-merger and criminal enforcement in the sample beginning in 1962, though this relationship is not statistically significant in the full sample. On the other hand, agency budget coefficients are individually insignificant after conditioning on the other predictors.[22]

Nevertheless, the variables we consider explain only a small share of the variation in the proxies. The regressions explain approximately 4–7 percent of proxy variation without lagged enforcement, rising to 27–30 percent for civil non-merger and criminal cases and 16–18 percent for civil merger cases when including the lagged proxies. However, allocating each filing across adjacent years mechanically creates serial dependence. Without this allocation, the specifications including lagged enforcement explain only 11–13 percent and 3–4 percent of variation, respectively.

Most proxy variation remains after accounting for common annual movements and persistent differences across sectors. Year fixed effects alone explain 4.5 percent of variation in the merger proxy and 6.9 percent in the civil non-merger and criminal proxy; industry fixed effects alone explain 17.8 and 10.4 percent, respectively. Together, these fixed effects explain 22.3 and 17.3 percent, leaving approximately four-fifths of variation available for the industry analysis.

We next use these proxies to estimate the sector-level effects of antitrust enforcement, accounting for the sources of predictability examined here.

5 The Sector-Level Impact of Antitrust Enforcement

We estimate the dynamic response of sector-level outcomes to changes in antitrust enforcement using panel local projections. At each horizon h, we estimate the following specification:

$$y_{i,t+h} - y_{i,t-1} = \delta^z_{t,h} + \psi^z_{i,h} + \beta^z_h \nu^z_{i,t} + \sum_{j=1}^{2} \left(\alpha^z_{j,h} y_{i,t-j} + \Gamma^z_{j,h} X_{i,t-j} + \phi^z_{j,h} \nu^z_{i,t-j} \right) + u_{i,t+h}. \qquad (10)$$

[22] If we exclude the trend, agency budgets are significant. Thus, the time variation in budgets is insufficient to tell their effects apart from a pure trend effect.

$y_{i,t}$ denotes an outcome for sector i, and $\nu^z_{i,t}$ is the enforcement proxy for civil non-merger and criminal cases or civil merger cases. We reverse the raw proxy's sign so that positive values denote stronger enforcement and divide it by its pooled standard deviation. The coefficient β^z_h thus measures the response to a one-standard-deviation increase in enforcement of category z.

The specification controls for the potential determinants of enforcement examined in Section 4.4. Year fixed effects, $\delta^z_{t,h}$, absorb common annual predictors, including agency budgets, merger activity, and the HSR regime. Sector fixed effects, $\psi^z_{i,h}$, absorb persistent sector differences. Regressions also include two lags of the enforcement proxy and the control vector $X_{i,t}$, which contains real value added, its deflator, average markups, markup dispersion, and concentration, using the transformations described below. We add two own-outcome lags when the outcome is not already among these controls. We apply the Herbst and Johannsen (2024) analytic correction for finite-sample bias in dynamic local projections, including Nickell-type bias arising from estimating sector fixed effects. We report Driscoll and Kraay (1998) standard errors with eight lags. The baseline LP figures report pointwise 68 and 90 percent confidence intervals.

Our annual panel covers 19 two-digit BEA/NAICS sectors for almost seven decades. The baseline sample begins in 1957 for non-Compustat outcomes and in 1962 for markups, markup dispersion, and revenue productivity. Most series extend through 2024, although coverage and the final estimation year vary by outcome and horizon. At horizon zero, the regressions contain between 1,105 and 1,256 sector–year observations, depending on the outcome. Appendix Table A.9 summarizes the data sources.

We examine measures of competition, production and input use, innovation, productivity, and allocative efficiency. BEA data provide real value added, its deflator, real gross output, real intermediate inputs, employment, and tangible investment. The BEA–BLS integrated production accounts provide hours worked, R&D capital, and KLEMS productivity. We complement these series with industry concentration (Appendix A.2.6), IRS profit margins (Appendix A.2.7), and patent grants (Appendix A.2.8). Average markups, revenue productivity, and markup dispersion are constructed from Compustat (Appendices A.2.2 for data and A.3.1 for estimation). Sector markups and their dispersion use firm observations with markups strictly above one.

All outcomes enter in logarithms except: profit margins, which enter in levels; markup dispersion, which enters as the within-sector standard deviation of logarithmic firm markups, equivalently the logarithm of their geometric standard deviation, and serves as an indicator of allocative efficiency; and patent grants, for which we use the logarithm of one plus the number of grants.[23] All responses are multiplied by 100 and represent approximate percentage changes for logarithmic outcomes, percentage-point changes for profit margins, and approximate percentage changes in the geometric standard deviation for markup dispersion.

5.1 Results

5.1.1 Civil Non-Merger and Criminal Antitrust Cases

Civil non-merger and criminal enforcement improves the competitive environment (Figure 9). Concentration declines immediately, with a response significant at the five percent level on impact and the ten percent level through year two. Profit margins fall more gradually, with significant declines at the five percent level at years three and five. Markups fall by about 0.9 percent on impact and remain lower throughout the five-year horizon. Their decline is significant at the one percent level through year four and the ten percent level at year five.

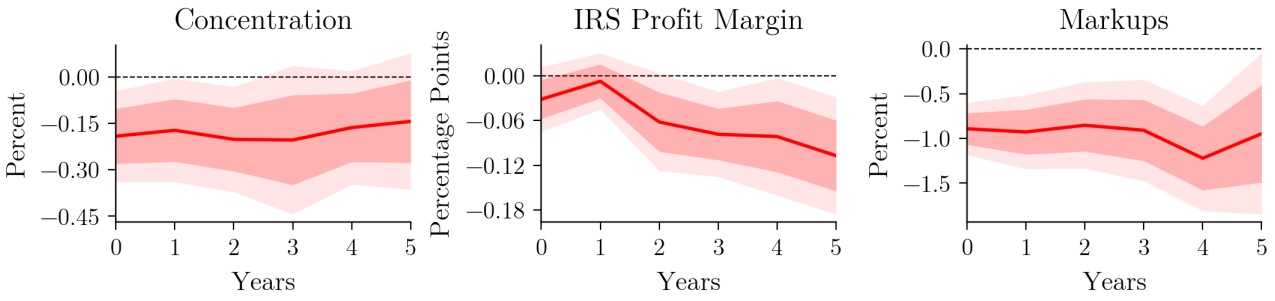


Figure 9: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Competition

Notes: Responses to a one-standard-deviation increase in the civil non-merger and criminal enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags.

Lower markups are accompanied by persistent increases in output and input use (Figure 10). After five years, real value added, employment, and hours worked are each about 1 percent

[23] We add one to retain sector–year observations with no patent grants.

higher, while real intermediate inputs are about 1.5 percent higher. These responses are significant at the one percent level over the entire forecast horizon, except for hours worked in year one, which is significant at the five percent level. In line with the fall in markups, the value added deflator falls significantly at the ten percent level in years one through three, although its response becomes less precisely estimated at later horizons. Tangible investment falls significantly at the ten percent level on impact and rises afterwards, but subsequent responses are statistically indistinguishable from zero.

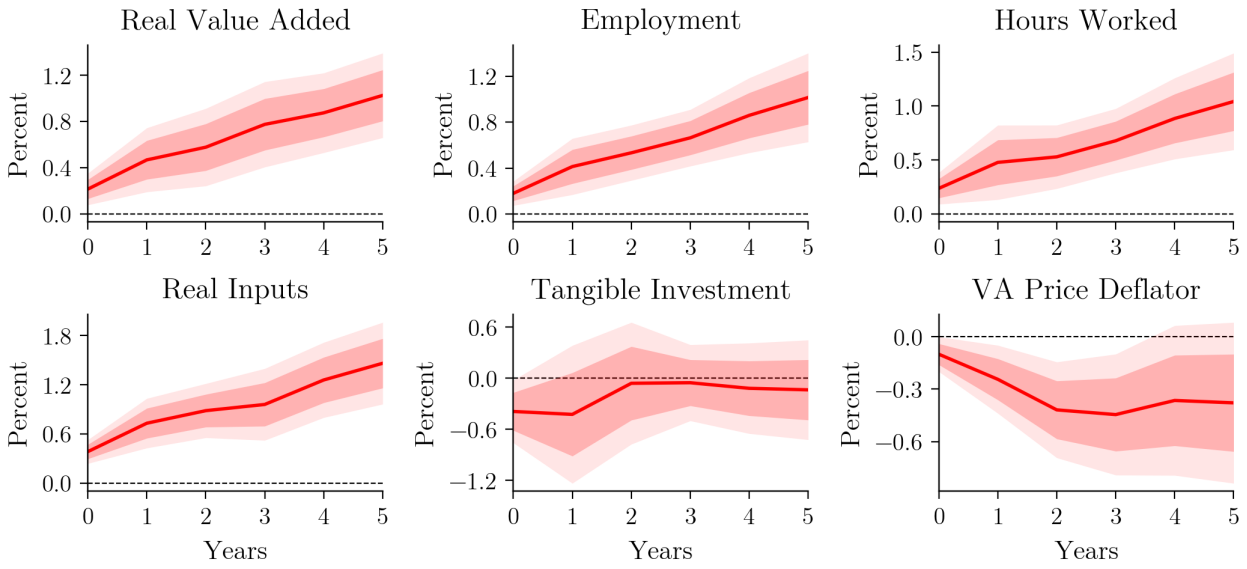


Figure 10: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Sector-Level Output, Inputs, Investment, and Prices

Notes: Responses to a one-standard-deviation increase in the civil non-merger and criminal enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags.

Innovation increases following stronger non-merger enforcement (Figure 11). R&D capital rises immediately and remains significantly higher at the one percent level throughout the horizon. Patent grants rise significantly at the five percent level from year three onward, reaching one percent significance at year five. Markup dispersion declines significantly at the one percent level from year two onward, consistent with improved allocative efficiency. Productivity responses are positive but imprecise. However, their interpretation requires care: revenue productivity also captures price effects, while KLEMS productivity does not adjust for market-power distortions.

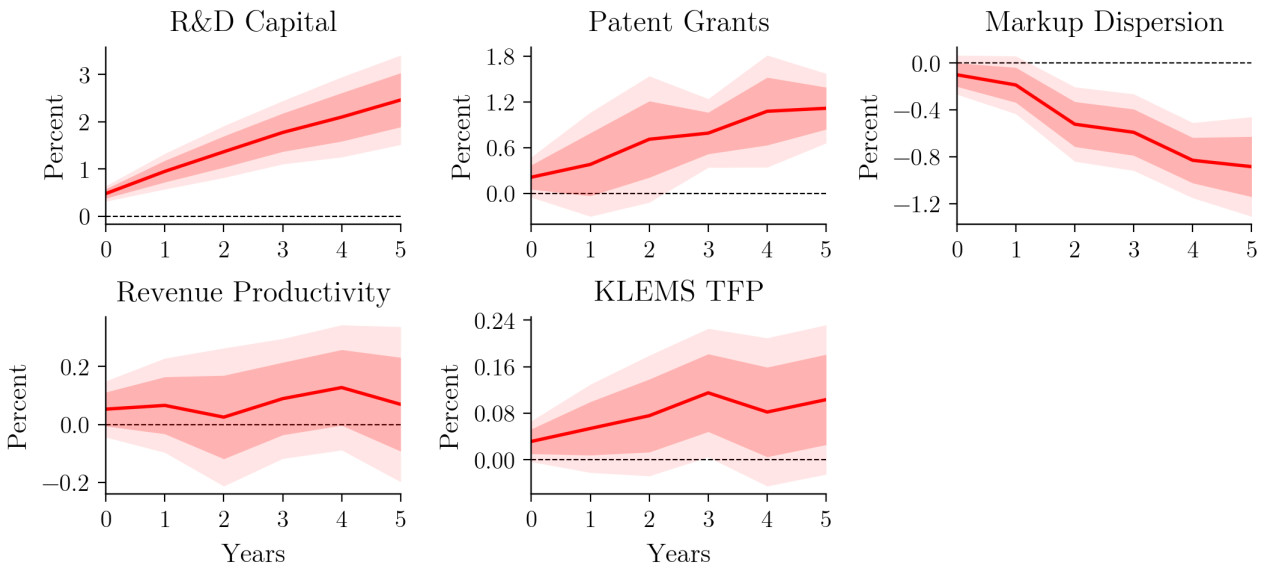


Figure 11: The Dynamic Impact of Civil Non-Merger and Criminal Cases on Sector-Level Innovation, Dispersion, and Productivity

Notes: Responses to a one-standard-deviation increase in the civil non-merger and criminal enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags.

The declines in industry markups and increases in output and input use are consistent with the firm-level evidence in Section 3, where defendants reduce markups while expanding sales and input expenditure.

5.1.2 Civil Merger Antitrust Cases

Civil merger enforcement also improves the competitive environment, although the concentration and profit-margin responses are less precisely estimated (Figure 12). Concentration initially declines, and profit-margin responses are generally negative. The 68 percent confidence intervals exclude zero over the entire forecast horizon for profit margins and at year two for concentration. Markups fall significantly at the five percent level on impact and the one percent level at every subsequent horizon.

Unlike civil non-merger and criminal enforcement, merger enforcement stimulates output without a corresponding expansion in input use (Figure 13). Real value added rises gradually and is about 0.8 percent higher after five years. The increase is significant at the ten percent level in year three and the one percent level in years four and five. In contrast, employment,

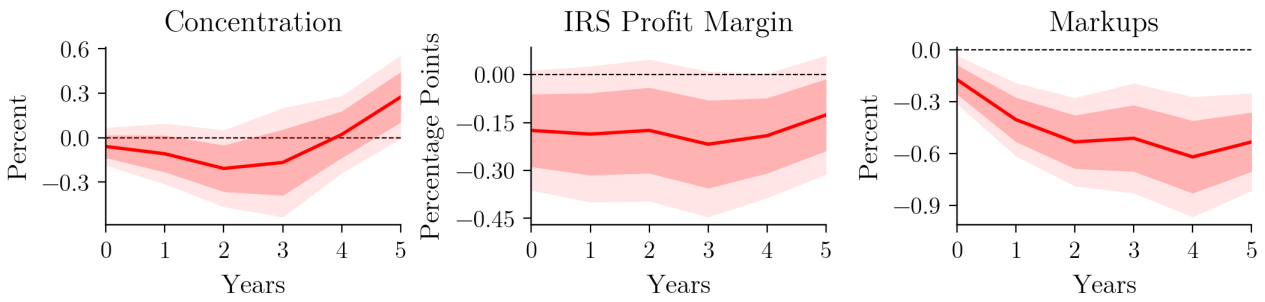


Figure 12: The Dynamic Impact of Civil Merger Cases on Competition

Notes: Responses to a one-standard-deviation increase in the civil merger enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags.

hours worked, real intermediate inputs, and tangible investment remain broadly flat. Prices decline at later horizons, but the estimates are imprecise.

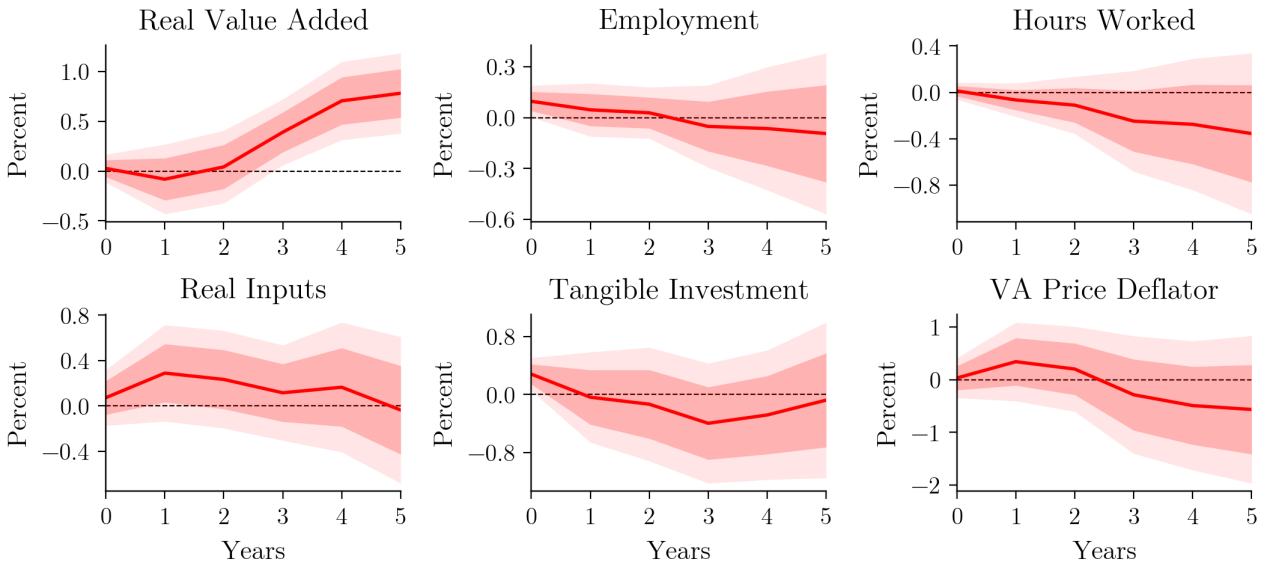


Figure 13: The Dynamic Impact of Civil Merger Cases on Sector-Level Output, Inputs, Investment, and Prices

Notes: Responses to a one-standard-deviation increase in the civil merger enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags.

Higher output without sustained increases in measured inputs suggests a role for productivity and resource allocation (Figure 14). We find that KLEMS productivity rises significantly at the ten percent level at years four and five, and the shape of its response mirrors that of

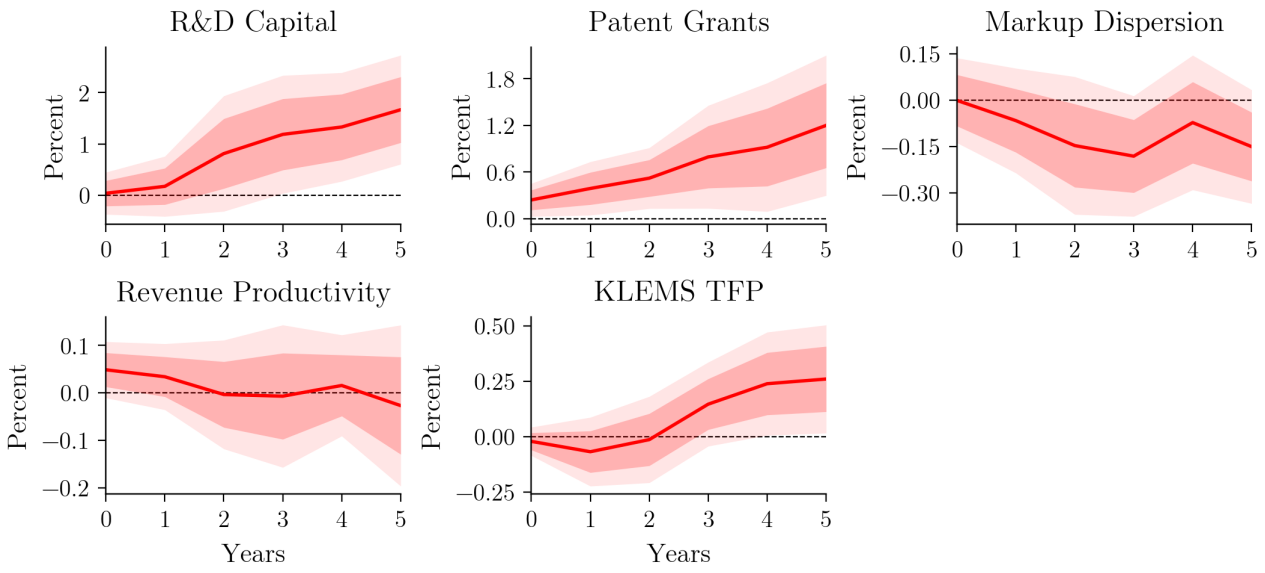


Figure 14: The Dynamic Impact of Civil Merger Cases on Sector-Level Innovation, Dispersion, and Productivity

Notes: Responses to a one-standard-deviation increase in the civil merger enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags.

real value added. Markup dispersion declines, indicating improved allocative efficiency, and its 68 percent confidence intervals exclude zero in years one through three and year five. R&D capital and patent grants also increase, with both responses significant at the five percent level at year five. These findings are particularly relevant to merger review, where potential gains from combining research capabilities must be weighed against the risk that reduced competition weakens incentives to innovate.

Our finding that civil merger enforcement stimulates output and innovation is consistent with Cavenaile, Celik and Tian (2021), who examine merger policy in a general equilibrium model with Schumpeterian innovation and oligopolistic competition. Aghion et al. (2005) document an inverted-U relationship between competition and innovation. Our estimates concern mergers selected for antitrust intervention because of their potentially anticompetitive effects. To the extent that authorities weigh potential innovation gains against competitive harm when deciding which transactions to challenge, this selection may favor cases in which protecting competition also promotes innovation.

5.2 Industry-Level Establishment and Job Dynamics

Next we examine whether enforcement changes establishment turnover and job creation. The Census Business Dynamics Statistics provide establishment entry and exit rates, together with job creation and destruction rates, for 1978–2023. We aggregate the three-digit data to the sector classification used above and estimate the same panel local projection specification. Responses are changes in annual rates, measured in percentage points. Appendix Table A.10 provides the data definitions.

Non-merger enforcement is followed by higher job creation (Figure 15, upper panels), consistent with the increase in employment shown in Figure 10. Job creation rises significantly at the one percent level on impact and at year one. This drives net job creation, which is significantly higher at the one percent level at year one and the ten percent level at year five. Establishment entry and exit both increase at some horizons, thereby indicating greater turnover, but net establishment entry is imprecisely estimated and not statistically different from zero.

In contrast, merger enforcement provides little evidence of an expansion through establishment entry or sustained employment growth (Figure 15, lower panels). Overall, responses of establishment entry and exit are muted and not statistically different from zero, though net establishment entry declines significantly at the one percent level at year three. On the other hand, job creation falls significantly at the one percent level from year three onward. In turn, net job creation rises initially and subsequently falls, with a decline significant at the five percent level at year three. These dynamics mirror the employment and hours responses in Figure 13, corroborating the evidence of output gains without sustained increases in employment.

5.3 Robustness

We examine the sensitivity of the results to outcome measurement, sample composition, case outcomes, the construction of the enforcement proxies, and the regression specification used.

First, we examine sensitivity to the production-function specification by replacing the translog with the more restrictive Cobb–Douglas form. The latter imposes constant output elasticities within each two-digit sector while allowing them to differ across sectors (equation (A.10), and see Table A.11 in Appendix A.3.1 for estimated elasticities and returns to

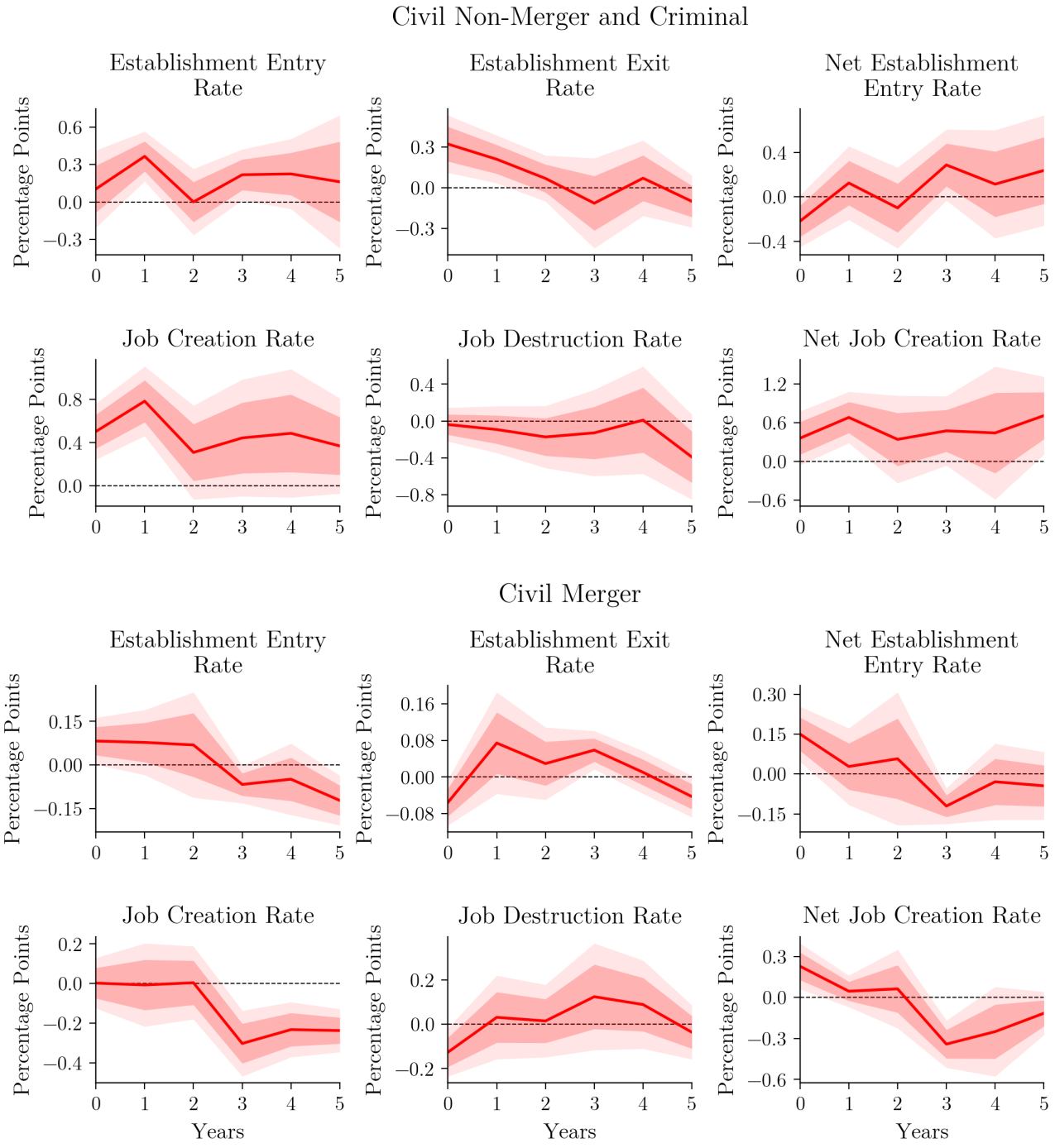


Figure 15: The Impact on Establishment and Job Turnover

Notes: Responses to a one-standard-deviation increase in the respective enforcement proxy, estimated using equation (10). Red lines show point estimates; dark and light shading denotes pointwise 68 and 90 percent confidence intervals based on Driscoll–Kraay standard errors with eight lags. The BDS sample covers 1978–2023. Responses are in percentage points.

scale). Markup and dispersion responses remain similar under the two specifications (Figure 16). Revenue productivity is more sensitive: the non-merger response becomes negative under Cobb–Douglas, whereas the translog estimate is mildly positive. Other outcome responses are largely unchanged. Figures A.9–A.10 in Appendix A.1.7 report the complete results.

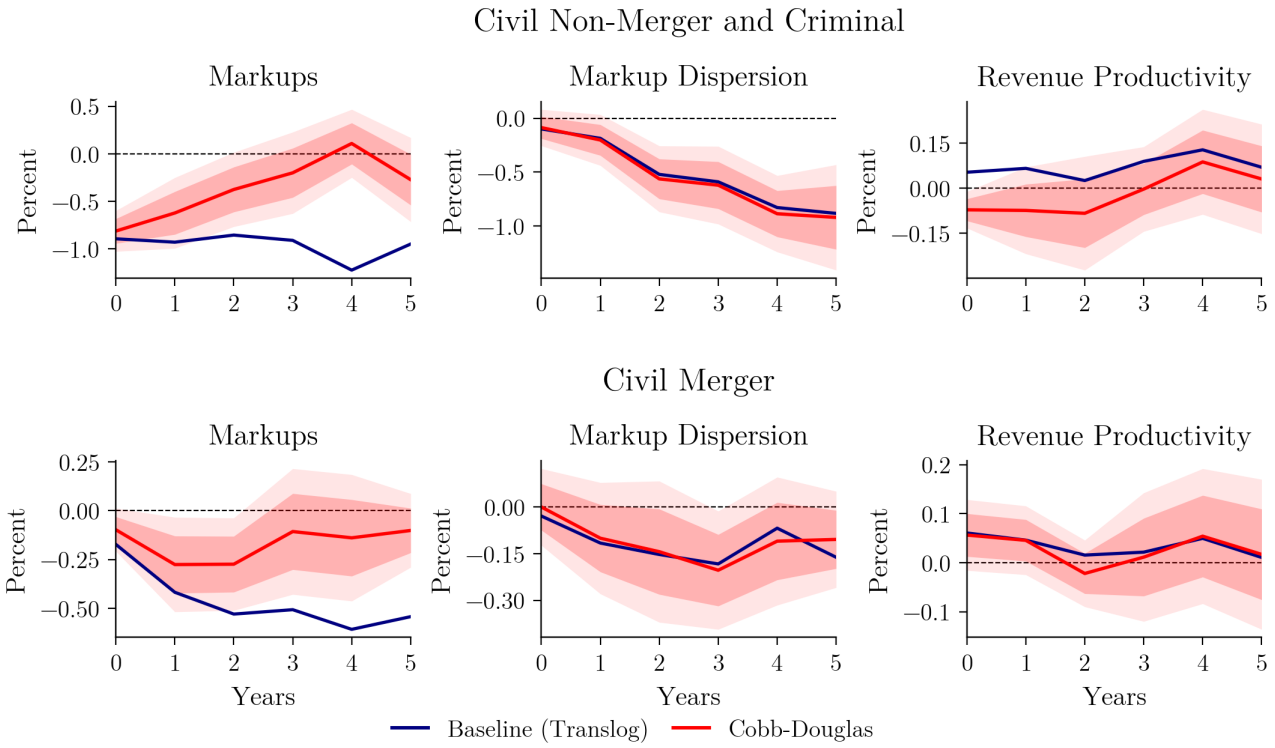


Figure 16: Sector-Level Results: Cobb–Douglas Production Function

Notes: Responses estimated using equation (10) with Cobb–Douglas rather than translog production functions. Civil non-merger and criminal cases appear in the top row and civil merger cases in the bottom row. The alternative specification is red and the baseline is navy. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the alternative, based on Driscoll–Kraay standard errors with eight lags.

Following de Loecker, Eeckhout and Unger (2020), we also check for robustness to excluding finance, insurance, and real estate (FIRE) sectors, where distinctive accounting practices complicate the measurement of output and production costs. We reestimate the responses of markups, markup dispersion, and revenue productivity on this restricted sample and find that they remain close to the baseline (Figure 17). The complete results appear in Appendix Figures A.11–A.12, which also show that excluding utilities leaves the results largely unchanged.

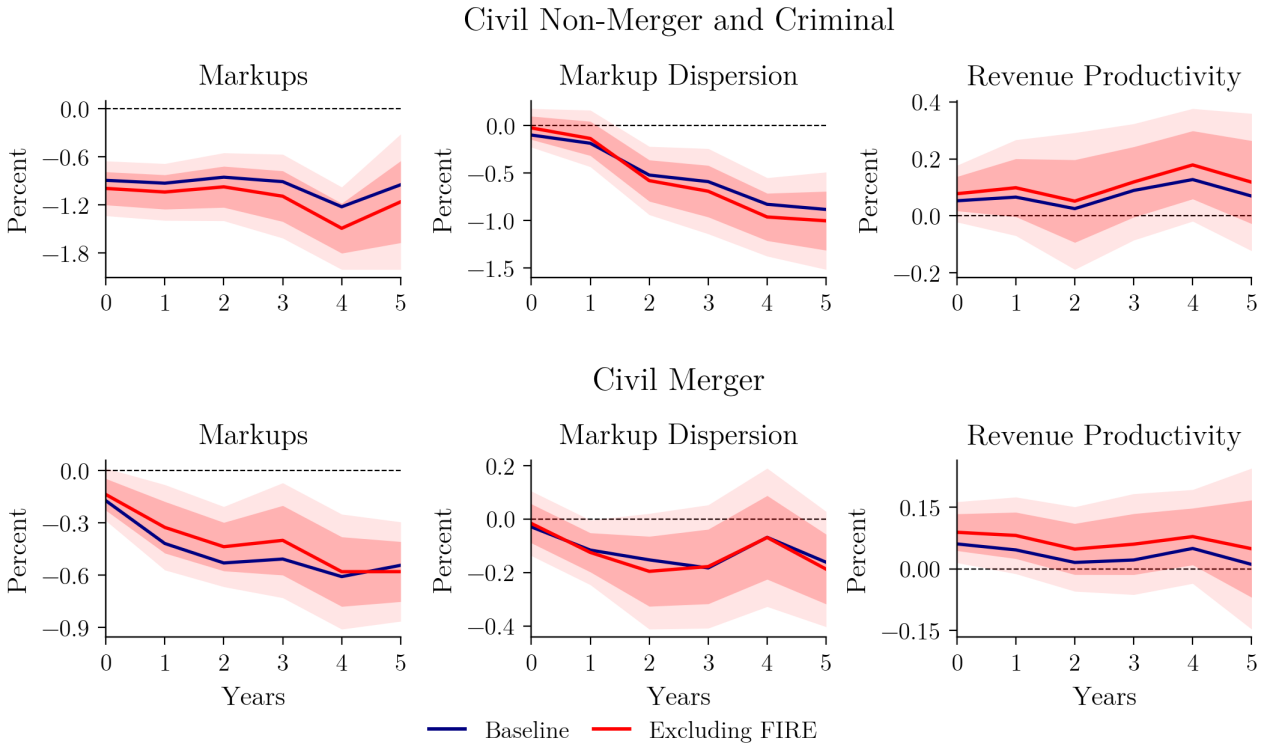


Figure 17: Sector-Level Results: Leaving Out FIRE

Notes: Responses estimated using equation (10), excluding finance, insurance and real estate. Civil non-merger and criminal cases appear in the top row and civil merger cases in the bottom row. The alternative specification is red and the baseline is navy. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the alternative, based on Driscoll–Kraay standard errors with eight lags.

The concentration in proxy variation, as shown in Figure 8, raises the concern that the results are dependent on a few cases. As a robustness exercise, we randomly omit 20 percent of contributing cases from each enforcement category, reconstruct the proxies, and reestimate the local projections over 10,000 replications. Median responses in the random case-omission exercise remain close to the baseline (Figure 18). The central 90 percent range generally excludes zero and implies similar magnitudes to the baseline responses. Appendix Figures A.13 and A.14 report the results for all outcomes.

We also examine sensitivity to excluding cases with recorded authority losses against every defendant, whether through dismissal, acquittal, or an adverse trial judgment. Cases with missing, pending, mixed, or otherwise unclassified outcomes remain in the sample. We retain all filings in the baseline to avoid conditioning on eventual case outcomes, which were unknown at filing. Enforcement proceedings may also deter anticompetitive conduct even when the

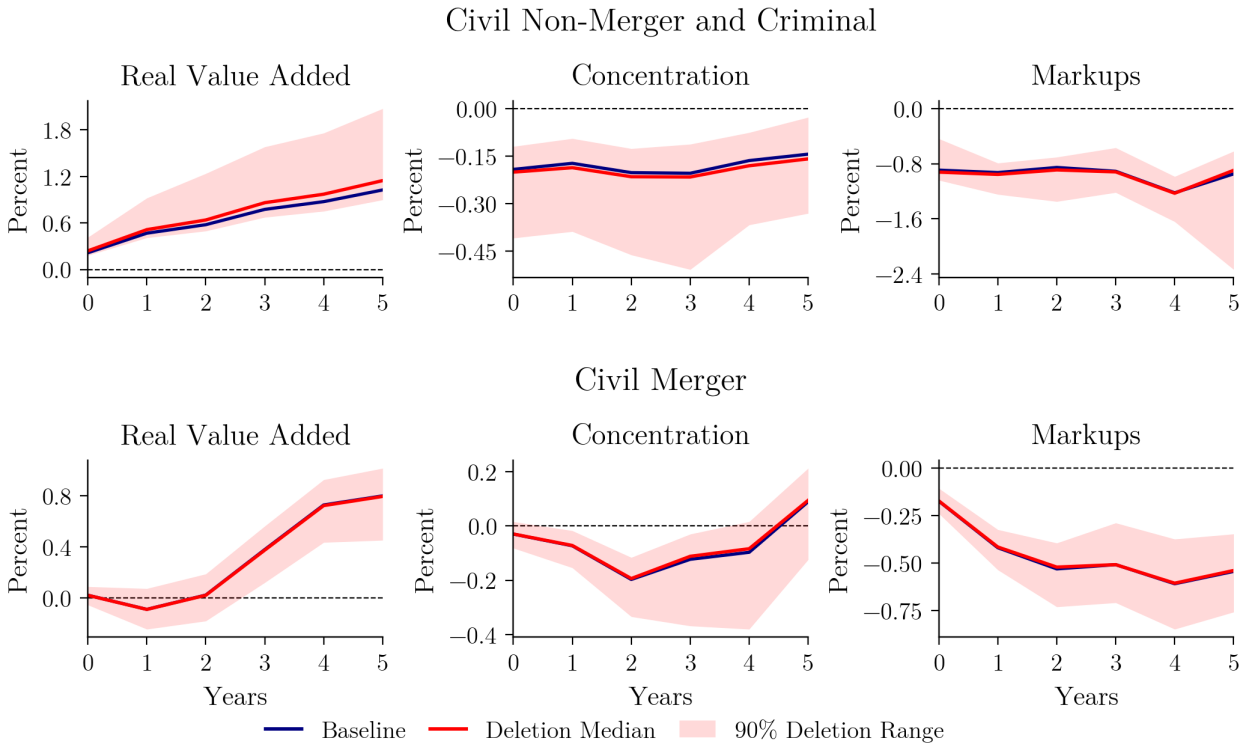


Figure 18: Sector-Level Results: Omitting 20 Percent of Cases

Notes: Responses estimated using equation (10). Each of 10,000 replications randomly omits 20 percent of contributing cases: 102 of 510 civil non-merger and criminal cases (top row), and 121 of 603 civil merger cases (bottom row). Navy lines show the baseline, red lines the median, and red shading the pointwise 5th–95th percentiles across replications. The shaded range describes sensitivity to case omissions.

authority is ultimately unsuccessful. Excluding cases with recorded authority losses preserves the main findings (Figures A.15–A.16). Responses for indicators of competition strengthen for merger enforcement and weaken somewhat for non-merger enforcement, but without changing any of the qualitative conclusions.

Another concern is that enforcement may respond to the past evolution of competition and economic activity within sectors. Therefore, we examine whether conditioning on a longer history changes the estimated responses, increasing the number of lags of the outcome, enforcement proxy, and all baseline controls from two to four. The main responses remain quantitatively very similar (Appendix Figures A.17 and A.18). These figures also compare the baseline with specifications using three lags, only one lag, or only lags of the outcome and the enforcement proxy. The responses remain very close to the baseline across all specifications.

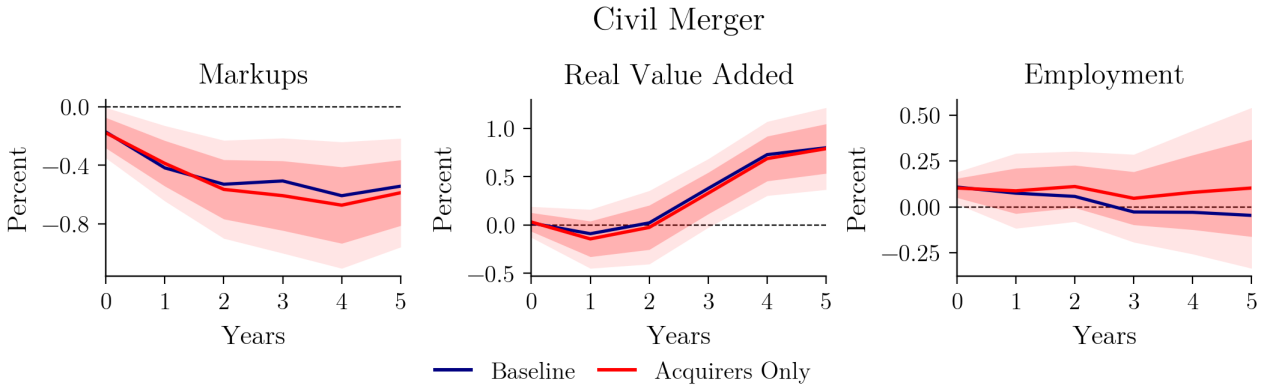


Figure 19: Sector-Level Results: Acquirer-Only Merger Proxy

Notes: Responses estimated using equation (10). Red lines use only defendants identified as acquirers, recomputing each case portfolio and its covariance adjustment; navy lines show the baseline. Dark and light red shading denotes pointwise 68 and 90 percent Driscoll–Kraay confidence intervals with eight lags. Both specifications use the same estimation samples and baseline proxy standard deviation.

A potential concern with the construction of the enforcement proxies is that equity-market losses may reflect expected fines or foregone merger premia, rather than expected changes in market power. We find, however, no statistically significant relationship between estimated cumulative abnormal returns and subsequent fines or civil monetary payments relative to defendants' pre-filing market capitalization, in either civil non-merger or criminal cases with recorded amounts. Appendix Figure A.7 plots the estimated valuation losses against fines (criminal cases) or civil payments (non-criminal cases) using logarithmic scales. For the vast majority of cases, valuation losses are orders of magnitude higher than the recorded fines or civil payments. Thus, our proxies primarily capture expected profit losses from antitrust enforcement, rather than anticipated fines.

To assess whether target shareholders' foregone takeover premia drive the merger results, we reconstruct the proxy using only defendants identified as acquirers. All responses remain quantitatively very similar to the baseline (Figure 19). Appendix Figure A.19 reports the complete set of responses and additional estimates based only on horizontal merger cases. Horizontal merger cases directly address the loss of competition from combining rival firms. In line with the expected procompetitive effects of such enforcement, we find larger declines in prices, concentration, and profit margins when restricting the proxy to these cases.

Finally, we examine sensitivity to the technical choices used to construct the proxies (Appendix Figures A.20 and A.21). Shortening the CAR cumulation window from $[-2, +10]$ to $[-2, +5]$ yields similar responses. Removing the negative-CAR filter preserves the broad qualitative pattern of the responses but generally attenuates their magnitude, consistent with greater contamination of the enforcement proxies by transitory daily stock return noise. Results using stricter negative-CAR filters are virtually unchanged and are therefore omitted for brevity.

5.4 The Distribution of Markups

The economic effects of enforcement depend not only on the decline in average markups, but also on changes in their distribution, which influences the allocation of resources across firms. We therefore estimate responses of COGS-weighted markup quantiles within each sector. The underlying firm data cover 1962–2024 and retain translog markups above one, positive sales, and positive COGS.

Let $q_{i,t}(p)$ denote the empirical quantile of logarithmic markups at rank p, with each firm's weight proportional to its COGS within sector i and year t.[24] We evaluate this function at 100 equally spaced midpoint ranks, $p_j = (j - 1/2)/100$, approximating its within-sector variation as

$$q_{i,t}(p_j) = \bar{q}_i(p_j) + \sum_{k=1}^{5} a_{i,t,k}\phi_k(p_j) + r_{i,t}(p_j), \tag{11}$$

where $\bar{q}_i$ is the sector's time-average quantile function. The basis functions ϕ_k are the five leading principal components of the covariance matrix of the sector-demeaned curves, giving each observed sector–year equal weight; $a_{i,t,k}$ are their coefficients and $r_{i,t}$ is the approximation residual. These components capture 92.4 percent of the within-sector variation in the quantile functions.

We apply the panel local projection specification described in equation (10) to each component, including lags of all five component coefficients and applying the same Herbst–Johannsen bias correction. Denoting the response of component k by $\widehat{b}^{z}_{h,k}$, we reconstruct the response at

[24] In the broad class of models studied by Edmond, Midrigan and Xu (2023), the aggregate markup is a cost-weighted average of firm-level markups. This motivates weighting the markup distribution by COGS, our measure of variable input expenditure.

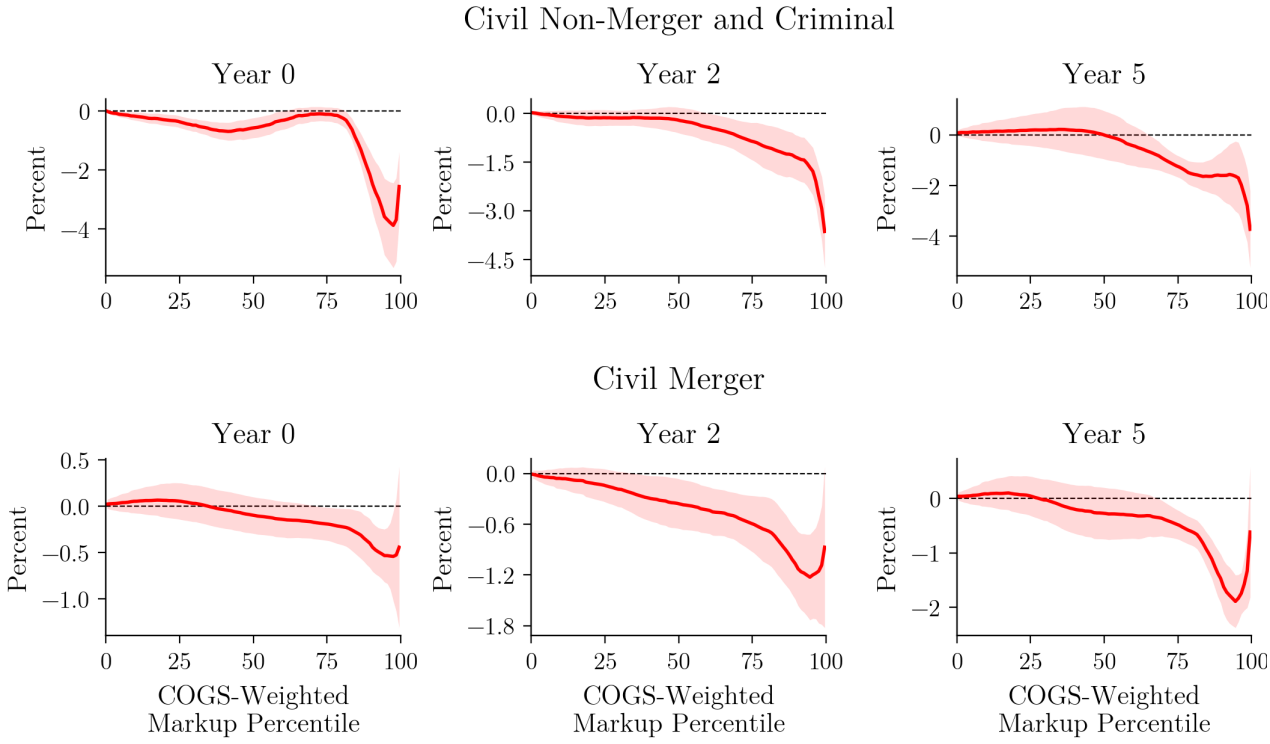


Figure 20: The Responses of Markups Across the Distribution

Notes: Responses of the COGS-weighted within-sector markup percentiles to a one-standard-deviation increase in the enforcement proxy at horizons zero, two and five, using five principal components. The graph displays all 100 estimated midpoint ranks, from the 0.5th to the 99.5th percentile. Firm observations have markups above one. Red lines are bias-corrected point estimates; shading denotes pointwise 90 percent Driscoll–Kraay confidence intervals.

percentile p as $100 \sum_{k=1}^{5} \widehat{b}^{z}_{h,k} \phi_k(p)$, giving the approximate percentage change in the markup at that percentile. We report pointwise 90 percent Driscoll–Kraay confidence intervals.

Enforcement reduces markups most strongly toward the upper end of the distribution, both on impact and at later horizons (Figure 20). This compression of the upper tail explains the declines in average markups and markup dispersion reported in Section 5.1. After five years, the responses near the 90th percentile are approximately −1.6 percent for civil non-merger and criminal cases and −1.5 percent for civil merger cases. Both declines are statistically significant at the one percent level, whereas the responses around the median are small and statistically indistinguishable from zero. Markup declines remain concentrated toward the upper tail when using seven principal components or Cobb–Douglas markups (Appendix Figure A.22). Seven-

component estimates closely match the baseline at year two, while Cobb–Douglas produces more positive responses in the lower and middle parts of the distribution.

6 The Causal Effects of Changes in Market Power

We use the antitrust proxies as instruments for sector-level average markups, allowing us to interpret the sectoral responses as the causal effects of enforcement-induced changes in market power. This connects our analysis with the literature documenting rising markups in the United States (de Loecker, Eeckhout and Unger, 2020; Döpper et al., 2025).

6.1 Approach

For each horizon h, we estimate the following specification using two-sample two-stage least squares (TS2SLS; see Inoue and Solon (2010)):

$$\text{First stage:} \qquad \mu^H_{i,t+1} = \psi^{\mu,z}_i + \delta^{\mu,z}_t + \pi^z_1 \nu^z_{i,t} + W'_{i,t} c^z + v^z_{i,t+1}, \tag{12}$$

$$\text{Second stage:} \quad y_{i,t+h} - y_{i,t-1} = \psi^{y,z}_{i,h} + \delta^{y,z}_{t,h} + \beta^{\mu,z}_h \widehat{m}^z_{i,t+1} + W'_{i,t} b^z_h + u^z_{i,t+h}, \tag{13}$$

where $\mu^H_{i,t+1}$ is the logarithm of the sales-weighted harmonic mean of translog firm markups in the year after enforcement, and $\widehat{m}^z_{i,t+1} = \widehat{\pi}^z_1 \nu^z_{i,t}$ is its instrument-predicted component. For each outcome, both stages use the same controls as the panel local projections in equation (10). We report first-stage F-statistics and pointwise 68 and 90 percent confidence intervals robust to weak instruments, obtained by inverting Anderson–Rubin-type tests, using Driscoll–Kraay covariance estimates with eight lags. Responses are normalized to a one percent increase in next-year average markups.

The two stages use different samples to retain early enforcement variation without relying on early Compustat coverage to estimate the first-stage slope. The first stage uses Compustat data from 1962 onward. The second stage retains available observations from 1957 for non-Compustat outcomes.[25] Compustat outcomes, including markup dispersion and revenue productivity, use the same 1962 sample cutoff as the first stage.

[25] The longer second-stage sample retains available pre-1962 values of the lagged controls, including average markups and markup dispersion.

The key identifying assumptions are that the antitrust proxies are exogenous conditional on the controls and correlated with average markups.[26] In particular, exogeneity requires enforcement to affect outcomes only through changes in the markup distribution. This assumption could be questioned for mergers, since such transactions may also influence productivity through synergies. Our approach assumes that transactions subject to antitrust intervention have predominantly anticompetitive effects. Additionally, because enforcement affects several features of the markup distribution, the estimates express their combined effects relative to the change in average markups, but do not isolate the contribution of the mean from other features of the distribution.

6.2 Results

Because the two enforcement proxies may induce different combinations of changes in the markup distribution, we report their normalized responses separately in Figures 21 and 22. First-stage F-statistics range from 14.1 to 18.9 for the non-merger proxy and from 8.6 to 19.9 for the merger proxy.[27]

As in the earlier analysis in Section 5, an increase in market power is accompanied by lower real value added under either instrument. A one percent increase in average markups reduces real value added after five years by approximately 1.4 percent using the civil non-merger and criminal proxy and 1.9 percent using the merger proxy. The civil non-merger and criminal response is significant at the one percent level at every horizon from impact through year five. The merger response develops more gradually, becoming significant at the ten percent level in year three and the one percent level in years four and five.

Using the civil non-merger and criminal proxy, employment, hours worked, and real intermediate inputs also decline. After five years, employment and hours are approximately 1.4 percent lower, close to the decline in real value added, while intermediate inputs fall by 2.1 percent. All three responses are significant at the one percent level at this horizon. These contractions are accompanied by higher prices, significant at the five percent level in years one to three,

[26] The first-stage relationship estimated on the later Compustat sample must also apply to the longer outcome sample.

[27] The F-statistics differ across outcomes because own-outcome lags and the available estimation samples can differ.

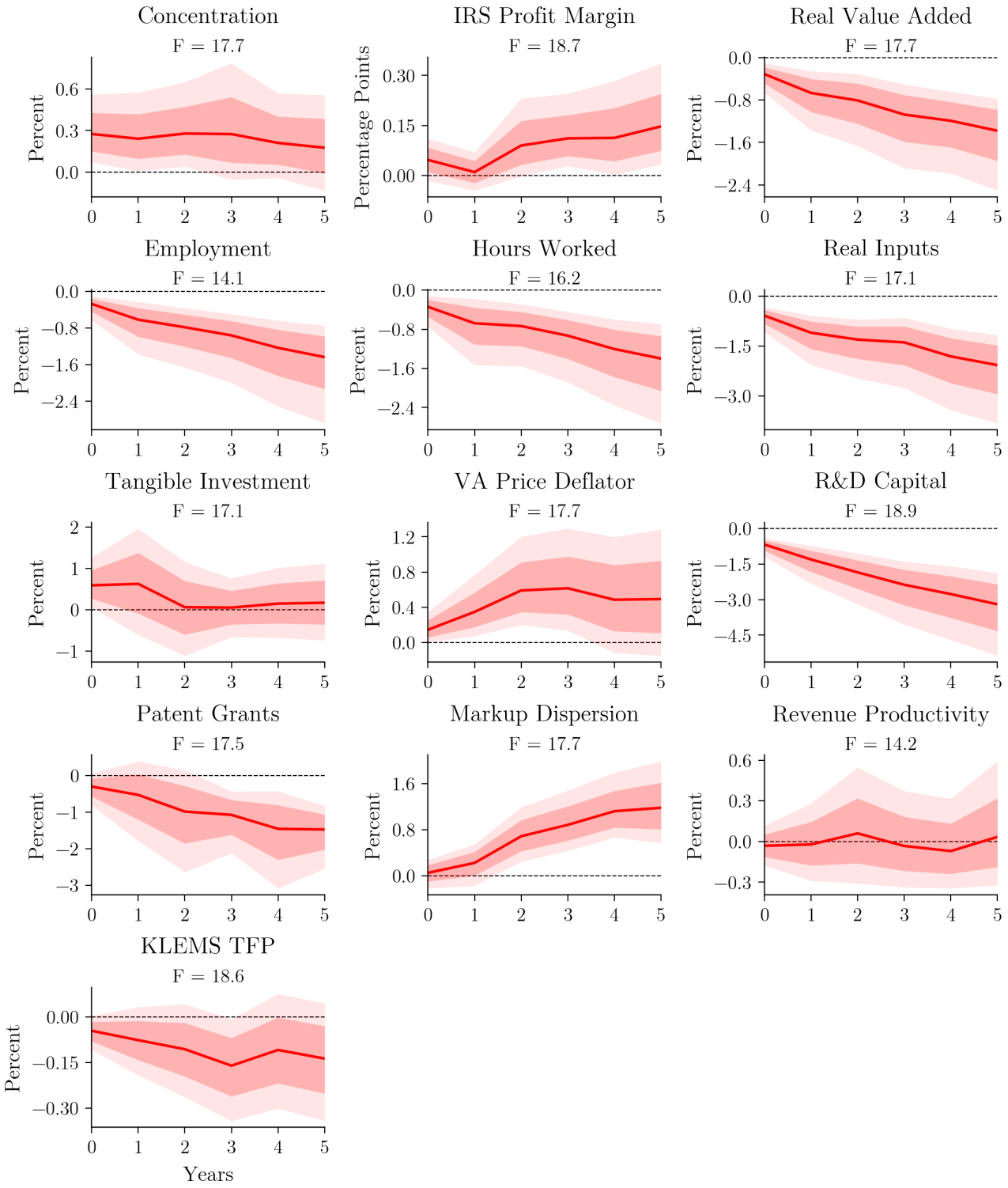


Figure 21: The Impact of Market Power: Civil Non-Merger and Criminal Cases

Notes: Responses normalized to a one percent increase in next-year average markups. Red lines show TS2SLS estimates; dark and light shading denotes pointwise 68 and 90 percent Anderson–Rubin-type confidence intervals. Panels report first-stage F-statistics.

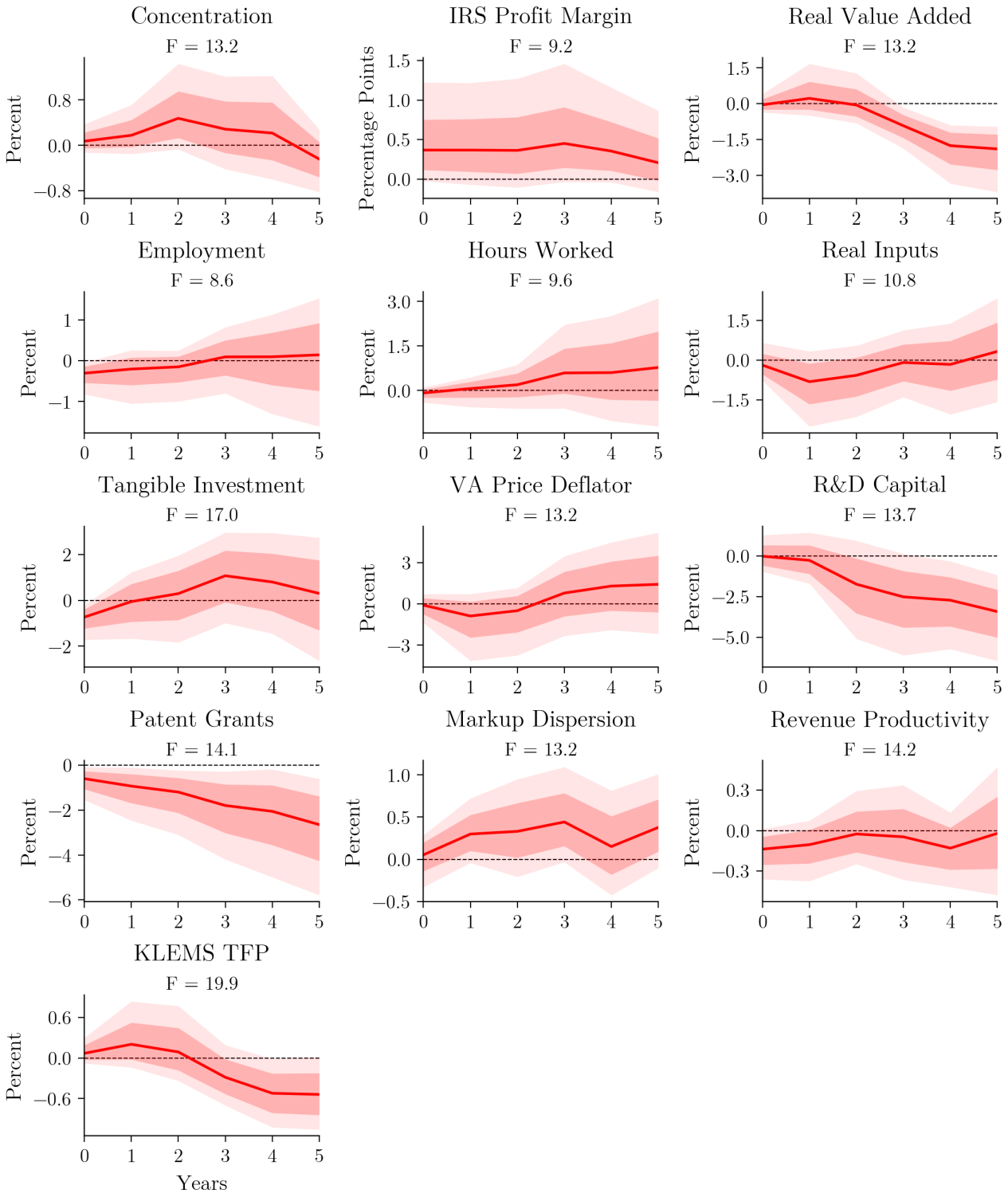


Figure 22: The Impact of Market Power: Civil Merger Cases

Notes: Responses normalized to a one percent increase in next-year average markups. Red lines show TS2SLS estimates; dark and light shading denotes pointwise 68 and 90 percent Anderson–Rubin-type confidence intervals. Panels report first-stage F-statistics.

and persistent declines in R&D capital and patenting, both significant at the one percent level after five years. Markup dispersion also increases significantly at the one percent level after five years, indicating greater misallocation, while KLEMS productivity declines persistently, with a response significant at the ten percent level in year three.

As in Section 5, the estimates using the civil merger proxy suggest a different transmission mechanism: lower output is accompanied by lower productivity and higher markup dispersion, while intermediate input spending and labor use show no sustained statistically significant changes. The decline in KLEMS productivity emerges with a delay that mirrors the output response and is significant at the ten percent level in year four. Higher markup dispersion points to greater misallocation, although this response is imprecisely estimated. For this instrument, we again find no trade-off between competition and innovation: patenting and R&D capital both decline following the rise in markups, with both responses significant at the five percent level after five years.

6.3 Antitrust and Rising Market Power

We now examine the extent to which antitrust enforcement has limited the rise in market power. Using a panel VAR, we compare observed aggregate markups with counterfactual paths obtained by setting the enforcement proxies to zero. We conduct separate simulations for civil non-merger and criminal enforcement and for civil merger enforcement.

The panel VAR contains the enforcement proxy and the five baseline control variables in equation (10). Writing their vector as $s^z_{i,t} = (\nu^z_{i,t}, X'_{i,t})'$, we estimate

$$s^z_{i,t} = \psi^z_i + \delta^z_t + \sum_{\ell=1}^{4} A^z_\ell s^z_{i,t-\ell} + B^z \varepsilon^z_{i,t}, \tag{14}$$

where ψ^z_i and δ^z_t are vectors of sector and year fixed effects. We estimate the system by least squares from 1962 onward and identify enforcement innovations through a Cholesky decomposition with the proxy ordered first: B^z is lower triangular and the structural innovations have identity covariance. We include four annual lags to allow richer dynamics and limit potential bias from extrapolating longer-horizon responses from a short lag history, in light of the bias–

variance trade-off discussed by Li et al. (2024). Overall, the impulse responses are close to the baseline local projections (Appendix Figures A.23 and A.24).

We construct a counterfactual in which the enforcement proxy equals zero in every sector and year from 1970 through 2024. Tildes denote counterfactual values. Because the proxy is ordered first, its equation is

$$\widetilde{\nu}^z_{i,t} = \underbrace{\widehat{\psi}^z_{i,1} + \widehat{\delta}^z_{t,1} + \sum_{\ell=1}^{4} \widehat{A}^z_{\ell,1\cdot} \widetilde{s}^z_{i,t-\ell}}_{\text{Predicted enforcement}} + \widehat{B}^z_{11} \widetilde{\varepsilon}^z_{i,t,1}, \tag{15}$$

where $A^z_{\ell,1\cdot}$ denotes the first row of A^z_ℓ. The first three terms predict enforcement given the counterfactual history. We set the enforcement innovation to offset this prediction exactly:

$$\widetilde{\varepsilon}^z_{i,t,1} = -\left[\widehat{\psi}^z_{i,1} + \widehat{\delta}^z_{t,1} + \sum_{\ell=1}^{4} \widehat{A}^z_{\ell,1\cdot} \widetilde{s}^z_{i,t-\ell}\right] / \widehat{B}^z_{11}, \tag{16}$$

such that $\widetilde{\nu}^z_{i,t} = 0$. We insert this innovation into the full VAR in equation (14), update all variables, and use their simulated values to construct the following year's counterfactual. Observed values for 1966–1969 provide the initial four lags. Estimated sector and year fixed effects and all other structural innovations remain unchanged. We aggregate observed and counterfactual sector markup levels using the same observed sales weights, taking the harmonic mean as in the baseline.

The counterfactual paths imply that antitrust enforcement has restrained aggregate markups (Figure 23, upper panels). Civil non-merger and criminal enforcement helps contain the rise in market power during the 1970s: observed markups remain broadly stable, while their counterfactual path rises. By 1979, aggregate markups would have been approximately 4.9 percentage points higher without such enforcement. In other words, the aggregate net markup would have been 22.3 percent, compared with the observed 17.4 percent. Consistent with the decline in antitrust activity aimed at civil non-merger and criminal offenses that we have documented, this contribution diminishes during the 1980s, with the counterfactual gap narrowing to 1.1 percentage points by 1990. The contribution of merger enforcement reaches its maximum in 2005, when aggregate markups would have been approximately 5.8 percentage points higher

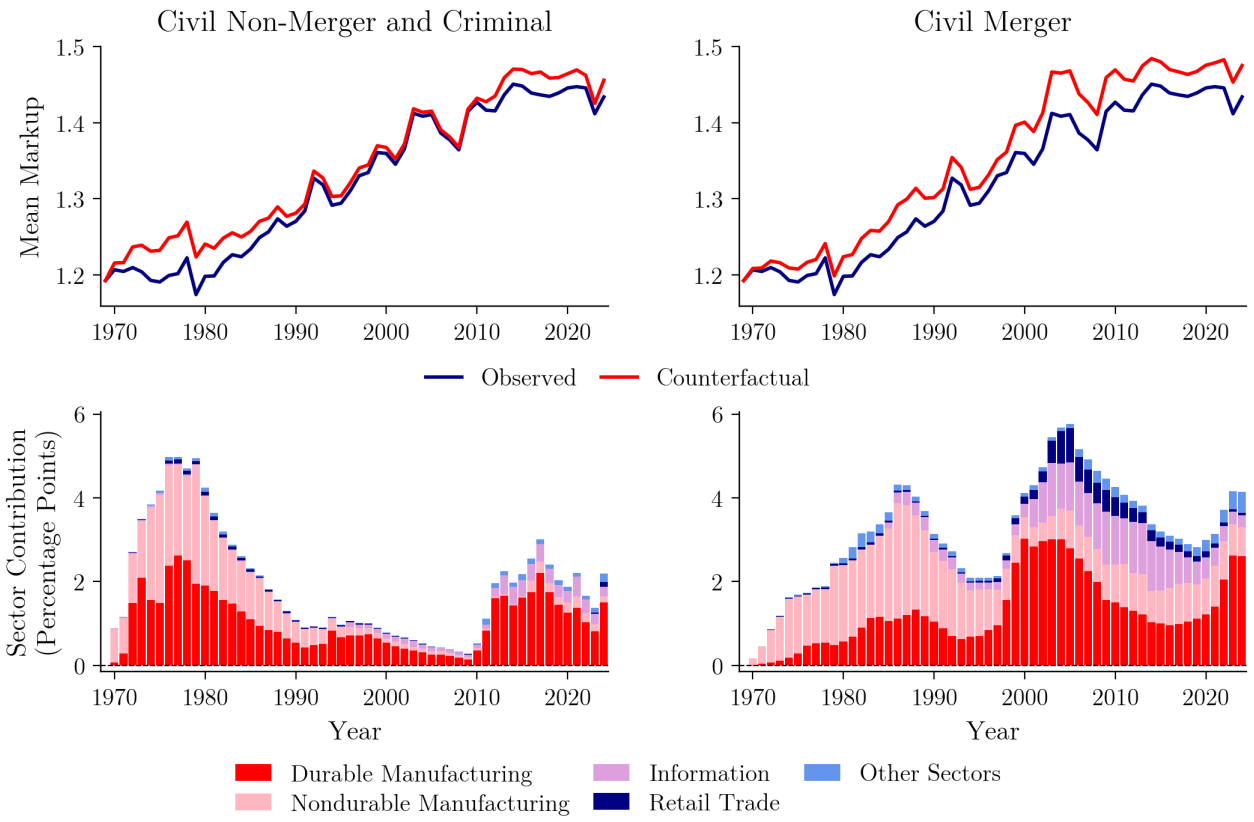


Figure 23: Antitrust Enforcement and Aggregate Markups

Notes: Upper panels show sales-weighted harmonic mean markups: observed values in navy and counterfactual values in red. The counterfactuals set the indicated enforcement proxy to zero from 1970, using the four-lag panel VAR in equation (14). Initial conditions, fixed effects, other structural innovations, and sales weights remain unchanged. Lower panels show sector contributions to the counterfactual-minus-observed markup gap, in percentage points. Each stack sums to 100 times the difference between the corresponding lines. The columns represent separate simulations.

without it. By 2024, the aggregate harmonic mean markup would have been approximately 2.2 percentage points higher without civil non-merger and criminal enforcement, and 4.1 percentage points higher without merger enforcement.[28]

Manufacturing accounts for most of the difference between observed and counterfactual markups (Figure 23, lower panels). In 1979, durable and nondurable manufacturing contribute 2.0 and 2.8 percentage points, respectively, to the 4.9-point gap for civil non-merger and criminal enforcement. For merger enforcement, the corresponding contributions are 0.5 and 1.9 percentage points. Manufacturing remains dominant later in the sample, accounting for 3.7 of

[28]Repeating the analysis with arithmetic mean markups yields corresponding differences of 4.2 and 8.9 percentage points, see Appendix Figure A.25.

the 5.8 percentage-point merger gap in 2005, while information and retail trade contribute a further 1.1 and 0.8 percentage points, respectively.

7 Conclusion

In this paper, we use federal antitrust filings since the mid-1950s to establish that curbing market power through antitrust enforcement stimulates economic activity. Firm-level estimates show that defendants facing civil non-merger and criminal proceedings reduce their markups while expanding sales and input expenditure. To examine the wider economic consequences of enforcement, we construct sector-level indicators of antitrust activity from defendants' stock market reactions to case filings, interpreting these as proxies for enforcement-induced changes in market power. The resulting estimates show that greater market power depresses output and innovation and worsens resource allocation.

The economic consequences of antitrust enforcement extend beyond price effects in individual markets. Our findings support a broader evaluation that incorporates effects on production, resource allocation, and innovation. The increases in R&D and patenting indicate that stronger competition need not reduce innovation, providing evidence against an inevitable trade-off between these policy objectives.

A further contribution is our narrative record of the universe of federal antitrust case filings in the United States since the mid-1950s. While our firm-level estimates and proxy construction rely on publicly listed defendants, the database also covers unlisted firms and individuals. We hope that the dataset will be of value to other researchers more broadly.

An important direction for further research is to connect these empirical findings to the mechanisms through which antitrust enforcement affects firms' pricing decisions and economic activity. Amorim Cabaço (2025) examines mergers using a structural firm-dynamics model and finds that their anticompetitive effects outweigh their productivity gains. Our ongoing work uses the estimates in this paper to inform a structural model of antitrust enforcement, with the aim of quantifying its aggregate and welfare consequences and evaluating alternative policies.

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A Appendix

A.1 Additional Results

A.1.1 Landmark Antitrust Cases

Year	Defendants	Agency	Type	Outcome
1950s				
1957	El Paso Natural Gas / Pacific Northwest Pipeline *Merger in natural-gas transmission*	DoJ	Merger	Divestiture ordered
1957	Procter & Gamble *Acquisition of Clorox (household bleach)*	FTC	Merger	Divestiture upheld (1967)
1958	Standard Oil (N.J.), et al. *Market allocation in petroleum products*	DoJ	Non-Merger	Consent decree
1959	Brown Shoe *Exclusive dealing in footwear*	FTC	Non-Merger	Cease-and-desist order; lifted in 1984
1960s				
1960	Westinghouse, General Electric, et al. *Price fixing and bid rigging: power transformers*	DoJ	Criminal	Guilty pleas
1960	Westinghouse, General Electric, et al. *Civil action: power-transformer cartel*	DoJ	Non-Merger	Consent decree
1961	Ford Motor / Electric Autolite *Acquisition of spark-plug and battery assets*	DoJ	Merger	Divestiture ordered against Ford
1962	Bethlehem Steel, U.S. Steel, et al. *Price fixing: open-die steel forgings*	DoJ	Criminal	No-contest pleas
1963	Bethlehem Steel, U.S. Steel, et al. *Price fixing: wrought-steel wheels*	DoJ	Criminal	No-contest pleas
1963	Blaw-Knox, Bethlehem Steel, et al. *Price fixing: steel castings*	DoJ	Criminal	No-contest pleas
1964	U.S. Steel, et al. *Bid rigging: fabricated structural steel*	DoJ	Criminal	No-contest pleas
1964	U.S. Steel, Bethlehem Steel, et al. *Price fixing: carbon-steel sheets*	DoJ	Criminal	No-contest pleas
1964	Standard Oil (N.J.) / Potash Co. of America *Merger in potash production*	DoJ	Merger	Merger blocked
1969	IBM *Monopolization of general-purpose computers*	DoJ	Non-Merger	DoJ withdrew case (1982)

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Table A.1 (Continued)

Year	Defendants	Agency	Type	Outcome
1969	Automobile Manufacturers Association, et al. *Coordination restricting emissions technology*	DoJ	Non-Merger	Consent decree
1970s				
1972	Xerox *Monopolization of plain-paper copiers*	FTC	Non-Merger	Consent order; patent licensing
1972	Kellogg, General Mills, General Foods, Quaker Oats *Shared monopoly in breakfast cereals*	FTC	Non-Merger	Dismissed (Quaker: 1978; others: 1982)
1973	Exxon and seven other oil companies *Monopolization of petroleum refining*	FTC	Non-Merger	Dismissed without reaching merits (1981)
1974	AT&T *Monopolization of telecommunications*	DoJ	Non-Merger	Consent decree; Bell System divestiture
1976	IBM *Interlocking directorates with New York Telephone*	FTC	Non-Merger	Consent order
1980s				
1983	American Airlines / Robert Crandall *Solicitation of airline price fixing*	DoJ	Non-Merger	Consent decree
1986	General Electric *Acquisition of RCA (electron tubes)*	DoJ	Merger	Consent decree; divestiture
1988	Vons / Safeway *Acquisition of supermarkets*	FTC	Merger	Consent order; store divestitures
1990s				
1992	Airline Tariff Publishing and eight airlines *Airfare coordination through fare announcements*	DoJ	Non-Merger	Consent decrees
1994	General Electric / De Beers *Price fixing: industrial diamonds*	DoJ	Criminal	GE acquitted; De Beers guilty plea, $10m fine
1994	Microsoft *Restrictive operating-system licensing*	DoJ	Non-Merger	Consent decree
1995	Microsoft / Intuit *Merger in personal-finance software*	DoJ	Merger	Transaction abandoned
1996	Archer Daniels Midland *Price fixing: lysine and citric acid*	DoJ	Criminal	Guilty plea; $100m fine
1997	Staples / Office Depot *Merger of office-supply superstores*	FTC	Merger	Injunction; transaction abandoned
1998	Microsoft *Operating-system monopoly and browser tying*	DoJ	Non-Merger	Monopoly liability upheld; consent decree
1998	Intel *Exclusion through microprocessor patent disputes*	FTC	Non-Merger	Consent order

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Table A.1 (Continued)

Year	Defendants	Agency	Type	Outcome
1999	Exxon / Mobil *Merger in petroleum refining and marketing*	FTC	Merger	Consent order; divestitures
2000s				
2000	Sotheby's *Price fixing: auction commissions*	DoJ	Criminal	Guilty plea; $45m fine
2000	AOL / Time Warner *Merger in internet access and cable television*	FTC	Merger	Consent order
2004	Infineon Technologies *Price fixing: DRAM memory chips*	DoJ	Criminal	Guilty plea; $160m fine
2005	Samsung Electronics / Samsung Semiconductor *Price fixing: DRAM memory chips*	DoJ	Criminal	Guilty pleas; combined $300m fine
2005	Procter & Gamble / Gillette *Merger in oral-care and personal-care products*	FTC	Merger	Consent order; divestitures
2009	Intel *Exclusion in processor markets*	FTC	Non-Merger	Consent order
2010s				
2010	Adobe, Apple, Google, Intel, Intuit, Pixar *Restrictions on employee recruitment*	DoJ	Non-Merger	Consent decree
2010	AU Optronics / AU Optronics America *Price fixing: LCD panels*	DoJ	Criminal	Trial convictions; $500m parent-company fine
2011	AT&T / T-Mobile / Deutsche Telekom *Merger in mobile telecommunications*	DoJ	Merger	Transaction abandoned
2012	Apple and book publishers *Price fixing: e-books*	DoJ	Non-Merger	Apple liable; publishers settled
2013	US Airways / AMR *Merger of US Airways and American Airlines*	DoJ	Merger	Consent decree; airport divestitures
2015	Sysco / US Foods *Merger in foodservice distribution*	FTC	Merger	Injunction; transaction abandoned
2015	Citicorp *Price fixing: euro–dollar foreign exchange*	DoJ	Criminal	Guilty plea; $925m fine
2015	JPMorgan Chase *Price fixing: euro–dollar foreign exchange*	DoJ	Criminal	Guilty plea; $550m fine
2015	Barclays *Price fixing: euro–dollar foreign exchange*	DoJ	Criminal	Guilty plea; $650m antitrust fine
2015	Royal Bank of Scotland *Price fixing: euro–dollar foreign exchange*	DoJ	Criminal	Guilty plea; $395m fine

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Table A.1 (Continued)

Year	Defendants	Agency	Type	Outcome
2015	Staples / Office Depot *Merger in business office supplies*	FTC	Merger	Injunction; transaction abandoned
2016	Halliburton / Baker Hughes *Merger in oilfield services*	DoJ	Merger	Transaction abandoned
2016	Anthem / Cigna *Merger in commercial health insurance*	DoJ	Merger	Merger blocked; affirmed on appeal
2016	Aetna / Humana *Merger in Medicare Advantage and health insurance*	DoJ	Merger	Merger blocked
2017	Qualcomm *Exclusion in cellular chips and patent licensing*	FTC	Non-Merger	FTC judgment reversed on appeal
2017	AT&T / DirecTV / Time Warner *Merger of television distribution and programming*	DoJ	Merger	DoJ lost at trial; affirmed on appeal
2020s				
2020	Pilgrim's Pride *Price fixing: broiler chickens*	DoJ	Criminal	Guilty plea; $107.9m fine
2020	Google *Monopolization of search and search advertising*	DoJ	Non-Merger	Liability and remedies; appeals pending
2020	Visa / Plaid *Merger in online debit payments*	DoJ	Merger	Transaction abandoned
2020	Meta Platforms (Facebook) *Monopolization of personal social networking*	FTC	Non-Merger	FTC lost at trial; appeal pending
2021	American Airlines / JetBlue *Northeast Alliance airline coordination*	DoJ	Non-Merger	Alliance enjoined; judgment upheld
2022	Meta Platforms / Within *Merger in virtual-reality fitness applications*	FTC	Merger	Injunction denied; FTC did not appeal
2022	Microsoft / Activision Blizzard *Merger in video games*	FTC	Merger	FTC complaint dismissed (2025)
2023	Google *Monopolization and tying in digital advertising*	DoJ	Non-Merger	Liability established; final remedy judgment pending
2023	JetBlue / Spirit *Merger of passenger airlines*	DoJ	Merger	Merger blocked

Table A.1: Landmark Antitrust Cases Against Publicly Listed Companies

Notes: A non-exhaustive selection from the database, with one proceeding per row. Defendant lists are abbreviated. Descriptions identify the challenged transaction or alleged conduct. The year refers to the first filing against the named defendants. Non-Merger denotes civil non-merger proceedings. Outcomes reflect the latest reviewed records available in October 2026; pending appeals are distinguished from pending trial proceedings.

A.1.2 Further Trends in Antitrust Activity

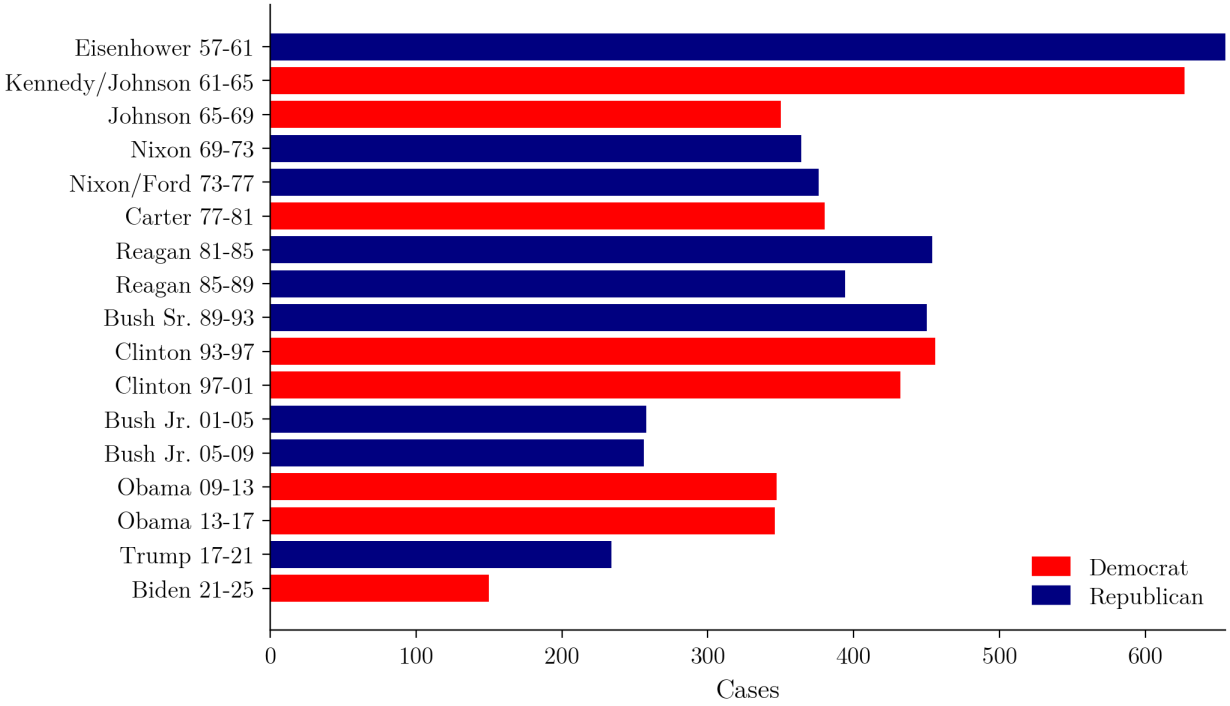


Figure A.1: Number of Antitrust Cases by Presidential Term

Notes: Counts of federal antitrust filings by four-year presidential term. Terms are calendar-year bins, including the first year shown and excluding the last. Red denotes Democrat presidents; navy denotes Republican presidents. Source: Own calculations.

	Defendants	Competitors	Difference	p-value
Equity Value	26,037.51	4,085.65	2.73	<0.001
Operating Income	2,638.49	432.94	2.69	<0.001
Capital Expenditure	1,066.38	182.71	3.28	<0.001
R&D Expenditure	889.53	153.45	2.54	<0.001
Number of Employees	47.83	8.64	2.95	<0.001
Market Share (%)	8.78	0.80	7.99†	<0.001
Sales/Employees	316.79	370.33	−53.54†	0.011
Sales/Fixed Assets	5.21	12.64	−7.44†	<0.001

Table A.2: Mean Characteristics of Defendant Firms and Their Competitors

Notes: "Competitors" are defined as listed firms operating in the same NAICS 3-digit sector as the defendant firms. Equity value, operating income, capital expenditure, and R&D expenditure are measured in millions of constant 2017 dollars, using the GDP deflator. Sales per employee are measured in thousands of constant 2017 dollars, and employment in thousands of employees. Equity value is the market value of the firm's equity. Operating income is operating income before depreciation. The market share is computed at the NAICS 3-digit level. The first two columns report means in the stated units. The Difference column compares mean logarithms, except for entries marked with †, which compare means in levels. The p-values refer to the corresponding tests.

A.1.3 Markup Levels and Dispersion

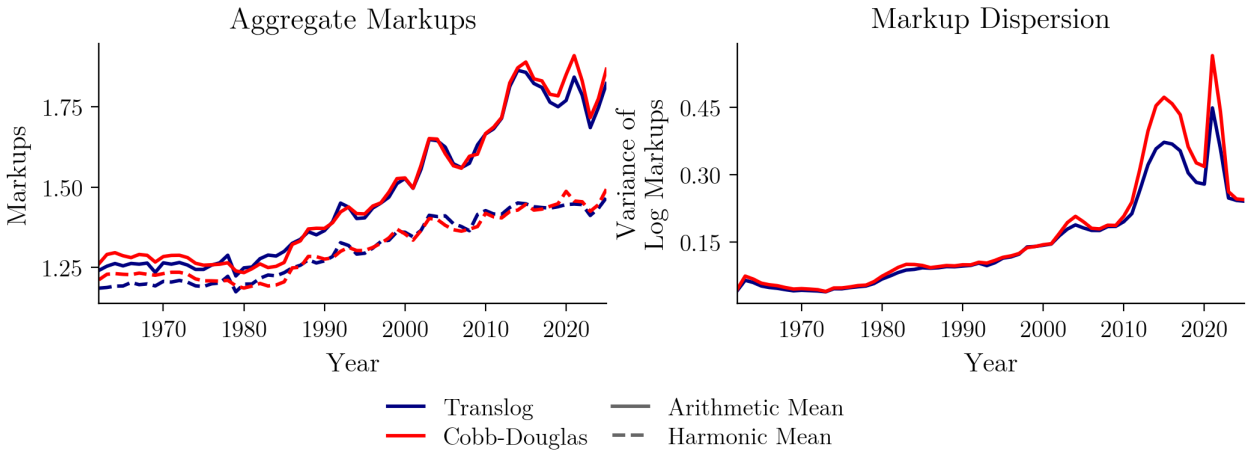


Figure A.2: Aggregate Markups and Markup Dispersion

Notes: Compustat firms from 1962 onward. Navy denotes translog and red denotes Cobb–Douglas production functions, estimated with common parameters within two-digit NAICS industries. Left: sales-weighted arithmetic (solid) and harmonic (dashed) means. Right: the cross-firm variance of logarithmic markups.

A.1.4 Additional Firm-Level Results

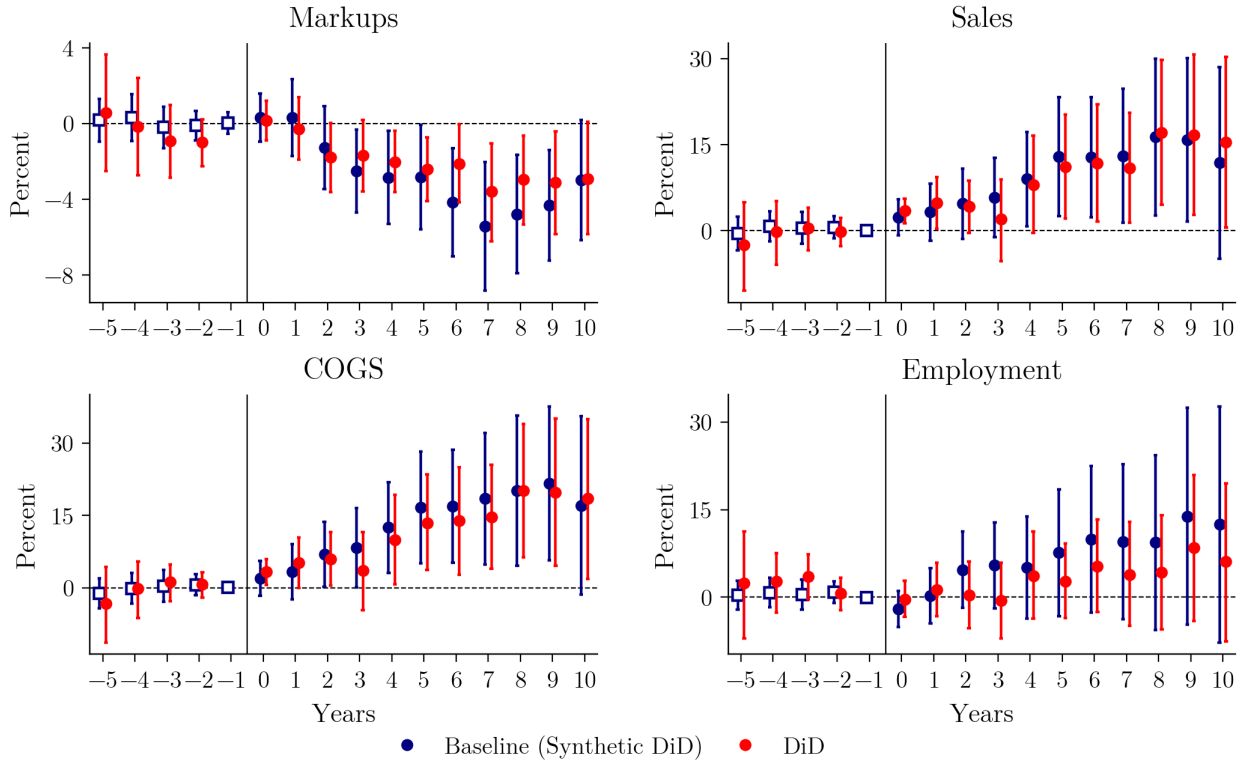


Figure A.3: Firm-Level Impact of Antitrust Case Filings: DiD and Synthetic DiD

Notes: Red denotes local-projection DiD and navy denotes synthetic DiD, retaining each estimator's baseline sample and weights. Year zero denotes the filing year. Whiskers are pointwise 95 percent confidence intervals. DiD uses inverted null-imposed wild-bootstrap tests with 10,000 filing-cohort draws and two-way firm/cohort studentization. Synthetic DiD uses normal intervals based on 1,000 global firm-pairs bootstrap refits; its pre-treatment points are fitted gaps. Estimates are shifted slightly horizontally for readability.

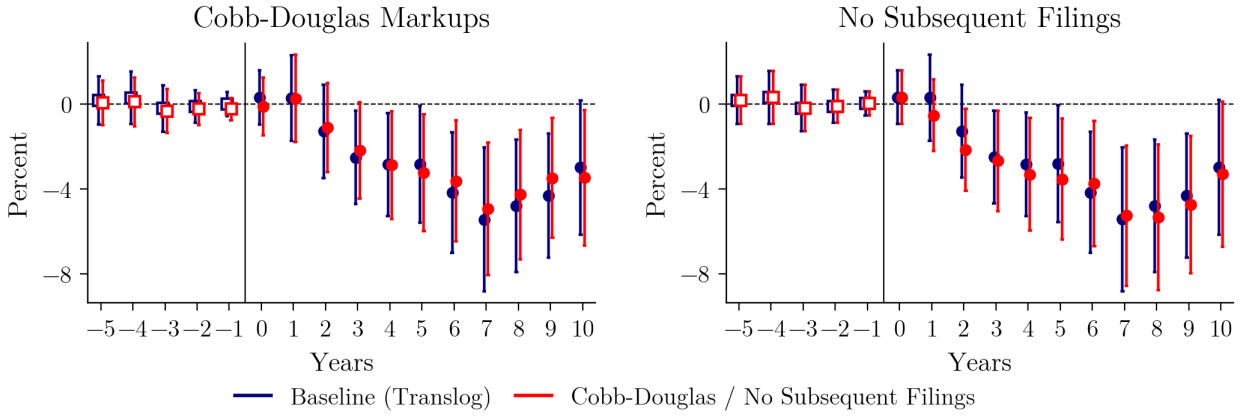


Figure A.4: Firm-Level Impact on Markups: Robustness

Notes: Synthetic DiD estimates for civil non-merger and criminal cases; see Appendix A.3.2. Navy denotes the baseline. Red denotes estimates using Cobb–Douglas markups (left) or excluding firms with an additional filing of any type in years $1, \ldots, h$ for the estimate at horizon h (right). Pre-filing points are fitted gaps. Whiskers are pointwise 95 percent normal confidence intervals from 1,000 global firm-pairs bootstrap refits.

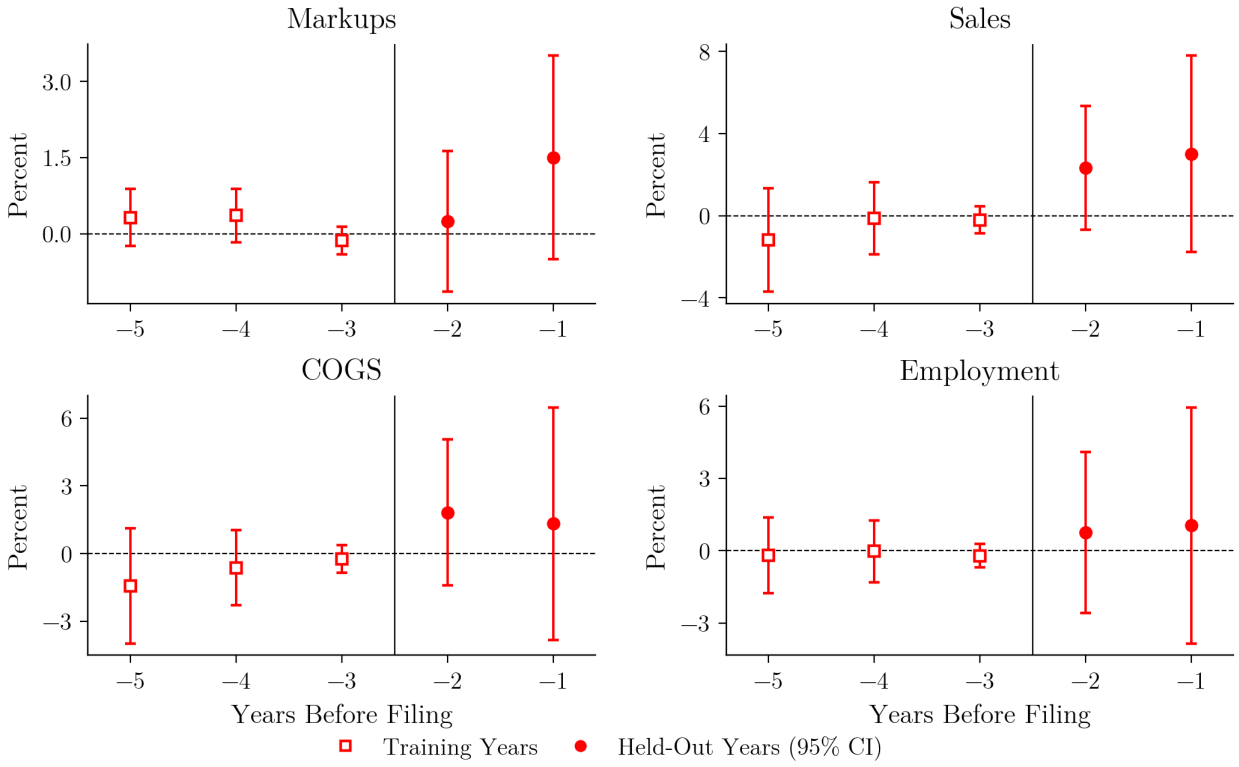


Figure A.5: Synthetic DiD: Held-Out Pre-Treatment Validation

Notes: Synthetic weights are fitted using years −5 through −3, with industry and sales-quintile cells defined at year −3. Open squares show fitted training gaps; filled circles show held-out prediction errors in years −2 and −1. The vertical line separates training and validation years. Whiskers are pointwise 95 percent normal confidence intervals from 1,000 global firm-pairs bootstrap refits. Neither held-out defendant outcome enters the weight estimation. Joint tests of zero prediction errors in years −2 and −1 have p-values of 0.15 for markups, 0.30 for sales, 0.46 for COGS and 0.89 for employment.

A.1.5 Additional Results for Abnormal Return Analysis

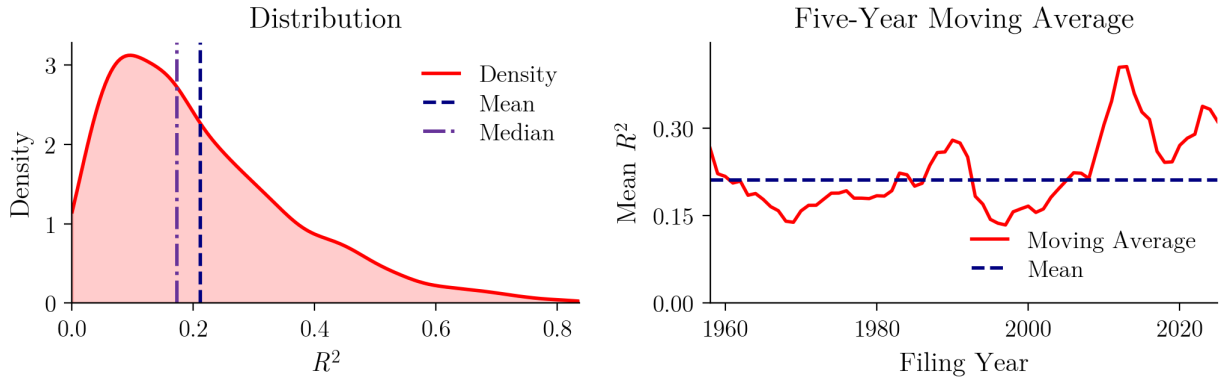


Figure A.6: Pre-Filing Factor-Model Fit: Distribution and Evolution

Notes: The left panel shows a kernel estimate of the density of the R^2 statistics from the pre-filing factor-model regressions in equation (4). The vertical lines show the mean (navy dashed) and median (purple dash-dotted). The right panel shows a five-year trailing moving average of annual mean R^2; its navy dashed horizontal line marks the full-sample mean. Annual means weight firm-filing events equally; the moving average weights years equally and requires five complete years. Both panels use the same baseline event-study sample after the exclusion of overlapping events.

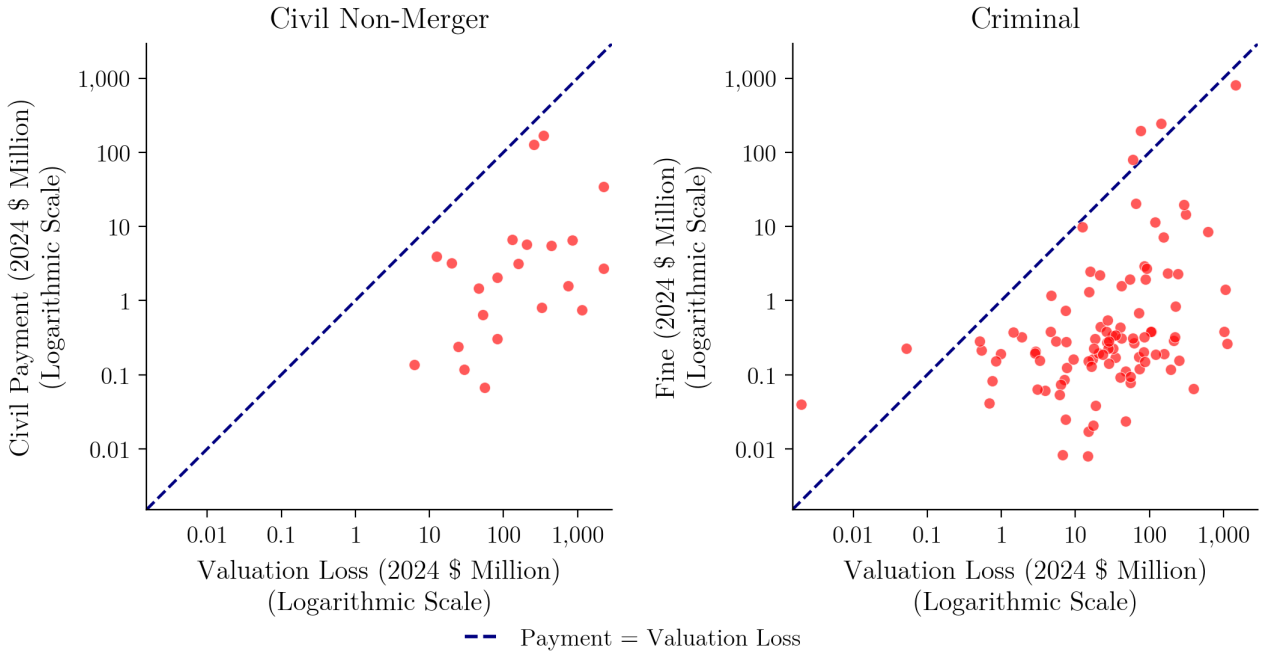


Figure A.7: Valuation Losses and Subsequent Monetary Payments

Notes: Each point is a firm-filing event with a negative CAR over trading days $[-2, +10]$ and complete recorded payments. Valuation losses are minus the CAR multiplied by pre-filing market capitalization. Civil non-merger payments include civil penalties and settlements; criminal payments are fines. Amounts are in millions of 2024 dollars, with losses deflated from the filing year and payments from the resolution year. Both axes are logarithmic, and the dashed navy line indicates equality. The panels contain 22 civil non-merger and 98 criminal observations and follow the baseline event-study exclusions.

	Department of Justice			Federal Trade Commission		
$[s_1, s_2]$	N	$\overline{CAR}$	p-value	N	$\overline{CAR}$	p-value
Panel A: All Cases						
$[-2, -1]$	1,372	−0.29	0.005	1,050	0.00	0.546
$[-2, 0]$	1,372	−0.82	<0.001	1,050	−0.42	<0.001
$[-2, 5]$	1,335	−1.55	<0.001	1,000	−0.88	<0.001
$[-2, 10]$	1,322	−1.52	<0.001	986	−1.08	<0.001
$[-2, 21]$	1,304	−1.83	<0.001	971	−1.30	<0.001
Panel B: Civil Merger						
$[-2, -1]$	579	−0.52	0.009	656	0.11	0.530
$[-2, 0]$	579	−1.10	<0.001	656	−0.21	0.156
$[-2, 5]$	542	−2.25	<0.001	606	−0.67	0.011
$[-2, 10]$	530	−2.41	<0.001	592	−0.91	0.010
$[-2, 21]$	513	−3.06	<0.001	577	−1.14	0.008

Table A.3: Cumulative Abnormal Returns by Enforcement Agency

Notes: Windows are trading days relative to filing. $\overline{CAR}$ reports equally weighted cumulative abnormal returns in percent; N is the number of firm-filing events contributing to each estimate. Reported p-values use the Boehmer, Musumeci and Poulsen (1991) test with the Kolari and Pynnönen (2010) adjustment for cross-sectional dependence. Cases involving both agencies enter both agency samples.

	Democrat President			Republican President		
$[s_1, s_2]$	N	$\overline{CAR}$	p-value	N	$\overline{CAR}$	p-value
$[-2, -1]$	1,191	−0.23	0.006	1,228	−0.10	0.339
$[-2, 0]$	1,191	−0.90	<0.001	1,228	−0.40	0.003
$[-2, 5]$	1,146	−1.26	<0.001	1,186	−1.27	<0.001
$[-2, 10]$	1,131	−1.43	<0.001	1,174	−1.23	<0.001
$[-2, 21]$	1,116	−1.96	<0.001	1,156	−1.25	<0.001

Table A.4: Cumulative Abnormal Returns by Presidential Party

Notes: Windows are trading days relative to filing. $\overline{CAR}$ reports equally weighted cumulative abnormal returns in percent; N is the number of firm-filing events contributing to each estimate. Reported p-values use the Boehmer, Musumeci and Poulsen (1991) test with the Kolari and Pynnönen (2010) adjustment for cross-sectional dependence. Categories refer to the president's party at filing.

	Horizontal			Mergers and Acquisitions			Monopolization, Exclusion, and Price Discrimination		
$[s_1, s_2]$	N	$\overline{CAR}$	p-value	N	$\overline{CAR}$	p-value	N	$\overline{CAR}$	p-value
$[-2, -1]$	567	−0.17	0.159	1,240	−0.20	0.075	292	−0.09	0.481
$[-2, 0]$	567	−0.57	0.004	1,240	−0.64	<0.001	292	−0.85	<0.001
$[-2, 5]$	567	−0.83	0.010	1,157	−1.47	<0.001	292	−1.76	<0.001
$[-2, 10]$	566	−0.54	0.213	1,131	−1.70	<0.001	292	−1.74	<0.001
$[-2, 21]$	565	−0.27	0.607	1,099	−2.15	<0.001	292	−1.98	<0.001

Table A.5: Cumulative Abnormal Returns by Violation Category

Notes: Windows are trading days relative to filing. $\overline{CAR}$ reports equally weighted cumulative abnormal returns in percent; N is the number of firm-filing events contributing to each estimate. Reported p-values use the Boehmer, Musumeci and Poulsen (1991) test with the Kolari and Pynnönen (2010) adjustment for cross-sectional dependence. Violation categories can overlap.

	Acquirer			Target		
$[s_1, s_2]$	N	$\overline{CAR}$	p-value	N	$\overline{CAR}$	p-value
$[-2, -1]$	796	−0.18	0.107	260	−0.08	0.450
$[-2, 0]$	796	−0.44	0.003	260	−1.13	0.012
$[-2, 5]$	795	−1.03	<0.001	177	−3.96	<0.001
$[-2, 10]$	795	−1.18	<0.001	153	−5.32	<0.001
$[-2, 21]$	790	−1.44	<0.001	128	−7.42	<0.001

Table A.6: Cumulative Abnormal Returns by Merger Role

Notes: Windows are trading days relative to filing. $\overline{CAR}$ reports equally weighted cumulative abnormal returns in percent; N is the number of firm-filing events contributing to each estimate. Reported p-values use the Boehmer, Musumeci and Poulsen (1991) test with the Kolari and Pynnönen (2010) adjustment for cross-sectional dependence. Roles refer to the listed security. Selling parents and other or unresolved roles are excluded.

	Competitors – Civil Non-Merger and Criminal			Competitors – Civil Merger		
$[s_1, s_2]$	N	$\overline{CAR}$	p-value	N	$\overline{CAR}$	p-value
$[-2, -1]$	89	0.00	0.677	264	−0.13	0.378
$[-2, 0]$	89	0.27	0.386	264	−0.43	0.112
$[-2, 5]$	89	−0.16	0.862	264	−0.18	0.659
$[-2, 10]$	89	−0.38	0.959	264	−0.69	0.317
$[-2, 21]$	89	2.68	0.051	264	−0.65	0.182

Table A.7: Cumulative Abnormal Returns of Defendants' Competitors

Notes: Windows are trading days relative to filing. $\overline{CAR}$ reports equally weighted cumulative abnormal returns in percent; N is the number of firm-filing events contributing to each estimate. Reported p-values use the Boehmer, Musumeci and Poulsen (1991) test with the Kolari and Pynnönen (2010) adjustment for cross-sectional dependence. Each defendant's closest eligible competitor is selected using the highest filing-year product-similarity score in Hoberg and Phillips (2025), available for 1988–2023. Firms with their own filing within 730 calendar days before or after the focal filing are excluded. See Appendix A.2.4.

Comparison	Cumulative Return Window			
	$[-2, 0]$	$[-2, 5]$	$[-2, 10]$	$[-2, 21]$
DoJ − FTC All Cases	−0.40 p =0.068	−0.68 p =0.036	−0.45 p =0.250	−0.53 p =0.293
DoJ − FTC Civil Merger	−0.89 p =0.007	−1.58 p <0.001	−1.50 p =0.009	−1.92 p =0.008
Merger − Horizontal	−0.07 p =0.810	−0.64 p =0.100	−1.16 p =0.012	−1.88 p =0.004
Monop./Excl./Price Discr. − Horizontal	−0.27 p =0.438	−0.93 p =0.074	−1.20 p =0.041	−1.70 p =0.033
Target − Acquirer	−0.69 p =0.158	−2.93 p =0.002	−4.13 p <0.001	−5.98 p <0.001

Table A.8: Differences in Cumulative Abnormal Returns

Notes: Each cell reports the first group's cumulative abnormal return minus the second group's, in percentage points, with the two-sided p-value below. Tests use 5,000 wild-bootstrap draws clustered by filing date, sharing bootstrap signs across groups and accounting for overlapping samples. Windows are trading days relative to filing. Tests are pointwise and unadjusted for multiple comparisons. Monop./Excl./Price Discr. denotes monopolization, exclusion, and price discrimination.

A.1.6 Agency Budgets and Merger Activity

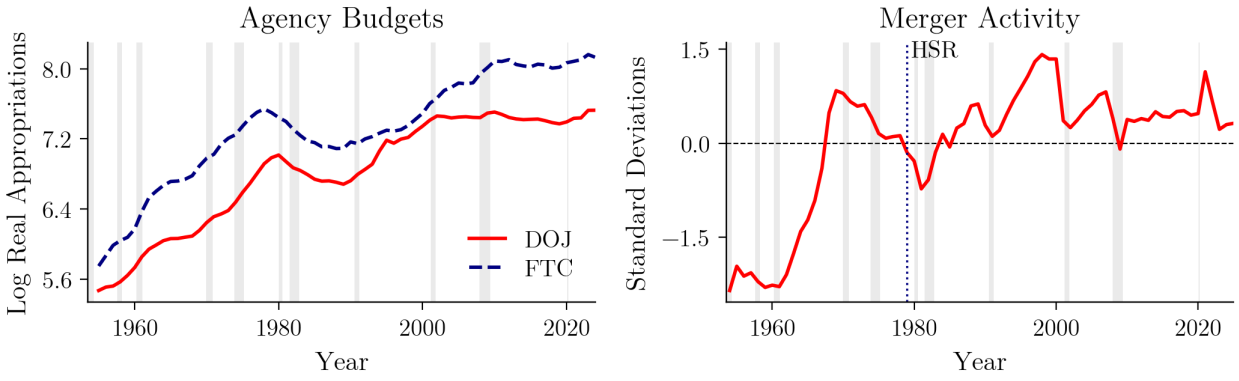


Figure A.8: Antitrust Agency Budgets and Aggregate Merger Activity

Notes: Left: logarithms of real calendar-year appropriations for the DoJ Antitrust Division (red) and FTC (navy), 1955–2024, deflated using the GDP deflator. Right: the filtered merger-activity index in standard-deviation units, 1954–2025. The dotted line marks the HSR regime from 1979; shading denotes NBER recessions. Appendix A.2.5 describes the construction of both series.

A.1.7 Additional Results: LP Estimates

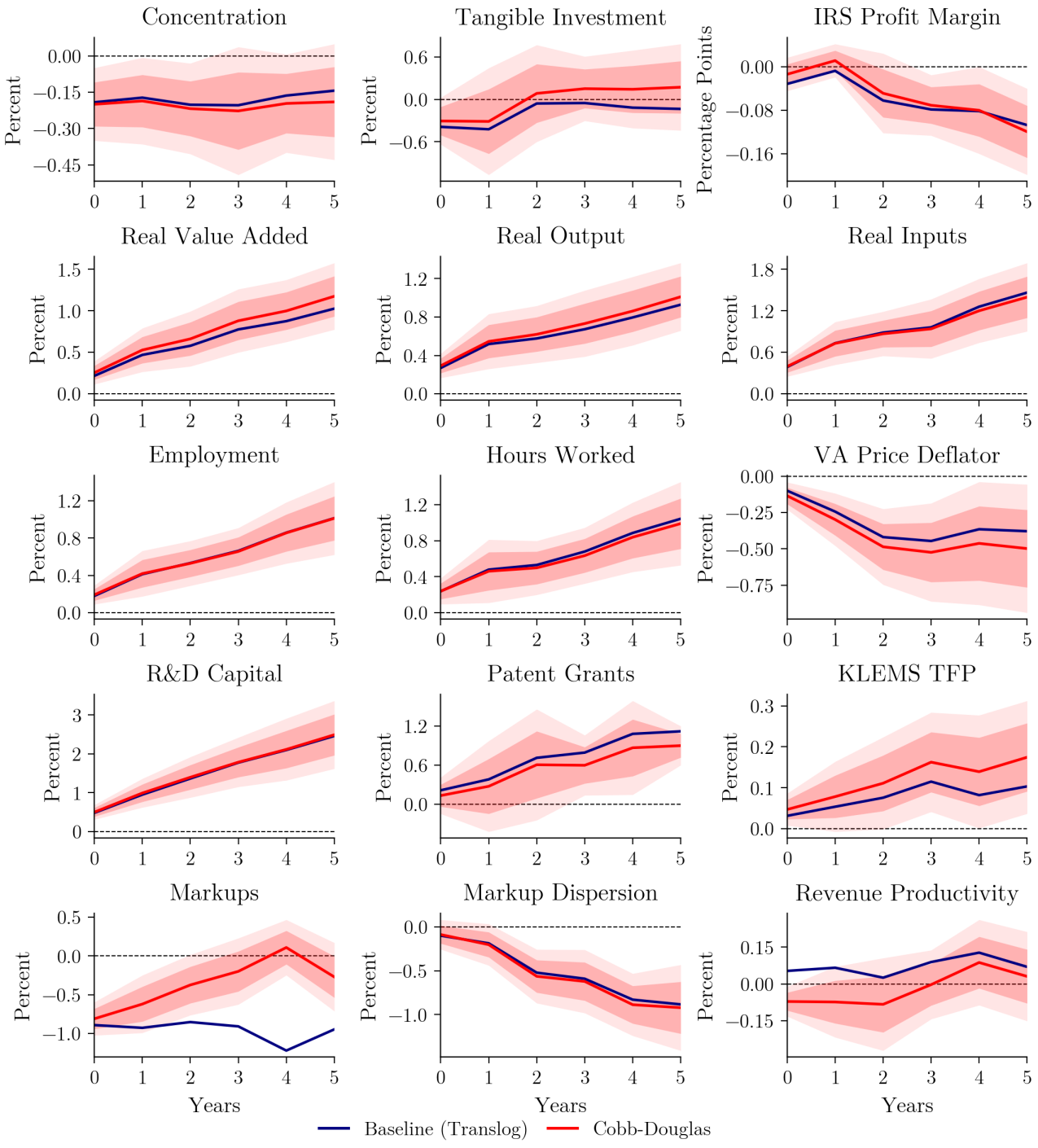


Figure A.9: Cobb–Douglas Technologies: The Dynamic Impact of Civil Non-Merger and Criminal Cases

Notes: Responses estimated using equation (10) with Cobb–Douglas production functions. The proxy covers civil non-merger and criminal cases. The alternative specification is red and the baseline is navy. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the alternative, based on Driscoll–Kraay standard errors with eight lags.

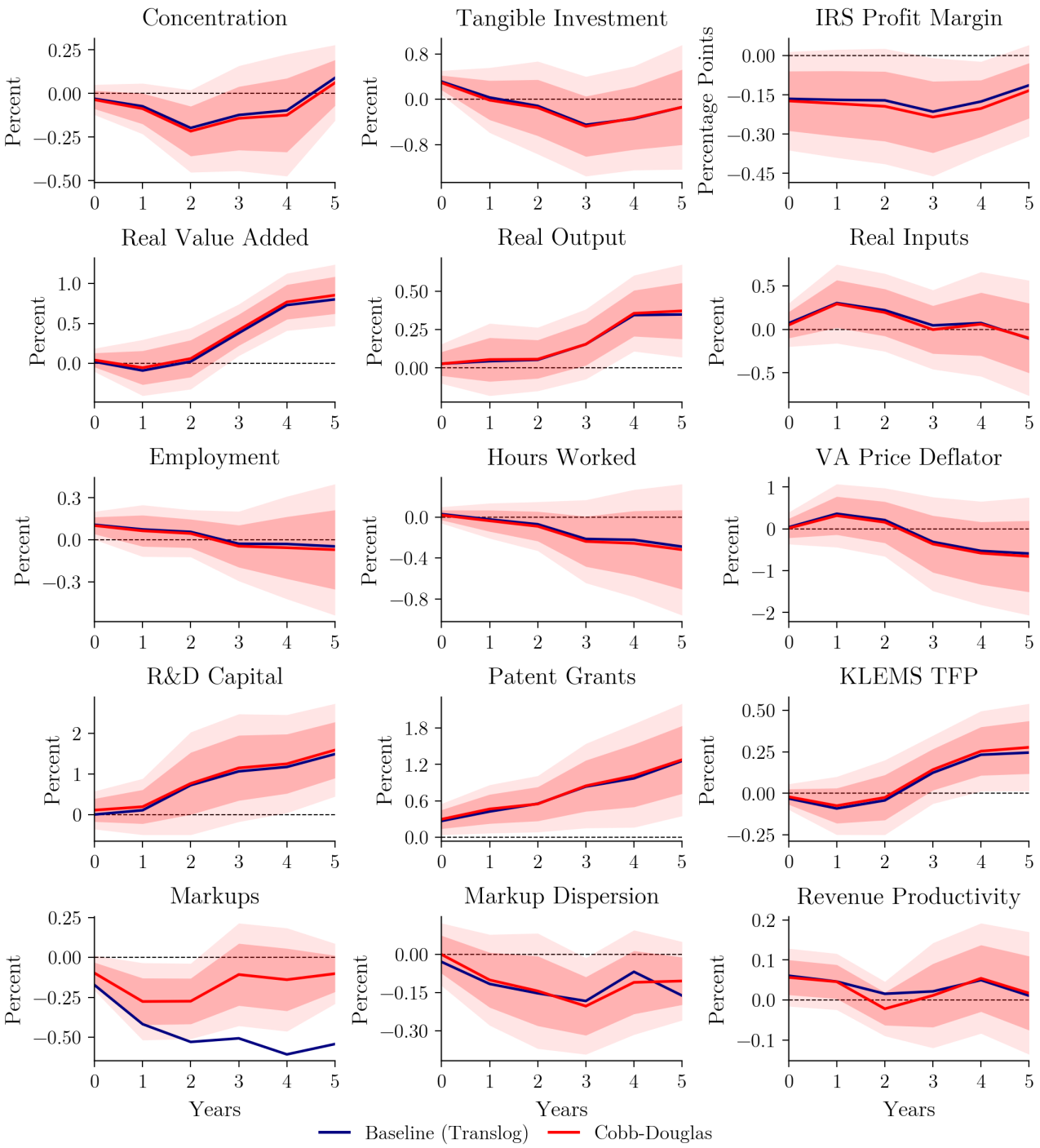


Figure A.10: Cobb–Douglas Technologies: The Dynamic Impact of Civil Merger Cases

Notes: Responses estimated using equation (10) with Cobb–Douglas production functions. The proxy covers civil merger cases. The alternative specification is red and the baseline is navy. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the alternative, based on Driscoll–Kraay standard errors with eight lags.

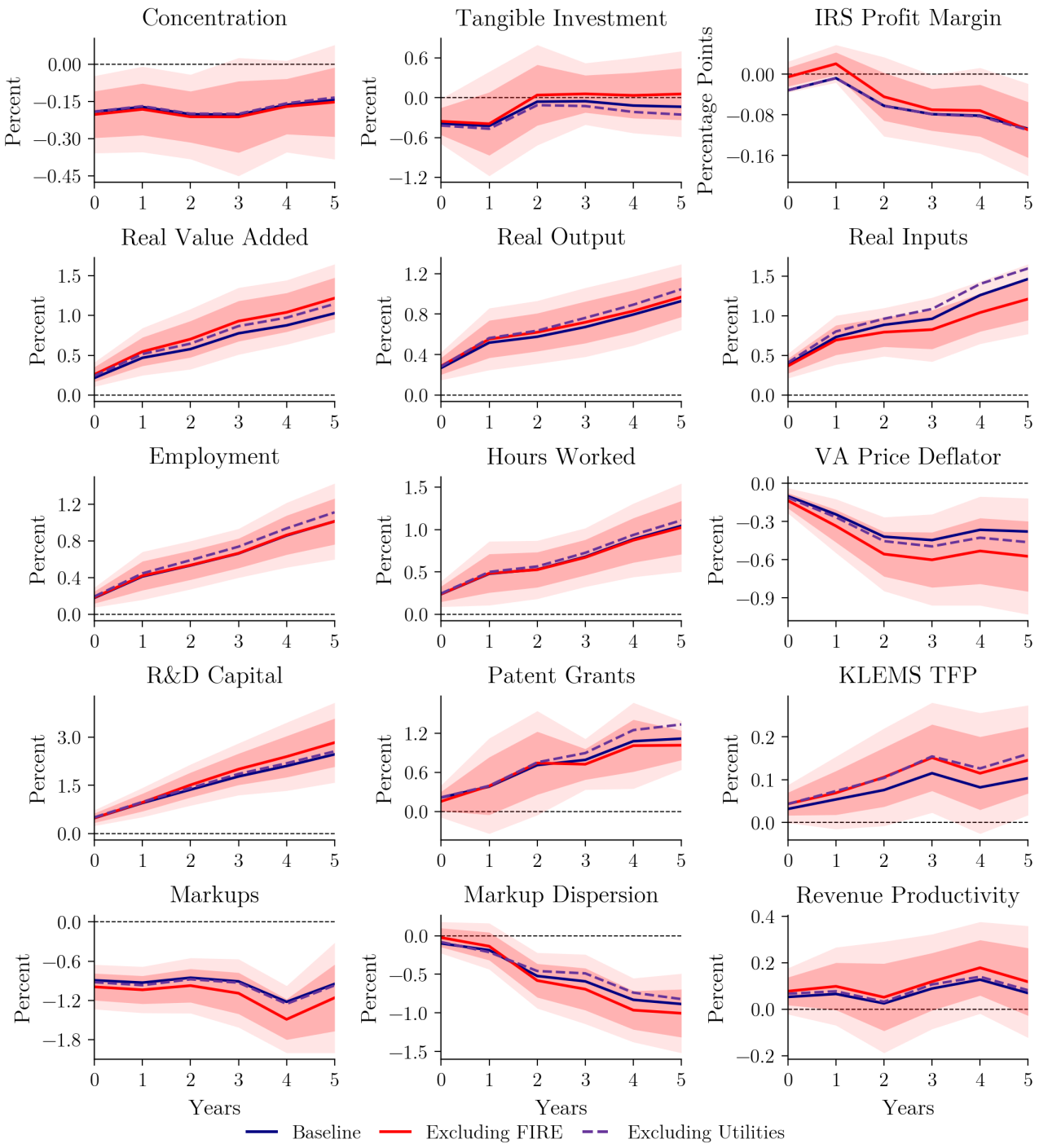


Figure A.11: Excluding FIRE or Utilities: Civil Non-Merger and Criminal Cases

Notes: Responses estimated using equation (10). Navy lines show the baseline; red lines exclude finance, insurance and real estate (FIRE); purple dashed lines exclude utilities. Each sector restriction is applied separately. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the FIRE exclusion only, based on Driscoll–Kraay standard errors with eight lags.

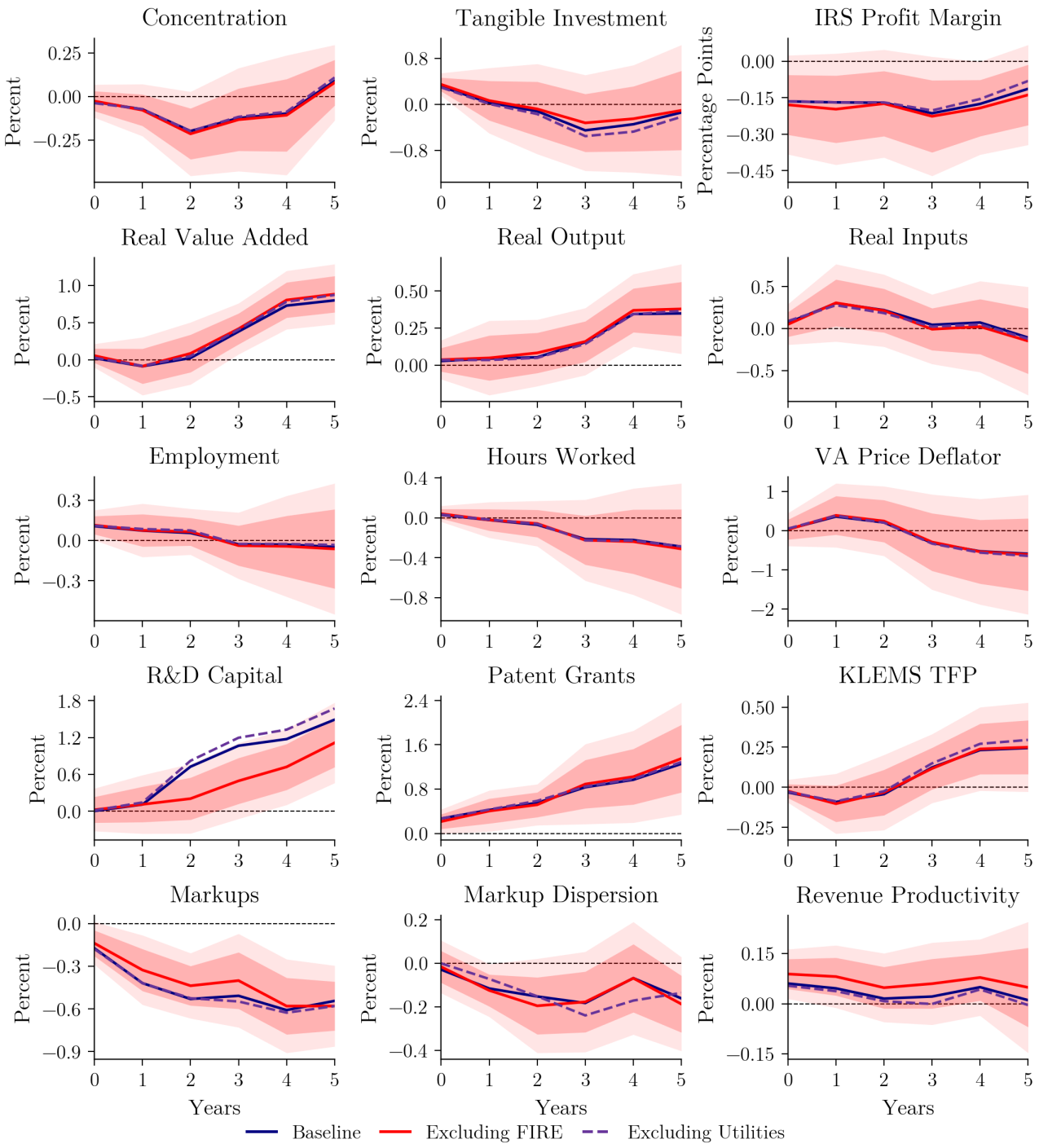


Figure A.12: Excluding FIRE or Utilities: Civil Merger Cases

Notes: Responses estimated using equation (10). Navy lines show the baseline; red lines exclude finance, insurance and real estate (FIRE); purple dashed lines exclude utilities. Each sector restriction is applied separately. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the FIRE exclusion only, based on Driscoll–Kraay standard errors with eight lags.

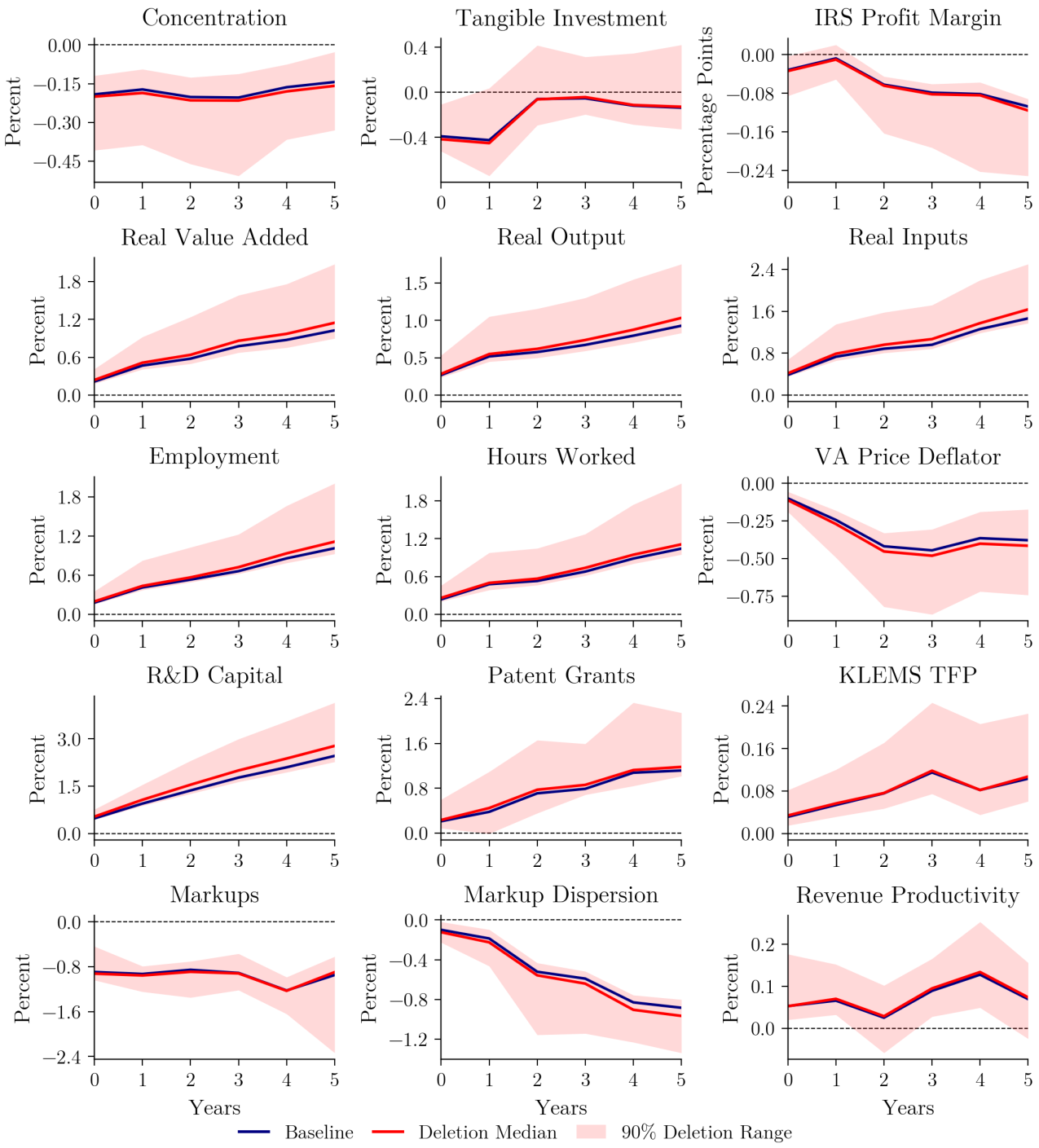


Figure A.13: Omitting 20 Percent of Cases: Civil Non-Merger and Criminal Cases

Notes: Each of 10,000 replications omits 102 of 510 contributing cases and reestimates equation (10), keeping outcome samples and the original proxy normalization fixed. Navy lines show the baseline; red lines and shading show the median and pointwise 5th–95th percentiles. Shading represents sensitivity to case omissions, not sampling confidence intervals.

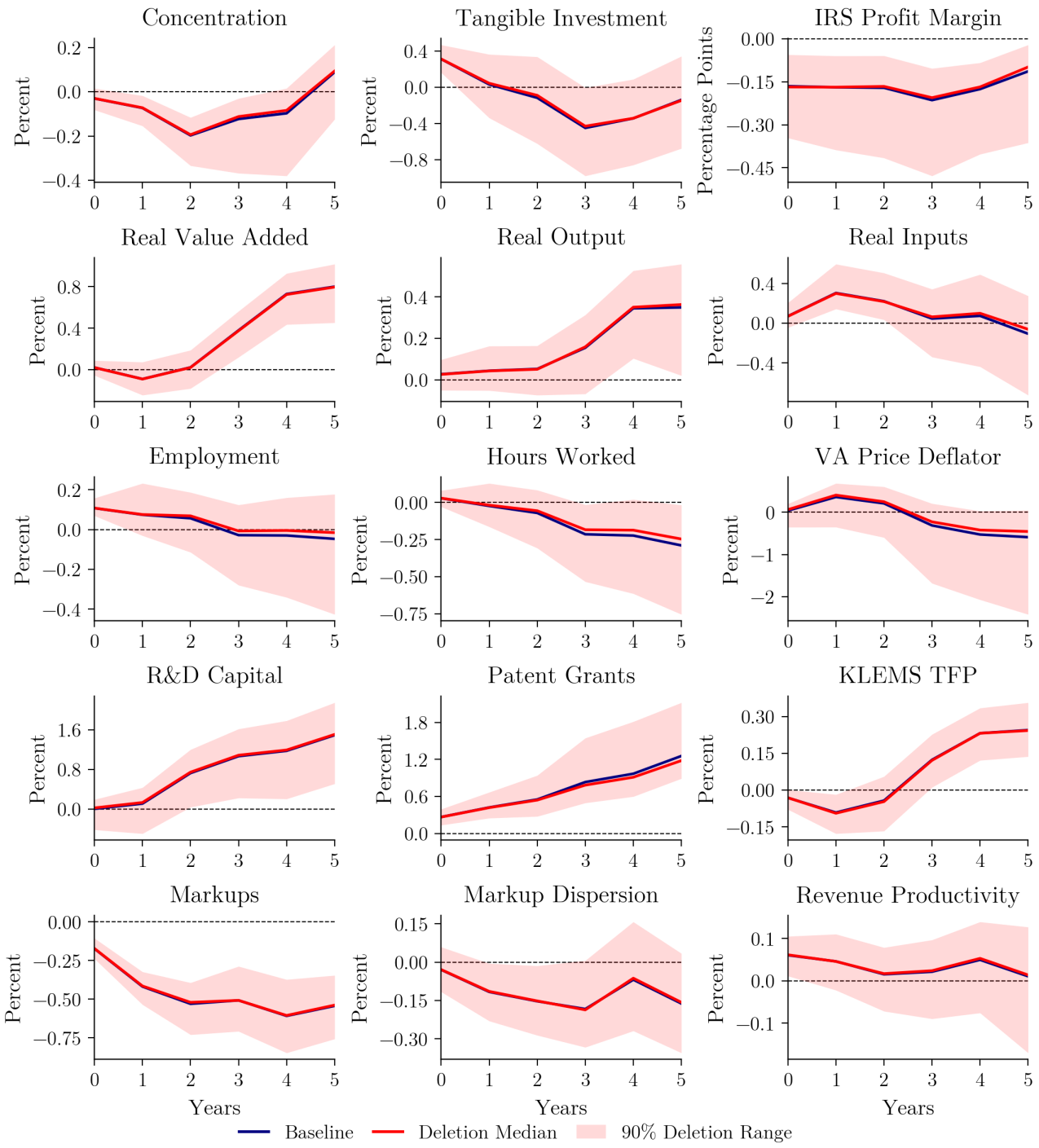


Figure A.14: Omitting 20 Percent of Cases: Civil Merger Cases

Notes: Each of 10,000 replications omits 121 of 603 contributing cases and reestimates equation (10), keeping outcome samples and the original proxy normalization fixed. Navy lines show the baseline; red lines and shading show the median and pointwise 5th–95th percentiles. Shading represents sensitivity to case omissions, not sampling confidence intervals.

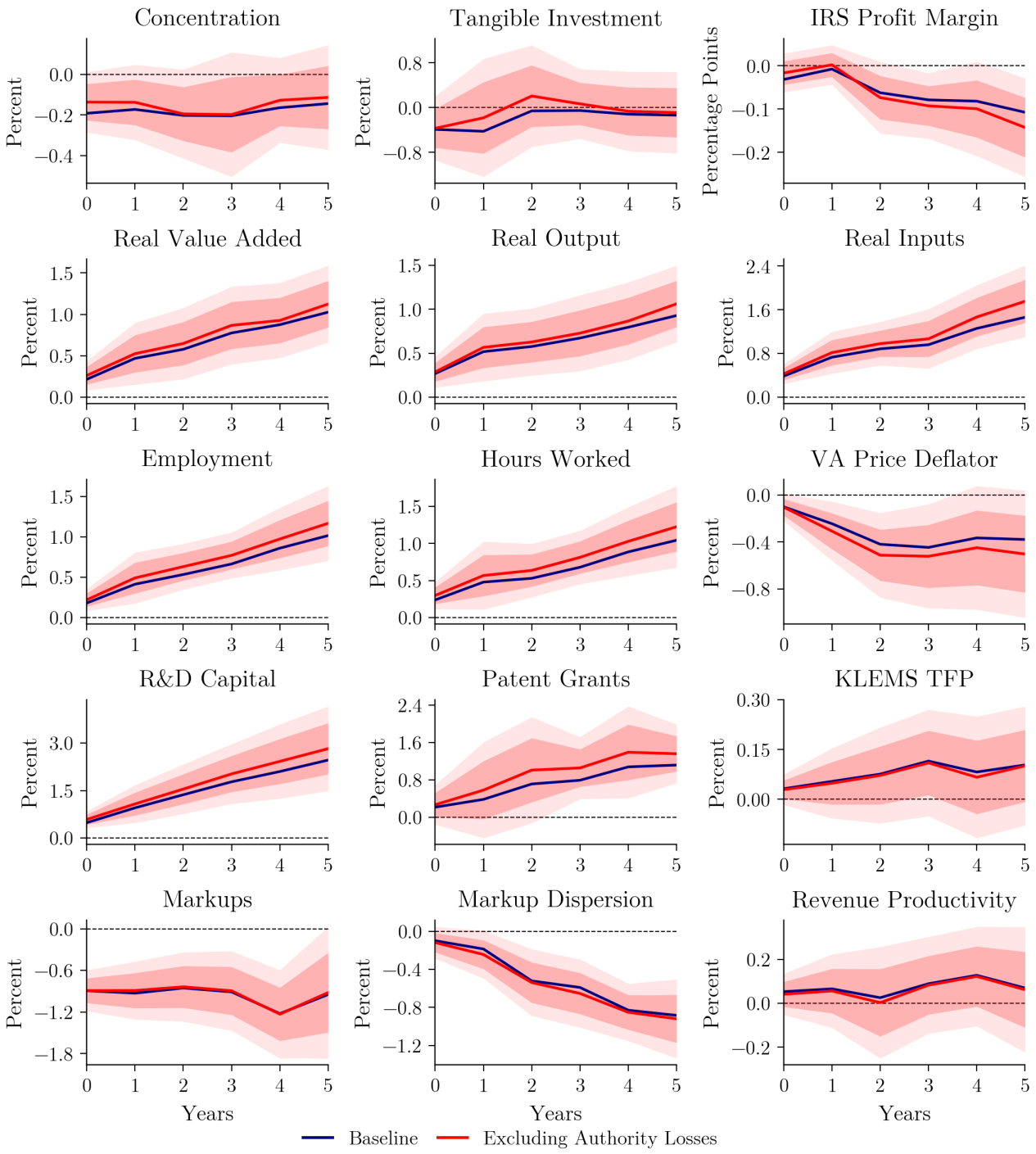


Figure A.15: Excluding Authority Losses: Civil Non-Merger and Criminal Cases

Notes: Responses estimated using equation (10), excluding recorded authority losses. The proxy covers civil non-merger and criminal cases. The alternative specification is red and the baseline is navy. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the alternative, based on Driscoll–Kraay standard errors with eight lags.

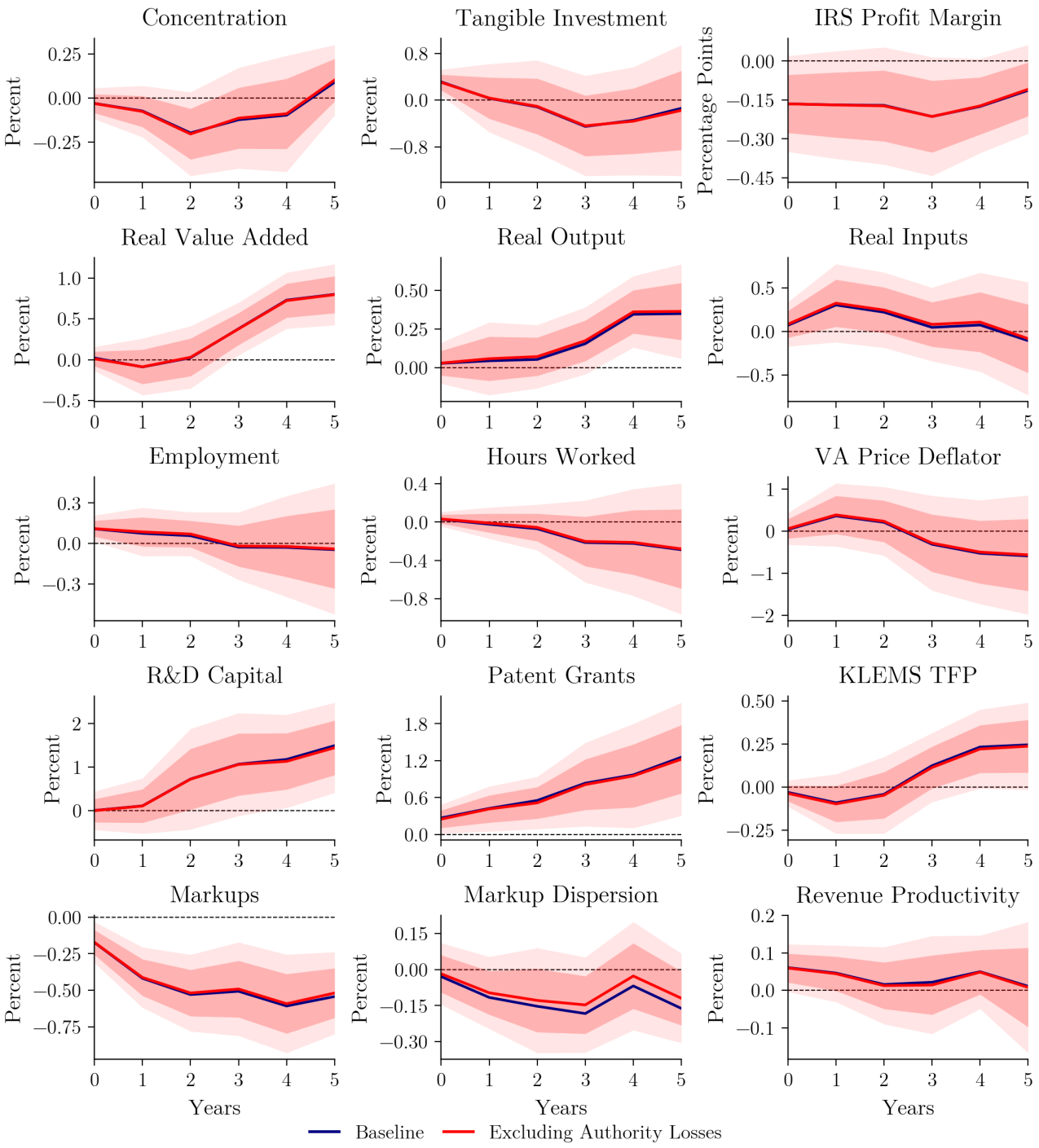


Figure A.16: Excluding Authority Losses: Civil Merger Cases

Notes: Responses estimated using equation (10), excluding recorded authority losses. The proxy covers civil merger cases. The alternative specification is red and the baseline is navy. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals for the alternative, based on Driscoll–Kraay standard errors with eight lags.

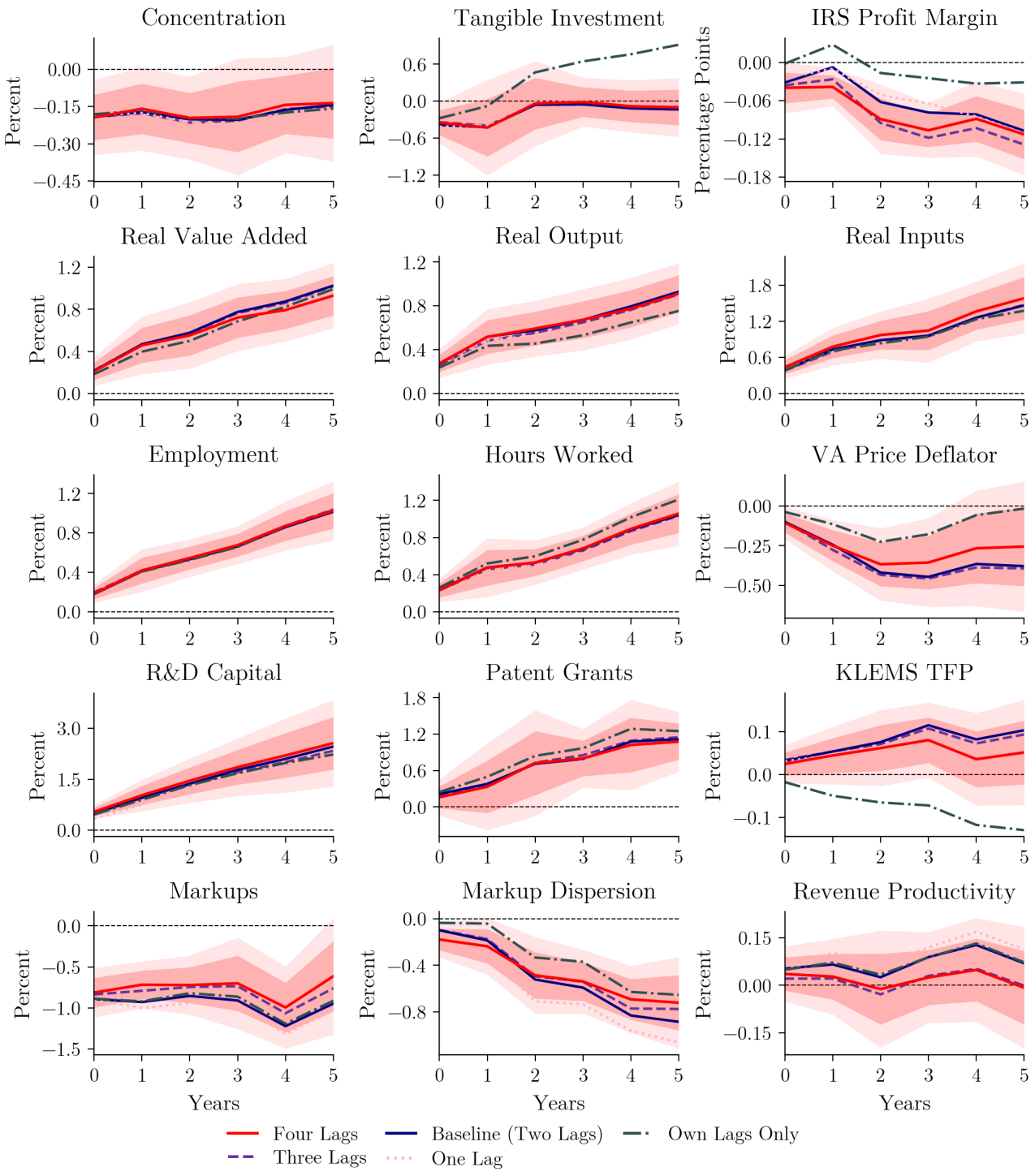


Figure A.17: Conditioning on Sectoral History: Civil Non-Merger and Criminal Cases

Notes: Four-lag responses from equation (10) appear in red, with pointwise 68 and 90 percent Driscoll–Kraay confidence intervals using eight lags (dark and light shading). Other lines show point estimates for three lags, two lags (baseline), one lag and own lags only. Lag orders apply to the outcome, proxy and baseline controls; own lags only retains two lags of the outcome and proxy. All specifications include sector and year fixed effects and use their available estimation samples with the baseline shock normalization.

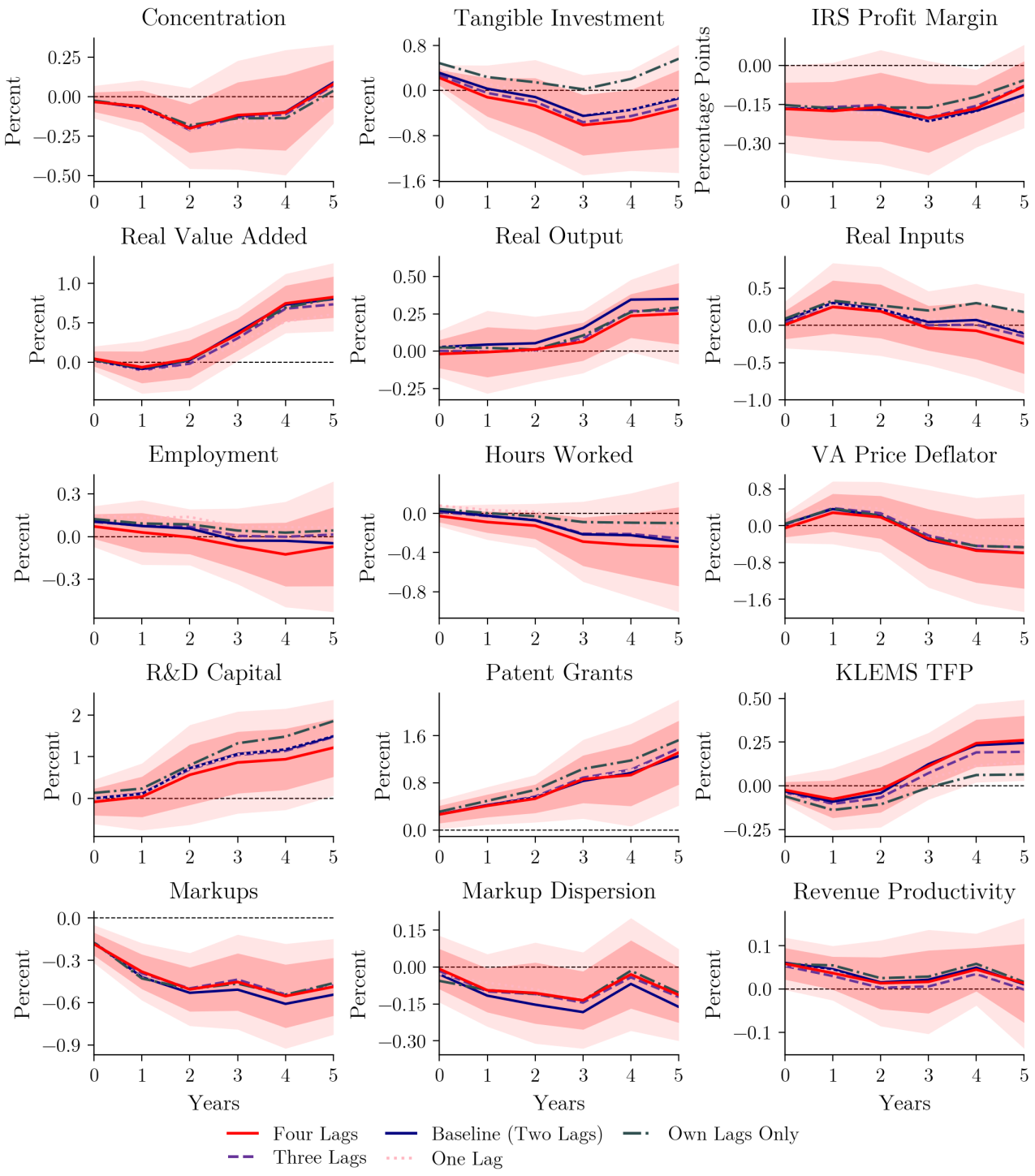


Figure A.18: Conditioning on Sectoral History: Civil Merger Cases

Notes: Four-lag responses from equation (10) appear in red, with pointwise 68 and 90 percent Driscoll–Kraay confidence intervals using eight lags (dark and light shading). Other lines show point estimates for three lags, two lags (baseline), one lag and own lags only. Lag orders apply to the outcome, proxy and baseline controls; own lags only retains two lags of the outcome and proxy. All specifications include sector and year fixed effects and use their available estimation samples with the baseline shock normalization.

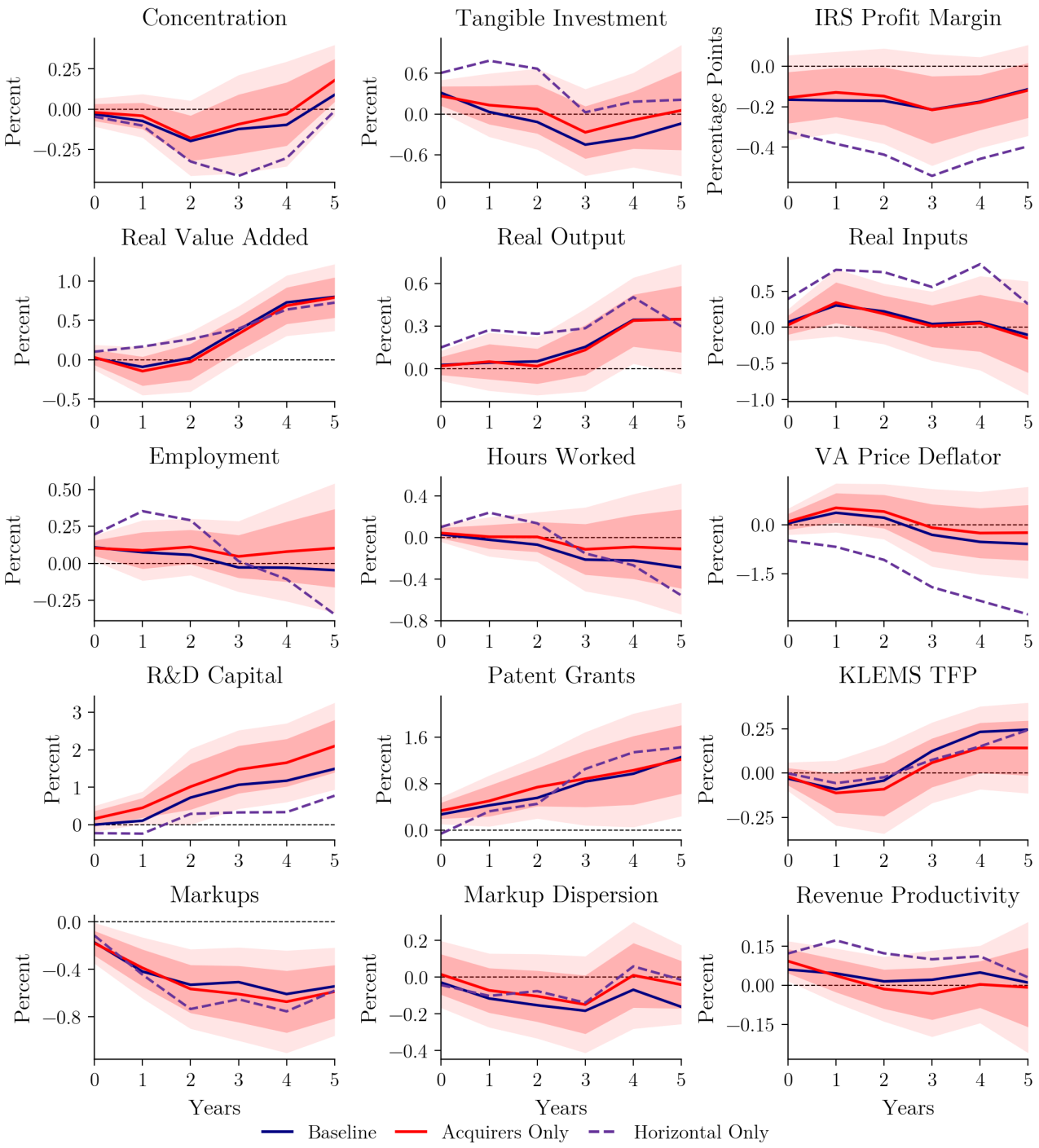


Figure A.19: Acquirer-Only and Horizontal-Only Proxies: Civil Merger Cases

Notes: Responses estimated using equation (10). Red lines use identified acquirers only; navy lines show the baseline. Purple dashed lines show point estimates using only cases classified exclusively as horizontal mergers, retaining all eligible defendants. For the acquirer-only proxy, case portfolios and covariance adjustments are recomputed on the retained defendants. All specifications use the same estimation samples and baseline proxy standard deviation. Dark and light red shading denotes pointwise 68 and 90 percent Driscoll–Kraay confidence intervals with eight lags for the acquirer-only estimates.

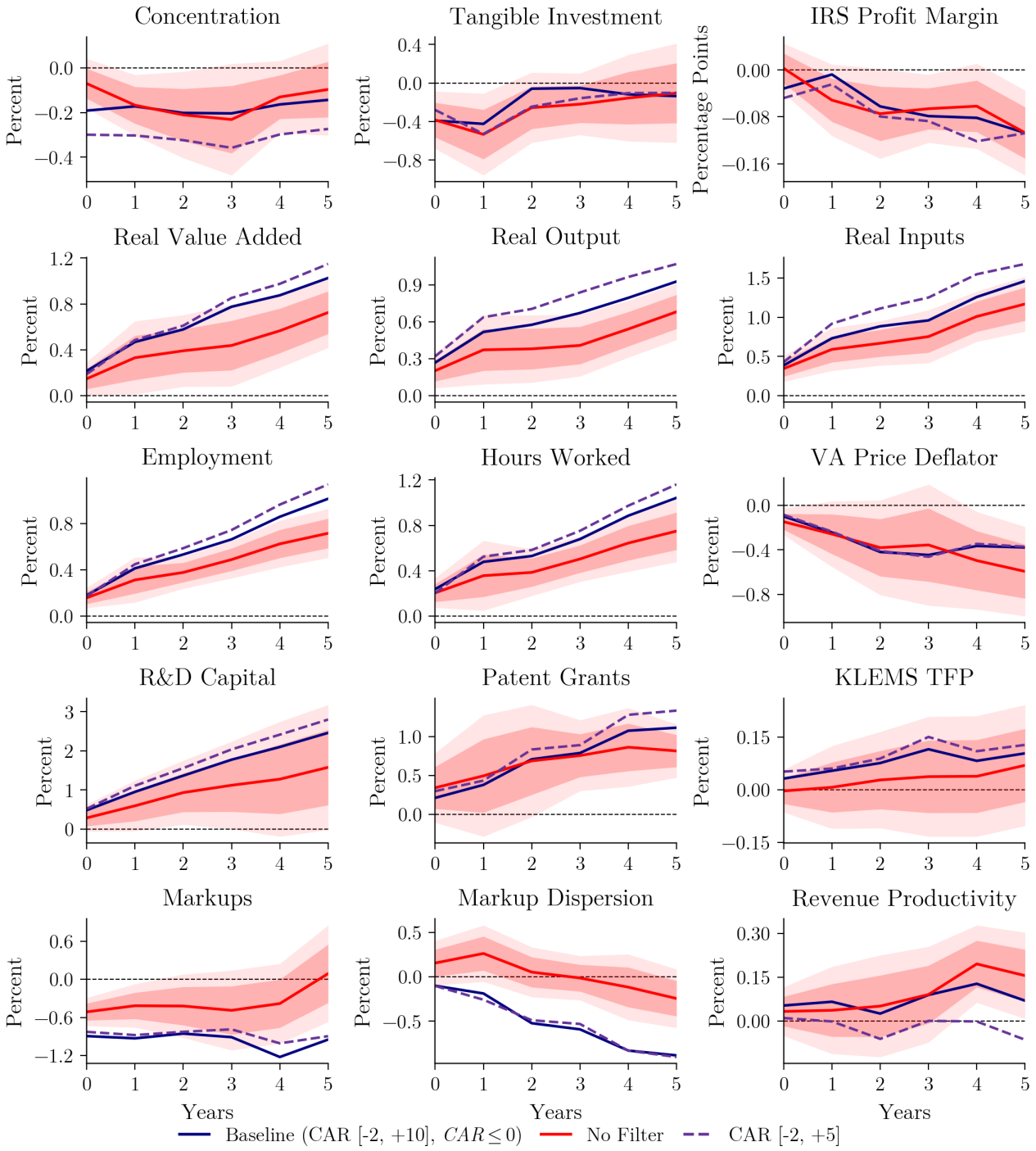


Figure A.20: Proxy Construction: Civil Non-Merger and Criminal Cases

Notes: Navy lines show the baseline, which cumulates abnormal returns over $[-2, +10]$ and retains defendants with non-positive CARs. Red lines remove this filter, retaining the baseline window. Purple dashed lines show point estimates for the shorter window $[-2, +5]$, retaining the non-positive-CAR screen evaluated at day +10. Case portfolios and covariance adjustments are recomputed on the eligible defendants. Each proxy is normalized by its own pooled standard deviation. Dark and light red shading denotes pointwise 68 and 90 percent Driscoll–Kraay confidence intervals with eight lags for the no-filter specification.

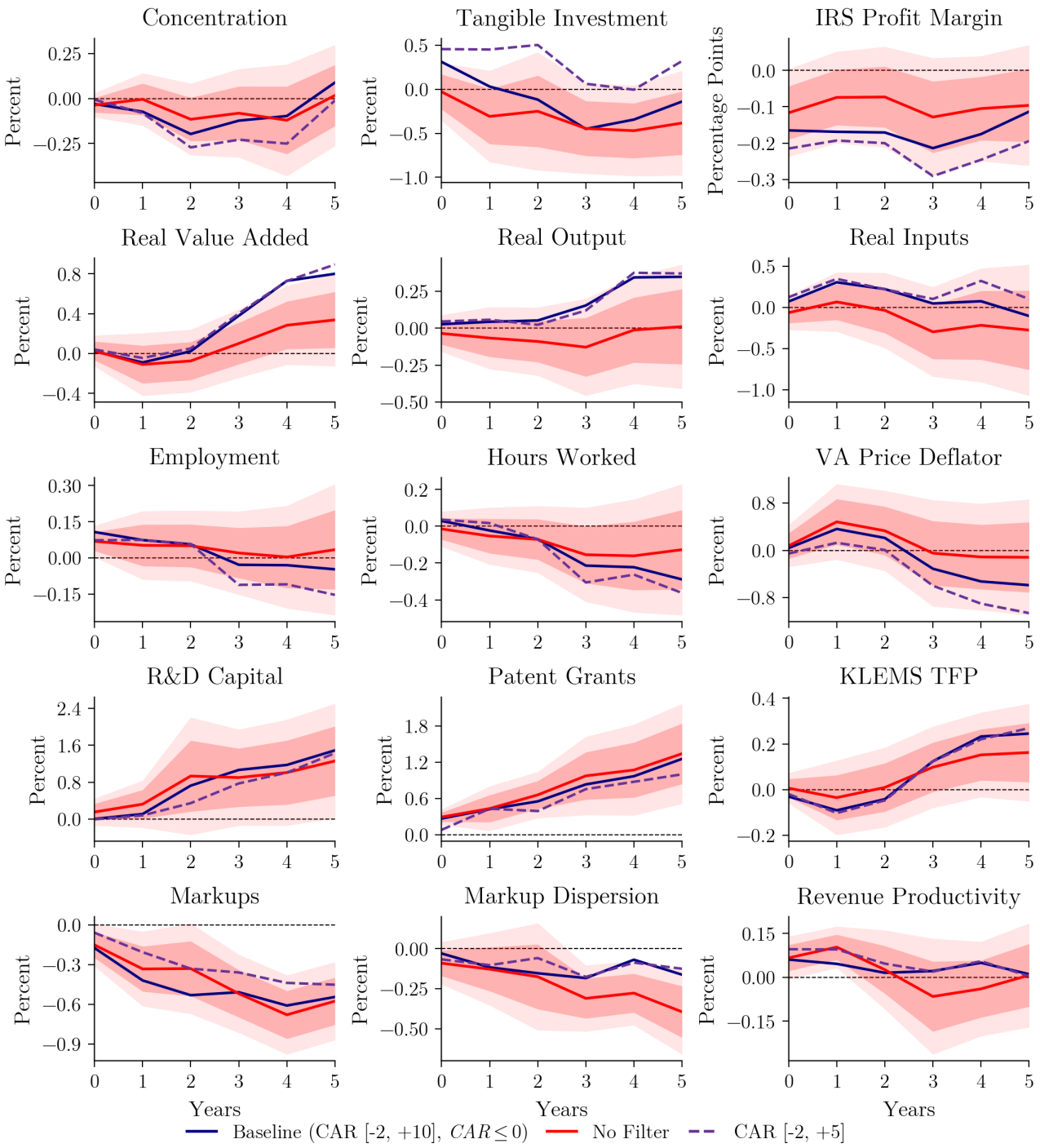


Figure A.21: Proxy Construction: Civil Merger Cases

Notes: Navy lines show the baseline, which cumulates abnormal returns over $[-2, +10]$ and retains defendants with non-positive CARs. Red lines remove this filter, retaining the baseline window. Purple dashed lines show point estimates for the shorter window $[-2, +5]$, retaining the non-positive-CAR screen evaluated at day +10. Case portfolios and covariance adjustments are recomputed on the eligible defendants. Each proxy is normalized by its own pooled standard deviation. Dark and light red shading denotes pointwise 68 and 90 percent Driscoll–Kraay confidence intervals with eight lags for the no-filter specification.

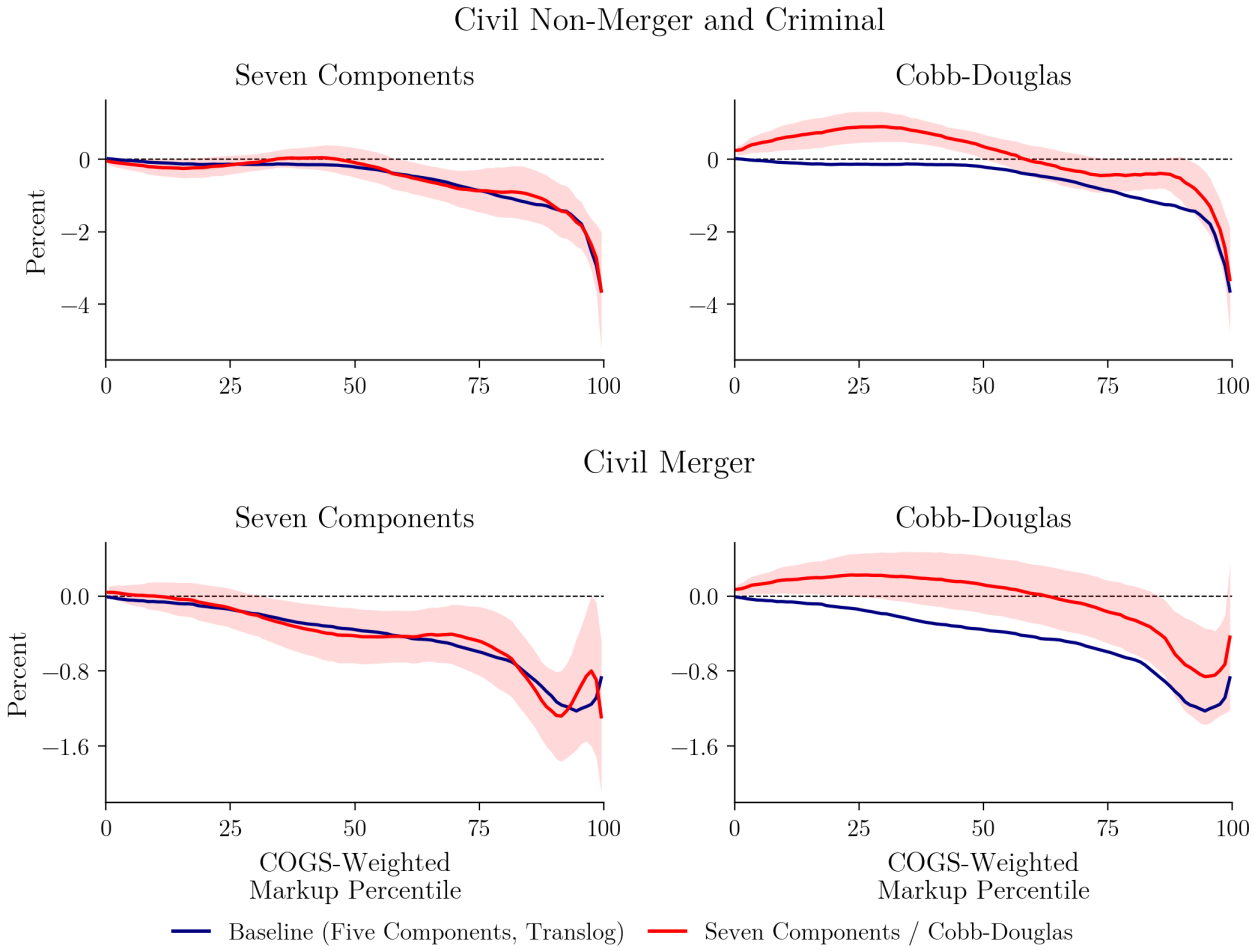


Figure A.22: Robustness of the Markup Distribution Responses at Year Two

Notes: Responses of COGS-weighted markup percentiles two years after a one-standard-deviation increase in the enforcement proxy. Navy lines show the baseline with five components and translog markups. Red lines use seven components on the left and Cobb–Douglas markups with five components on the right. Under Cobb–Douglas, the distributions and markup controls are reconstructed using the alternative markup estimates. Each specification retains firms with markups above one and positive sales and COGS. Responses use the Herbst–Johannsen correction. Red shading shows pointwise 90 percent Driscoll–Kraay confidence intervals with eight lags for the alternative specification.

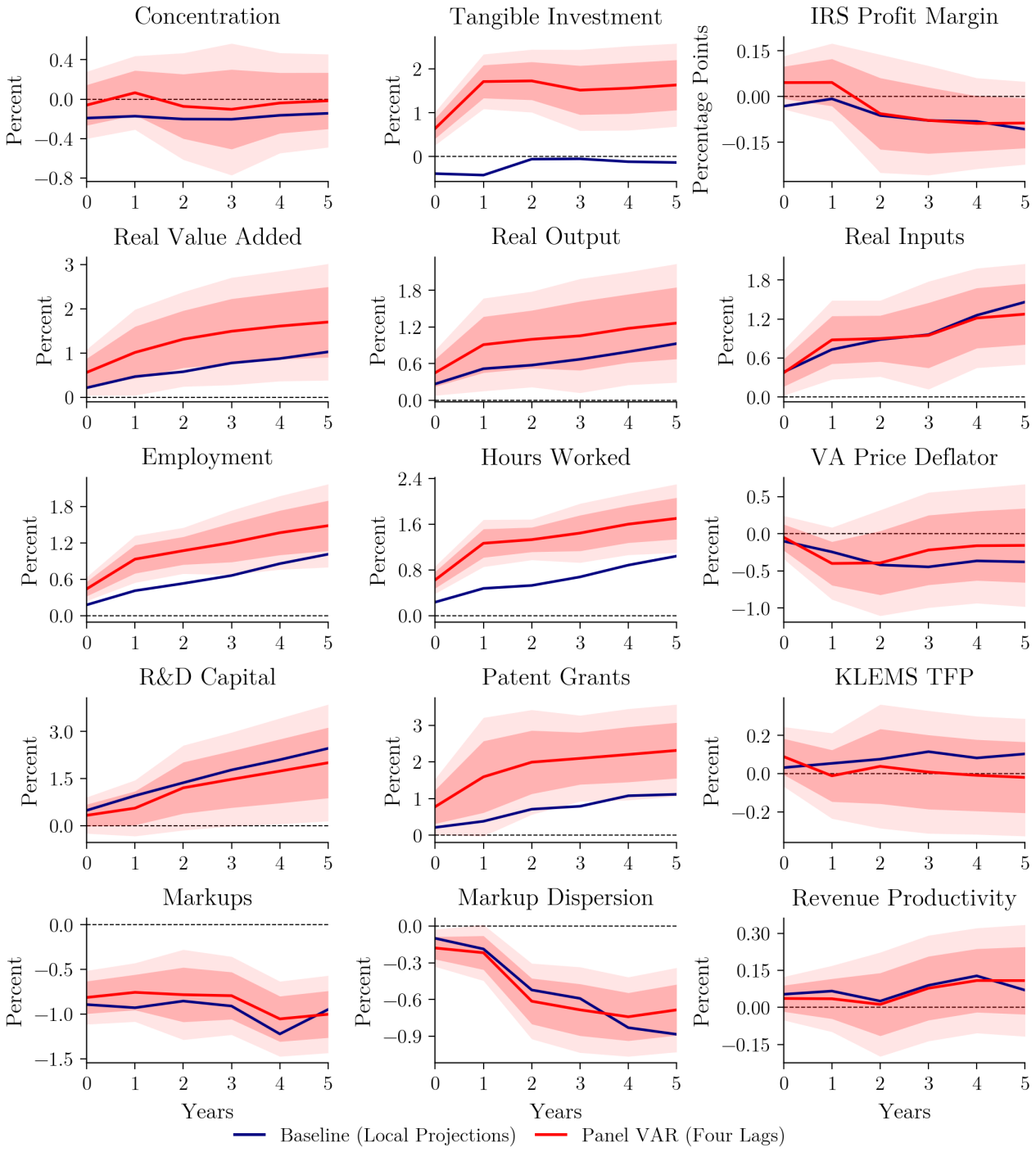


Figure A.23: Panel VAR and Baseline Local Projections: Civil Non-Merger and Criminal Cases

Notes: Responses to a one-standard-deviation increase in enforcement. Navy lines show the baseline local projections; red lines show the four-lag panel VAR in equation (14). Each additional outcome enters a separate system alongside the core variables. VAR samples begin in 1962; local projections retain their baseline samples. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals, using the delta method and joint Driscoll–Kraay covariance with eight lags.

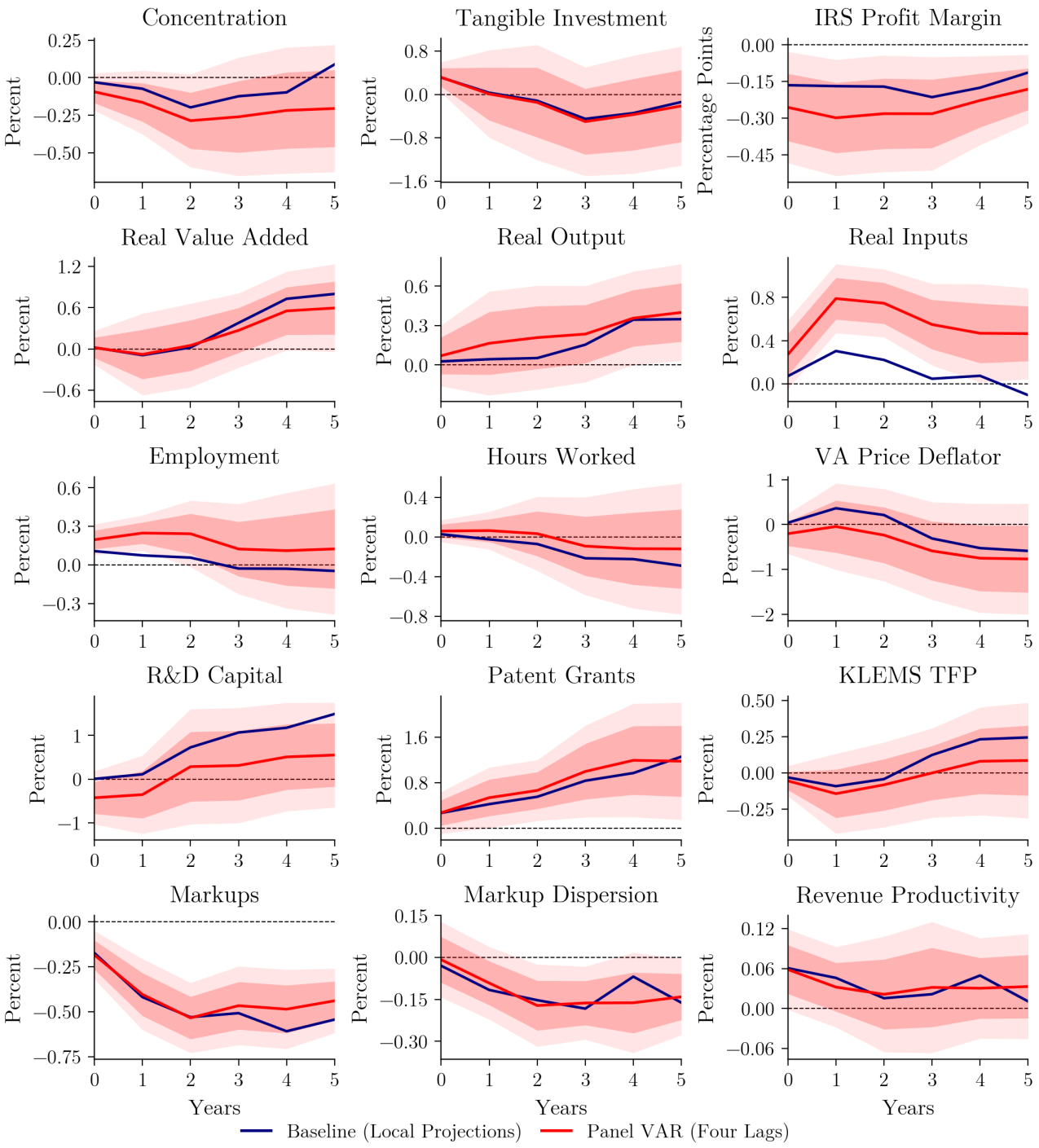


Figure A.24: Panel VAR and Baseline Local Projections: Civil Merger Cases

Notes: Responses to a one-standard-deviation increase in enforcement. Navy lines show the baseline local projections; red lines show the four-lag panel VAR in equation (14). Each additional outcome enters a separate system alongside the core variables. VAR samples begin in 1962; local projections retain their baseline samples. Dark and light red shading denotes pointwise 68 and 90 percent confidence intervals, using the delta method and joint Driscoll–Kraay covariance with eight lags.

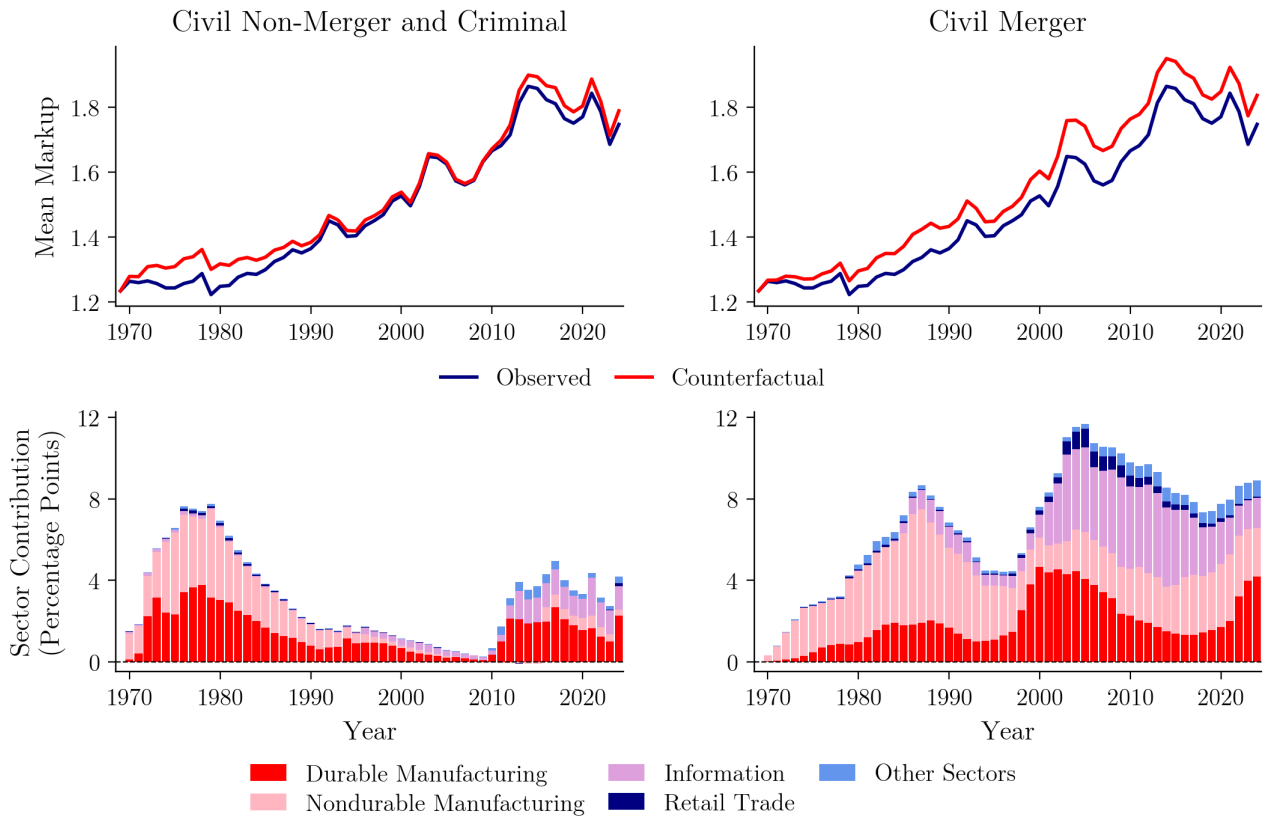


Figure A.25: Antitrust Enforcement and Aggregate Arithmetic Mean Markups

Notes: The counterfactual exercise in Section 6.3, replacing harmonic with sales-weighted arithmetic mean markups in estimation and aggregation. Upper panels show observed values in navy and counterfactual values in red. Lower panels show sector contributions to the counterfactual-minus-observed markup gap, in percentage points; each stack sums to 100 times the difference between the corresponding lines. Each column sets one enforcement proxy to zero from 1970, retaining observed initial conditions, fixed effects, other structural innovations, and sales weights.

A.2 Data

A.2.1 Data Sources and Definitions

Table A.9 summarizes the sources used in the firm-level and industry analyses. The following subsections describe the construction of the non-BEA series in detail. We harmonize industry identifiers to nineteen private BEA sectors. Manufacturing is split into durable and nondurable goods, retail trade and transportation each combine their constituent two-digit NAICS industries, and government and management of companies are excluded from the estimation panel. The underlying industry panel covers 1954–2024. Reduced-form regressions begin in 1957 for

non-Compustat outcomes and in 1962 for Compustat outcomes. Source-specific endpoints and projection horizons determine each equation's available sample.

A.2.2 Compustat: Accounting Data and Industry Classification

We obtain annual accounts from Compustat North America through WRDS. The extract contains consolidated, standardized statements for the domestic population reporting in U.S. dollars. We retain industrial and financial statement formats, preferring the industrial record when both are available for the same firm and fiscal year. Firms are identified by GVKEY and time by the Compustat fiscal-year label. We estimate production functions using Compustat accounts through 2025. The firm-level analysis uses accounts from 1962 through 2025, while Compustat-based industry outcomes cover 1962–2024. The starting date reflects limited earlier coverage.

Sales, costs of goods sold, employment, and net property, plant and equipment are measured by Compustat items `sale`, `cogs`, `emp`, and `ppent`, respectively. Selling, general and administrative expenses are measured by `xsga`. Sales, costs of goods sold, SG&A, and the other income and expenditure items are deflated using the GDP deflator. Property, plant and equipment are deflated using the gross private domestic investment price index. We obtain these indices from FRED, using series `GDPDEF` and `A008RD3Q086SBEA`. Employment is the number of employees reported by Compustat, in thousands. We fill an isolated missing observation in real sales, real COGS, real SG&A, or real net property, plant and equipment by the average of the preceding and following values only when all three fiscal years are consecutive. We do not create observations for absent fiscal years or interpolate longer gaps.

We obtain SIC and NAICS classifications from Compustat's company names table. For a firm without NAICS, we use the modal four-digit NAICS among firms with its SIC code. We exclude firms whose NAICS remains missing and those assigned to sector 99. Finance and real estate are retained in the baseline. The NAICS–BEA concordance maps firms to the industry definitions used in the panel analysis, first using a four-digit match, then a three-digit match, and finally a sector-level fallback.

The construction sample requires strictly positive real sales, COGS, and SG&A. We then remove observations below the first percentile or above the ninety-ninth percentile of the annual

Description	Source	Series Code / Construction
Panel A: Cases, Firms, and Equity Returns		
Antitrust filings	Authors' DoJ and FTC database	Filing dates, defendants, allegations, case outcomes, and recorded remedies
Firm sales, costs, and employment	Compustat	Annual accounts; monetary flows deflated by the GDP deflator; employment in thousands
Markups and revenue productivity	Compustat; estimated	Production-function elasticities and residuals; sales-weighted harmonic markups and arithmetic revenue productivity
Markup dispersion	Compustat; estimated	Unweighted within-sector variance of logarithmic markups; its square root enters the industry LPs
Daily equity returns and market value	CRSP	Ordinary common shares; daily returns, pre-filing firm market values, and total market value
Daily return factors	Kenneth French Data Library	Market excess return, SMB, HML, and risk-free rate
Product-market competitors	Hoberg–Phillips ETNIC	Annual product-description similarity, 1988–2023
Panel B: Industry Outcomes		
Real value added and prices	BEA GDP by Industry	Logarithms of real value added and the value added deflator
Tangible investment	BEA Fixed Assets	Equipment and structures; chained quantity index
Employment and hours	BEA; BEA–BLS	Full-time and part-time employees; integrated production-account hours
R&D capital and productivity	BEA–BLS	R&D-capital quantity and gross-output total factor productivity from the integrated production account
Concentration	Economic Census; CBP; QCEW	Annual filtered detailed-market HHI, aggregated and linked to BEA sectors
Corporate profit margin	IRS Statistics of Income	Net income less deficit divided by total receipts, 1954–2022
Patent grants	USPTO; PatentsView	Logarithm of one plus fractional utility grants; CPC-to-NAICS allocation, 1954–2024
Establishment and job dynamics	Census BDS	Entry, exit, and job flows, 1978–2023; definitions in Table A.10
Panel C: Common Inputs and Predictability Controls		
Prices, interest rates, and GDP	FRED	GDP and investment deflators, federal funds rate, and nominal GDP
Antitrust agency budgets	DoJ; FTC; appropriation laws	Real calendar-year appropriations, with the 1976 transition quarter treated separately
Merger activity	FTC; Mergerstat; CRSP; HSR; SDC	Standardized filtered common factor of annual merger-count measures

Table A.9: Data Sources

Notes: CRSP denotes the Center for Research in Security Prices; BEA, the Bureau of Economic Analysis; BLS, the Bureau of Labor Statistics; CBP, County Business Patterns; QCEW, the Quarterly Census of Employment and Wages; IRS, the Internal Revenue Service; and BDS, Business Dynamics Statistics. Compustat and CRSP are obtained through WRDS. HSR denotes Hart–Scott–Rodino transaction reports, and SDC denotes the LSEG transaction database. The construction and weighting of the enforcement proxy are described in Section 4.3.

sales-to-COGS distribution. We apply the same annual tail exclusions sequentially to the variable-input cost shares,

$$s_{jt}^{V} = \frac{C_{jt}}{C_{jt} + u_t K_{jt}}, \qquad s_{jt}^{V,SGA} = \frac{C_{jt}}{C_{jt} + A_{jt} + u_t K_{jt}}, \tag{A.1}$$

where C_{jt} denotes real COGS, A_{jt} real SG&A, and K_{jt} real net property, plant and equipment. The annual user cost u_t equals the federal funds rate less GDP-deflator inflation plus 0.12. Missing or zero cost shares are also excluded. After estimation, we set markups at or below one to missing separately for each measure.

A.2.3 CRSP: Security Returns, Firm Links, and Market Values

We match listed defendants to CRSP securities using the identifiers recorded in the antitrust database. The event-study sample retains ordinary common equity: the CRSP CIZ security type is `EQTY`, security subtype `COM`, and share type `NS`. We use daily returns from CRSP's daily stock file and require the matched security to have an eligible classification on the filing date or during the preceding 30 calendar days. PERMNO identifies the security and PERMCO identifies the company. For the link to Compustat, we use the CRSP–Compustat Merged link table, retaining link types LC or LU and primary links P or C, and require the filing date to lie within the link's validity interval.

The U.S. Fama–French three-factor series and risk-free return come from Kenneth French's Data Library. The common trading calendar is defined by the factor observations. A filing on a nontrading day is assigned to the first subsequent market trading day, allowing a shift of at most five calendar days. The implementation transforms stock returns, factor returns, and the risk-free return to logarithmic returns before estimation.

We retain CRSP's return-quality and trading-status flags to distinguish nontrading intervals from unresolved missing data. A short internal gap is assigned a zero total return only when it is bracketed by observed returns no more than ten trading days apart, the security remains classified as actively trading, and the missing-return flags do not indicate an unresolved data problem. The subsequent trade-to-trade return is retained on its observed date. Returns outside an active spell, suspended securities, and longer or otherwise unresolved gaps remain missing.

A multi-day trade-to-trade return can enter a cumulative window but is excluded from the single-day abnormal-return average. Cumulative returns require a complete resolved window.

The market value used to weight a defendant's contribution to the enforcement proxy sums its eligible ordinary-common securities by PERMCO. For each security, market value is `dlycap` multiplied by 1,000 and by the ratio of the cumulative share-adjustment factor to the cumulative price-adjustment factor. We take the latest available company market value on or before five business days before the filing, with a 30-calendar-day lookback from that target date. The five-day lead avoids weighting cases by a market value already affected by the filing. Industry proxy normalization uses the annual mean of CRSP's total market value index, scaled to one in January 2000; Section 4.3 describes the remaining proxy construction.

A.2.4 Product-Market Competitors

We identify competitors using the embeddings-based text network industry classification of Hoberg and Phillips (2025). The annual network covers 1988–2023 and measures the similarity of product descriptions in firms' 10-K filings. We match each defendant to the network for the calendar year of its case filing and rank its other linked firms by their similarity score. The main competitor event study uses the highest-ranked eligible competitor. Competitors are linked to CRSP using valid CRSP–Compustat links at the filing date and are subject to the same ordinary-common-share restriction as defendants. Self-links are excluded, as are competitor securities with an observed own antitrust filing within 730 calendar days before or after the focal filing.

A.2.5 Agency Budgets and Merger Activity

The predictability regressions use annual appropriations for the DoJ Antitrust Division and the FTC. We convert fiscal-year appropriations into calendar-year amounts using the fraction of each fiscal year falling in that calendar year. Before 1976 this gives equal weights to fiscal years t and $t+1$; from 1977 onward the weights are three quarters and one quarter. Calendar year 1976 combines one half of fiscal-year 1976 appropriations, the separate July–September transition-quarter appropriation, and one quarter of fiscal-year 1977 appropriations. The transition amounts include base and supplemental appropriations under Public Laws 94-121 and

94-303: $5.834 million for the Antitrust Division and $12.543 million for the FTC. We deflate calendar-year appropriations using the GDP deflator and enter their logarithms with a one-year lag in the predictability regressions. The budget figure dates appropriations to their own calendar year.

The merger-activity index combines five measures: FTC large acquisitions in manufacturing and mining, Mergerstat announcements, CRSP public-company merger delistings, Hart–Scott–Rodino (HSR) transactions, and LSEG/SDC merger announcements. CRSP targets are ordinary-common-equity companies on the NYSE, AMEX, or Nasdaq; multiple securities are collapsed to a single company-date merger. SDC announcements have a U.S. target, acquirer, or identified ultimate parent, exclude rumor and non-acquisition records, and satisfy the contemporaneous minimum HSR transaction-value threshold. The SDC measurement enters from 1981. HSR fiscal-year transaction counts are allocated to calendar years using three-quarter and one-quarter weights, and the incomplete first reporting year is excluded.

Each positive count series is logged and standardized over its observed years. The HSR series is standardized separately before and after the 2001 change in its reporting threshold. For these transformed measurements $x_{s,t}$, we estimate the common-factor model

$$x_{s,t} = \lambda_s f_t + e_{s,t}, \qquad f_t = \rho f_{t-1} + \eta_t, \tag{A.2}$$

with serially uncorrelated, mutually independent measurement errors and a stationary first-order autoregressive factor. Missing source observations do not enter the measurement update. The index is the filtered factor, oriented to rise with merger activity and standardized to unit variance over 1954–2025. Filtering uses measurements through year t, conditional on parameters and normalizations estimated over the full sample.

A.2.6 Industry Concentration

We construct annual concentration measures from the Economic Census and the business-universe employment and establishment data. The concentration series covers 1954–2024. The underlying markets are detailed SIC industries through 1996 and six-digit NAICS industries from 1997. Most SIC markets are identified at the four-digit level; where the source publishes

only a broader market, we retain that published definition. We map these markets to the BEA sectors using the Census SIC–NAICS concordance and the NAICS–BEA crosswalk. Ambiguous SIC mappings are resolved using the industry's activity description and broader SIC group. The SIC and NAICS histories are treated as separate markets and joined after aggregation.

The Census observations combine the compilation of historical Economic Census tables assembled by Keil (2017) with the 2017 and 2022 Economic Census concentration tables. We use the published HHI, where available, together with sales and payroll shares accounted for by the 4, 8, 20 and 50 largest firms. The recent direct HHI observations refer to the 50 largest firms. All HHI values and concentration ratios are expressed on the unit scale. Annual employment and establishment counts come from the national Quarterly Census of Employment and Wages from 1975 onwards. For earlier years, we hold the detailed 1975 market structure fixed and scale its employment and establishment counts by sector-specific exponential trends fitted to the digitized County Business Patterns data of Eckert et al. (2022), using sector totals for 1959–1973.[29]

The Census measures are available only at periodic intervals. We therefore estimate annual concentration with a state-space model. Let H_{mt} denote latent concentration in market m and let $z_{mt} = \log[H_{mt}/(1 - H_{mt})]$. Its transition equation is

$$z_{mt} = \mu_m + \rho(z_{m,t-1} - \mu_m) + \beta' \widetilde{x}_{mt} + \eta_{mt}, \qquad \eta_{mt} \sim N(0, q), \tag{A.3}$$

where $\widetilde{x}_{mt}$ contains changes in the logarithms of establishments and employment per establishment, and a common linear trend. These variables are winsorized at the first and ninety-ninth percentiles and standardized over the pooled panel. Missing values are set to zero after standardization. The transition coefficients and innovation variance are common across markets. The market-specific mean μ_m is the average of its observed Census logit HHI; for markets without a direct HHI observation, we use the average of the concentration-ratio prediction described below, or the pooled median direct observation when neither is available.

[29] For either employment or establishments, the historical value for market m in sector i is $B_{m,1975} \exp[\widehat{g}_i(t - 1974)]$, where $\widehat{g}_i$ is estimated by regressing the logarithm of the sector total on a time trend. When a sector lacks sufficient historical observations, its 1975 market structure is held constant.

We use two measurement equations:

$$y^H_{mt} = z_{mt} + e^H_{mt}, \qquad e^H_{mt} \sim N(0, r_H), \tag{A.4}$$

$$y^C_{mt} = z_{mt} + e^C_{mt}, \qquad e^C_{mt} \sim N(0, r_C + v_C). \tag{A.5}$$

The first measurement is the published Census HHI in logit units. The second is a prediction of that variable from eight sales and payroll share bands: the shares accounted for by firms ranked 1–4, 5–8, 9–20 and 21–50. We estimate this prediction by a ridge regression, with penalty parameter one, on market-waves with both a direct HHI and sales-share information, weighting observations by Census market sales. Missing share-band inputs are replaced by their calibration-sample medians before standardization. The variance v_C is the sales-weighted mean squared prediction error obtained by leaving out one Census wave at a time; r_C is an additional estimated measurement variance. The observation model treats the two measurement errors as independent of one another and of the state innovations. When both observations are available, both enter the filter.

The initial state has a normal prior in 1954. Its mean is μ_m when the concentration-ratio prediction is missing; otherwise it averages μ_m and that initial prediction with relative weights one and four. The initial variance is two in the first case and one in the second. Markets introduced subsequently evolve from this initial state without measurement updates until their data become available. We estimate ρ, β, q, r_H and r_C by Gaussian maximum likelihood, using the Kalman filter and pooling the prediction-error likelihood across markets. The annual market estimate is the inverse-logit transformation of the filtered state, $\widehat{H}_{mt} = \mathrm{expit}(\widehat{z}_{mt|t})$.

Sector concentration is the employment-weighted average of these detailed-market estimates,

$$\widehat{H}^{\mathrm{raw}}_{it} = \sum_{m \in i} \frac{E_{mt}}{\sum_{n \in i} E_{nt}} \widehat{H}_{mt}, \tag{A.6}$$

where the sum includes mapped markets with positive employment. This measures average concentration within detailed industries, rather than the HHI obtained by treating an entire BEA sector as a single market. We link the logarithmic sector series across the source and classification changes in 1975, 1978, 1988 and 1997. At each break, the observed logarithmic

change is replaced by the median annual logarithmic change in the surrounding five years on either side, excluding the break itself. This is implemented by adding a constant to the preceding segment, working backwards from the latest break. The post-1997 segment therefore determines the level of the linked series. The industry regressions use its logarithm. In validation exercises that omit one Census wave at a time, predicted detailed-market HHIs have a correlation of 0.86 with the omitted observations.

A.2.7 Corporate Profit Margins

We measure industry profit margins using the IRS Statistics of Income tabulations of active corporate income-tax returns for tax years 1954–2022. Unlike the Compustat measures, these tabulations include both publicly listed and unlisted corporations. The source changes over time: historical Corporation Income Tax Return reports supply Table 1 for 1954–1964, 1981–1984 and 1991–1993; machine-readable Corporation Source Book records held by the National Archives supply 1965–1980 and 1985–1990; and official IRS workbooks and flat files supply 1994–2022. We extract the historical tables from the printed reports, retaining the source-year industry definitions and disclosure indicators.

We select non-overlapping published industry totals and map them to the BEA sectors. When a published subtotal already includes its detailed industries, the constituent industries are not counted separately. Manufacturing industries are allocated between durable and nondurable goods. Historical SIC definitions require several further reclassifications: publishing and communications enter information; eating and drinking places are removed from retail trade and included in accommodation and food services; and holding-company activities are distinguished from finance wherever the published detail permits. Broad historical service categories that cannot be allocated are left unassigned. The resulting panel is unbalanced, and we do not interpolate missing sector-years. In particular, administrative and support services become separately available only in 1998, while education and health services begin in 1963.

For each sector and year, we first aggregate the published monetary amounts and then compute

$$\text{Profit Margin}_{it} = \frac{\text{Net Income}_{it} - \text{Deficit}_{it}}{\text{Total Receipts}_{it}}. \tag{A.7}$$

The numerator includes losses reported by deficit corporations. The denominator is total receipts, and the margin can be negative. At the final aggregation stage, a sector total is missing if an applicable selected component is missing or suppressed. The historical transcription by Kwon, Ma and Zimmermann (2024) provides an independent comparison; it supplies the return-count field omitted from the 1972 archive file but does not replace the official receipts or profit amounts used in this outcome. Our profit margins correlate above 0.99 with margins constructed from their data in each of the three sectors with directly comparable definitions: agriculture, mining, and construction.

A.2.8 Patent Grants

We measure patenting using U.S. utility patents granted between 1954 and 2024. Data on grants through 2014 come from the USPTO historical patent master files; those from 2015 onward come from PatentsView. We retain granted patents with numeric patent identifiers and exclude withdrawn grants in PatentsView. The series is dated by the grant year and does not restrict inventors or assignees to U.S. residents.

To maintain a common technological classification, we assign patents throughout the sample to their primary four-character Cooperative Patent Classification (CPC) code in the USPTO master classification file dated 1 January 2025. Where the designated primary inventional symbol is unavailable, we use the first available inventional symbol in the source record. This retrospective classification avoids changing from USPC to CPC at the 2015 boundary between the grant-date sources. Patents without a usable CPC-to-industry link are not allocated to industries.

We map CPC codes to three-digit 2007 NAICS industries using the probabilistic concordance of Goldschlag, Lybbert, and Zolas (2016), and then aggregate the allocation probabilities to the two-digit BEA level. We retain all links in the released concordance, which already applies the authors' probability threshold and coverage rule. If ω_{ci} is the resulting weight of sector i for technology class c, annual patenting is

$$P_{i,t} = \sum_{j \in \mathcal{G}_t} \omega_{c(j),i}, \qquad \sum_i \omega_{ci} = 1, \tag{A.8}$$

where $\mathcal{G}_t$ contains the mapped utility grants in year t. A patent can therefore contribute fractionally to several industries without being counted more than once across the allocation universe. The nineteen-sector estimation panel subsequently omits management of companies, as for the other outcomes; its patent allocation is not redistributed to the remaining sectors.

On the common validation sample, our sectoral patent counts have a correlation above 0.99 with counts constructed using the historical USPC classification.

A.2.9 Business Dynamics Statistics

Description	**Source**	**Series Code / Construction**
Establishment entry rate	Census BDS	Establishment entries divided by the average establishment stock in years t and $t-1$. The denominator is recovered exactly from the published count–rate pair for each three-digit industry and summed within sector; true zero-count cells use the adjacent-year midpoint stock directly.
Establishment exit rate	Census BDS	Establishment exits divided by the average establishment stock, constructed as for the entry rate.
Net establishment entry rate	Census BDS	Establishment entry rate less the establishment exit rate, using the same average establishment stock in the denominator.
Job creation rate	Census BDS	Gross job creation divided by the Davis–Haltiwanger–Schuh denominator (the average of employment in years t and $t-1$), with counts summed within sector before the rate is formed.
Job destruction rate	Census BDS	Gross job destruction divided by the Davis–Haltiwanger–Schuh denominator, constructed as for the job creation rate.
Net job creation rate	Census BDS	Net job creation (gross creation less gross destruction) divided by the Davis–Haltiwanger–Schuh denominator.

Table A.10: Business Dynamics Outcomes: Definitions and Sources

Notes: The balanced panel covers 19 sectors over 1978–2023, using the 2023 release of the Census BDS three-digit NAICS data. We map industries to BEA2 sectors and sum counts before calculating rates, rather than averaging published industry rates. Suppressed cells are treated as missing, not zero. Rates are expressed as fractions of their respective denominators. Source: U.S. Census Bureau, Business Dynamics Statistics, https://www.census.gov/programs-surveys/bds.html.

A.3 Methodological Details

A.3.1 Production Function and Markup Estimation

We estimate the production function in equation (1) separately for each two-digit NAICS sector using the control-function approach of Ackerberg, Caves and Frazer (2015). Appendix A.2.2 describes the inputs, deflators, and data filters. The baseline translog technology is given by:

$$\begin{aligned} f_l(x^d_{j,t}, x^v_{j,t}; \theta_l) = \theta_{1,l} x^d_{j,t} + \theta_{2,l} x^v_{j,t} + \theta_{3,l} (x^d_{j,t})^2 \\ + \theta_{4,l} (x^v_{j,t})^2 + \theta_{5,l} x^d_{j,t} x^v_{j,t}. \end{aligned} \tag{A.9}$$

The Cobb–Douglas specification restricts this function to:

$$f_l(x^d_{j,t}, x^v_{j,t}; \theta_l) = \theta_{1,l} x^d_{j,t} + \theta_{2,l} x^v_{j,t}. \tag{A.10}$$

The technology intercept is absorbed into the level of productivity. The persistent component $z^p_{j,t}$ is observed by the firm when choosing variable inputs and follows a first-order Markov process,

$$z^p_{j,t} = g_l(z^p_{j,t-1}) + \zeta_{j,t}, \qquad \mathbb{E}[\zeta_{j,t} \mid \mathcal{I}_{j,t-1}] = 0, \tag{A.11}$$

where $\mathcal{I}_{j,t-1}$ is the firm's information set at time $t-1$. The transitory component $z^{tr}_{j,t}$ includes measurement error and is mean-independent of the variables entering the first stage. Capital $x^d_{j,t}$ is predetermined (chosen prior to t) and therefore uncorrelated with the productivity innovation $\zeta_{j,t}$. Variable inputs $x^v_{j,t}$ are correlated with lagged variable inputs $x^v_{j,t-1}$ through serial correlation in market shares, but $x^v_{j,t-1}$ is uncorrelated with $\zeta_{j,t}$.

Because firms adjust variable inputs to productivity, OLS does not identify the technology coefficients. Instead, we assume variable-input demand takes the form $x^v_{j,t} = v_t(x^d_{j,t}, z^p_{j,t}, s_{j,t})$, where $s_{j,t}$ contains the firm's Compustat sales shares within 2-, 3-, and 4-digit NAICS industries. Following de Loecker, Eeckhout and Unger (2020), the market shares help account for differences in markups and prices when quantities and firm-level prices are not separately observed.[30] Under strict monotonicity in productivity, one can invert $v_t()$ to obtain $z^p_{j,t} = \phi_t(x^v_{j,t}, x^d_{j,t}, s_{j,t})$.

[30] De Ridder, Grassi and Morzenti (2026) discuss the omitted-price problem and the conditions under which production-function methods applied to financial data yield informative markup estimates.

Substituting this control function into the production equation gives a composite function of inputs and market shares.

In the first stage, we approximate this composite function with a cubic polynomial in the two logarithmic inputs, with market shares and year fixed effects entering additively:

$$q_{j,t} = P_{3,l}(x^v_{j,t}, x^d_{j,t}) + \gamma_l' s_{j,t} + \delta_{l,t} + z^{tr}_{j,t}. \tag{A.12}$$

The polynomial includes a constant and all input terms of total degree up to three. Estimating this equation by OLS yields the fitted polynomial component, corresponding to output purged of the first-stage controls and transitory error:

$$\widetilde{q}_{j,t} = q_{j,t} - \widehat{z}^{tr}_{j,t} - \widehat{\gamma}_l' s_{j,t} - \widehat{\delta}_{l,t} = \widehat{P}_{3,l}(x^v_{j,t}, x^d_{j,t}). \tag{A.13}$$

In the second stage, each candidate parameter vector implies control-adjusted productivity $\widetilde{z}^p_{j,t}(\theta_l) = \widetilde{q}_{j,t} - f_l(x^d_{j,t}, x^v_{j,t}; \theta_l)$. We fit a linear AR(1) with an intercept to this series by OLS, recovering the innovation $\widehat{\widetilde{\zeta}}_{j,t}(\theta_l)$. Consistent with the input-timing assumptions, identification requires this innovation to be orthogonal to current capital and lagged variable inputs. We estimate the technology parameters by minimizing

$$\widehat{\theta}_l = \arg\min_{\theta_l} n_l \left\| \frac{1}{n_l} \sum_{(j,t)\in\mathcal{S}_l} Z_{j,t} \widehat{\widetilde{\zeta}}_{j,t}(\theta_l) \right\|_2^2, \tag{A.14}$$

where $\mathcal{S}_l$ is the industry estimation sample and n_l its number of firm–year observations. For the translog specification,

$$Z_{j,t} = \left(x^v_{j,t-1},\ x^d_{j,t},\ (x^v_{j,t-1})^2, x^v_{j,t-1}x^d_{j,t},\ (x^d_{j,t})^2\right)'. \tag{A.15}$$

For Cobb–Douglas, only the first two instruments enter. Estimated revenue productivity is the difference between purged output and the estimated technology.

Following de Loecker and Warzynski (2012), we calculate markups using equation (2). The expenditure share uses observed nominal sales and COGS from the cleaned accounts. For the translog specification, $\widehat{\theta}^v_{j,t} = \widehat{\theta}_{2,l} + 2\widehat{\theta}_{4,l}x^v_{j,t} + \widehat{\theta}_{5,l}x^d_{j,t}$; under Cobb–Douglas it equals $\widehat{\theta}_{2,l}$.

Industry	Translog			Cobb–Douglas		
	$\bar{\theta}_v$	$\bar{\theta}_d$	$\bar{\xi}$	$\bar{\theta}_v$	$\bar{\theta}_d$	$\bar{\xi}$
11. Agriculture, Forestry, Fishing and Hunting	0.82	0.13	0.94	0.98	0.04	1.02
21. Mining, Quarrying, and Oil and Gas Extraction	0.70	0.28	0.99	0.78	0.20	0.99
22. Utilities	0.79	0.17	0.96	0.84	0.12	0.96
23. Construction	0.95	0.02	0.98	0.76	0.11	0.86
31. Food and Bev., Textiles and Clothing	0.83	0.14	0.97	0.93	0.07	1.00
32. Wood, Paper, Petroleum, Chemicals etc.	0.81	0.14	0.95	0.82	0.12	0.94
33. Machinery and Equipment, Computers, Transportation	0.85	0.10	0.95	0.86	0.10	0.96
42. Wholesale Trade	0.91	0.06	0.96	0.89	0.07	0.96
44. Retail Trade	0.83	0.12	0.95	0.82	0.12	0.95
45. Retail Trade	0.87	0.09	0.96	0.86	0.09	0.95
48. Transportation	0.85	0.13	0.98	0.85	0.11	0.96
49. Postal, Courier, and Warehousing Services	0.94	0.05	0.99	0.91	0.06	0.97
51. Information	0.77	0.16	0.92	0.75	0.17	0.92
52. Finance and Insurance	0.66	0.29	0.95	0.72	0.25	0.97
53. Real Estate	0.79	0.11	0.90	0.77	0.11	0.88
54. Professional, Scientific, and Technical Services	0.80	0.15	0.95	0.78	0.15	0.94
56. Adm. and Support and Waste Managem. and Remediation Services	0.85	0.10	0.95	0.85	0.10	0.95
61. Educational Services	0.70	0.18	0.88	0.80	0.14	0.94
62. Health Care and Social Assistance	0.85	0.07	0.92	0.77	0.10	0.88
71. Arts, Entertainment, and Recreation	0.86	0.08	0.94	0.79	0.11	0.89
72. Accommodation and Food Services	0.85	0.12	0.97	0.89	0.09	0.98
81. Other Services (not Public Admin.)	0.76	0.09	0.85	0.72	0.11	0.83

Table A.11: Production Function Estimates

Notes: The table reports estimated output elasticities and returns to scale for the translog and Cobb–Douglas specifications described in Appendix A.3.1. The elasticities with respect to the variable input (COGS) and capital are denoted by $\bar{\theta}_v$ and $\bar{\theta}_d$, respectively, and returns to scale by $\bar{\xi} = \bar{\theta}_v + \bar{\theta}_d$. Translog entries are unweighted averages of firm-year elasticities within each two-digit NAICS industry; Cobb–Douglas elasticities are constant within each industry. Retail trade is estimated separately for NAICS prefixes 44 and 45.

Let $\mathcal{J}_{l,t}$ contain firms with an eligible markup in industry l and year t, and let $w_{j,l,t}$ be a firm's share of sales within this set. We construct the sales-weighted harmonic and arithmetic means as

$$M^H_{l,t} = \left(\sum_{j \in \mathcal{J}_{l,t}} \frac{w_{j,l,t}}{\mu_{j,t}} \right)^{-1}, \qquad M^A_{l,t} = \sum_{j \in \mathcal{J}_{l,t}} w_{j,l,t} \mu_{j,t}. \tag{A.16}$$

The baseline industry outcome is the harmonic mean.[31] Dispersion is the unweighted cross-sectional sample variance of logarithmic markups among the same eligible firms; the industry local projections use its square root. The revenue-productivity outcome is the logarithm of the sales-weighted arithmetic mean of firm revenue productivity, with weights calculated among

[31] Arguments for using the sales-weighted harmonic mean of sector or aggregate markups can be found in Grassi (2017), Baqaee and Farhi (2020), Edmond, Midrigan and Xu (2023), and Hasenzagl and Pérez (2023).

firms with a nonmissing productivity estimate. Economy-wide descriptive series apply the same aggregation rules across all eligible firms in a year.

A.3.2 Details of the Synthetic DiD Estimator

This appendix describes the weights and inference for equation (3), using the treatment definition and sample restrictions in Section 3.2.[32] For a comparison cell c with filing year g and horizon h, let $\mathcal{T}_{c,h}$ and $\mathcal{C}_{c,h}$ contain the eligible defendants and controls, with respective counts $n^T_{c,h}$ and $n^C_{c,h}$. Write $P_g = \{g-5, \ldots, g-1\}$ for the pre-filing years and define $\overline{y}^T_{t,c,h} = (n^T_{c,h})^{-1} \sum_{j \in \mathcal{T}_{c,h}} y_{j,t}$.

Sales quintiles are formed from ranks among eligible firms in sector–years with at least ten firms. Sector and quintile assignments are fixed at $g-1$ across horizons. Eligibility is assessed separately for each outcome, without requiring the other outcomes to be observed.

The control-firm weights in equation (3) solve

$$(\widehat{a}_{c,h}, \widehat{\boldsymbol{\omega}}_{c,h}) = \arg\min_{a_{c,h}, \boldsymbol{\omega}_{c,h}} \frac{1}{5} \sum_{t \in P_g} \left(\overline{y}^T_{t,c,h} - a_{c,h} - \sum_{j \in \mathcal{C}_{c,h}} \omega_{j,c,h} y_{j,t} \right)^2 + \rho^{\omega}_{c,h} \sum_{j \in \mathcal{C}_{c,h}} \omega^2_{j,c,h}, \qquad \omega_{j,c,h} \geq 0, \quad \sum_{j \in \mathcal{C}_{c,h}} \omega_{j,c,h} = 1. \tag{A.17}$$

The pre-filing time weights solve

$$(\widehat{d}_{c,h}, \widehat{\boldsymbol{\lambda}}_{c,h}) = \arg\min_{d_{c,h}, \boldsymbol{\lambda}_{c,h}} \frac{1}{n^C_{c,h}} \sum_{j \in \mathcal{C}_{c,h}} \left(y_{j,g+h} - d_{c,h} - \sum_{t \in P_g} \lambda_{t,c,h} y_{j,t} \right)^2 + \rho^{\lambda}_{c,h} \sum_{t \in P_g} \lambda^2_{t,c,h}, \qquad \lambda_{t,c,h} \geq 0, \quad \sum_{t \in P_g} \lambda_{t,c,h} = 1. \tag{A.18}$$

In equation (3), each defendant receives firm weight $1/n^T_{c,h}$, and year $g+h$ receives time weight one.

Here $\boldsymbol{\omega}_{c,h} = (\omega_{j,c,h})_{j \in \mathcal{C}_{c,h}}$ and $\boldsymbol{\lambda}_{c,h} = (\lambda_{t,c,h})_{t \in P_g}$ collect the weights, and $a_{c,h}$ and $d_{c,h}$ are unrestricted intercepts. Let $\widehat{\sigma}_{c,h}$ be the sample standard deviation of controls' changes between adjacent pre-filing years, pooled across firms and dates. The penalties are $\rho^{\omega}_{c,h} = \sqrt{n^T_{c,h}}\, \widehat{\sigma}^2_{c,h}$ and

[32] The non-merger category includes source records classified as civil non-merger, criminal, or mixed.

$\rho^{\lambda}_{c,h} = 10^{-12}\widehat{\sigma}^2_{c,h}$. The former limits concentration of control-firm weights; the latter regularizes the time-weight optimization. No additional covariates enter these objectives.

For cell c and horizon h, the difference in year t between defendants' average outcome and the weighted average outcome of controls is

$$G_{t,c,h} = \overline{y}^{T}_{t,c,h} - \sum_{j \in \mathcal{C}_{c,h}} \widehat{\omega}_{j,c,h} y_{j,t}. \tag{A.19}$$

The estimated effect of a filing within this cell is the difference at $g+h$ minus the time-weighted average of the five pre-filing differences:

$$\widehat{\tau}_{c,h} = G_{g+h,c,h} - \sum_{t \in P_g} \widehat{\lambda}_{t,c,h} G_{t,c,h}. \tag{A.20}$$

The final synthetic difference-in-differences estimate at horizon h averages these cell estimates, weighting each by its number of eligible defendants. Writing $\mathcal{K}_h$ for the eligible comparison cells,

$$\widehat{\tau}_h = \sum_{c \in \mathcal{K}_h} \frac{n^{T}_{c,h}}{\sum_{c' \in \mathcal{K}_h} n^{T}_{c',h}} \widehat{\tau}_{c,h}. \tag{A.21}$$

Reported responses and standard errors are multiplied by 100. The pre-filing points in Figure 6 use $G_{t,c,0}$ from the $h = 0$ fit, subtract its time-weighted pre-filing average, and aggregate with the same defendant-count weights.

For pointwise inference at each horizon, we draw $B = 1{,}000$ bootstrap samples from the distinct firms appearing in its comparison cells. Each selected firm contributes its full observation history. If a firm is sampled more than once, the same number of copies is used in every comparison cell where it appears. We refit both sets of synthetic weights and aggregate using the resampled defendant counts, holding the original penalties and cell assignments fixed. Cells with no resampled defendant contribute nothing. A draw is repeated if a cell with a resampled defendant has fewer than two control copies, or if no cell remains. The 95 percent intervals are $\widehat{\tau}_h$ plus or minus 1.96 times the bootstrap standard deviation, calculated with divisor B.

For Table 3, we estimate the average response

$$\widehat{\bar{\tau}} = \frac{1}{11}\sum_{h=0}^{10}\widehat{\tau}_h. \tag{A.22}$$

Eligible firms may differ across horizons. We draw B additional bootstrap samples from the union of firms across the eleven fits. Within each draw, the number of times a firm is sampled is held fixed across all horizons and comparison cells. We refit the weights as above and repeat a draw across all horizons if it fails the sampling conditions at any one horizon. With $\widehat{\bar{\tau}}^{*(b)}$ denoting the average in draw b, the two-sided p-value for a zero average response is

$$\widehat{p} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{\left|\widehat{\bar{\tau}}^{*(b)} - \widehat{\bar{\tau}}\right| \geq \left|\widehat{\bar{\tau}}\right|\right\}}{B+1}. \tag{A.23}$$

The standard error is the standard deviation of the bootstrap averages, using divisor $B-1$.

The held-out exercise in Appendix Figure A.5 uses years $g-5, g-4, g-3$ for training and $g-2, g-1$ for validation. Sector and sales-quintile assignments are measured at $g-3$. The control-firm objective averages over the three training years, replacing $1/5$ by $1/3$ in (A.17). In the time-weight objective, the control outcome at $g+h$ is replaced by its mean over the two validation years. The control-firm penalty becomes $\sqrt{2n_c^T}\,\widehat{\sigma}_c^2$, where n_c^T is the number of eligible defendants in the validation cell and $\widehat{\sigma}_c$ is computed from the training years. Neither held-out defendant outcome enters the weight estimation. Training observations satisfy the baseline eligibility rules; validation observations require only an observed positive value of the relevant outcome. Joint validation tests compare a Wald statistic based on the bootstrap covariance matrix with its centered bootstrap distribution.